Retirement Reforms That Pay for Themselves*

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Abstract

Retirement policies change the return to work decades before benefits are paid, act differently on workers in good and poor health, and spill over to income taxes, disability insurance, and means-tested transfers. We compute the marginal value of public funds (MVPF) of 66 retirement reforms in a life-cycle model of labor supply, saving, and Social Security claiming with health, disability, mortality, and wage risk, estimated on US panel data. For the 1920–1940 birth cohorts, the ranking turns on whether a reform rewards later claiming or lowers benefits at every claiming age. Raising the delayed retirement credit from the cohort-average 4.5 percent, or steepening the adjustment of benefits around the normal retirement age, raises welfare and lowers net government spending: these reforms pay for themselves. Raising the normal retirement age saves money but lowers welfare. Among benefit expansions, raising the bottom replacement rate of the benefit formula delivers the most welfare per dollar, and expanding Medicare coverage the least.

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*Ndiaye: New York University (an1490@nyu.edu). Yu: Louisiana State University. The quantitative life-cycle analysis of retirement policy in this paper first appeared in an earlier version of our companion paper, Ndiaye and Yu (2026), which studies the constrained-efficient design of taxes and retirement benefits in a dynamic Mirrlees economy estimated on the same data. This paper develops the positive and fiscal analysis separately.

1 Introduction

The marginal value of public funds (MVPF) has become a common currency for comparing government policies. It divides the beneficiaries' willingness to pay for a policy by the policy's net cost to the government, where the net cost includes the fiscal externalities created by the behavioral responses the policy induces (Hendren 2016; Hendren and Sprung-Keyser 2020). Because it puts a tax cut, a health insurance subsidy, and a training program on the same scale, the MVPF identifies which policies deliver the most welfare per dollar and which pay for themselves. Most existing MVPFs rest on quasi-experimental estimates of behavioral responses, and their long-run fiscal externalities are projected rather than observed (Hendren and Sprung-Keyser 2020; Finkelstein and Hendren 2020).

Life-cycle policies fit this approach less easily, and retirement policy is the leading example. Social Security benefits depend on a worker's highest 35 years of earnings, so a change in the benefit formula changes the return to work at age 35 as well as at age 65. Rules that tie benefits to the claiming age change when people stop working, and therefore change income and payroll tax revenue and the outlays of disability insurance and means-tested transfers. Who responds depends on health: whether a worker can postpone retirement depends on how costly work becomes as health deteriorates. A natural experiment that shifts benefits for one cohort identifies a local response of claiming or employment. It does not identify the present value of the fiscal externalities that accumulate over a lifetime, nor the insurance value of a reform to a worker who does not yet know whether their health will hold up.

Retirement policy also carries large fiscal stakes. Population aging puts pressure on the financing of public pensions (Board of Trustees, OASDI 2024; Congressional Budget Office 2024), and the most common response is to raise the age at which full benefits are paid. The Social Security Amendments of 1983 raised the normal retirement age (NRA) in the United States from 65 to 67, beginning with workers born in 1938. They also raised the delayed retirement credit (DRC), the increase in annual benefits for each year a worker postpones claiming beyond the NRA, gradually to 8 percent per year (Svahn and Ross 1983; Diamond and Gruber 1999). Cross-country simulations of pension reform likewise contrast increases in eligibility ages with changes in the actuarial adjustment of benefits (Gruber and Wise, eds 2007). This paper asks which retirement reforms deliver the most welfare per dollar of public funds, and which pay for themselves.

We answer this question in an estimated life-cycle model of retirement. Individuals face health shocks with three states—good health, poor health, and disability—that lower wages, raise out-of-pocket medical expenses, and raise mortality. They face persistent

wage risk, choose consumption, saving, labor force participation, and hours, and decide when to claim Social Security benefits. Their preferences over consumption and leisure are nonseparable, as in French and Jones (2011), and working entails a fixed time cost that is larger in poor health, so that labor supply is nonconvex and retirement emerges as a choice. Reentering the labor market after an exit is costly. The model incorporates the main features of the US system: the Social Security benefit formula with its claiming-age adjustments and earnings test, disability insurance, employer pensions, Medicare, a progressive and age-dependent income tax, payroll taxes, and a means-tested consumption floor. We estimate it in two steps on the Panel Study of Income Dynamics for the cohorts born between 1920 and 1940, with out-of-pocket medical expenses from the Medical Expenditure Panel Survey. Health transitions, mortality, medical expenses, and wages are estimated directly from the data, and preferences are estimated by the method of simulated moments to match life-cycle profiles of participation and hours by health, of mean and median assets, and of labor market reentry. The estimated model reproduces the main features of the participation and hours profiles: high and stable participation in the thirties and forties, an earlier decline of participation among workers in poor health or with a disability, and the sharp exits of the sixties. It understates wealth after the early fifties and generates no reentry.

The estimates imply that working requires a fixed time cost of 791 hours per year in good health and 868 hours in poor health or with a disability—ratios of about 18 and 19 percent to the estimated annual time endowment of 4,491 hours—and that returning to work after an exit costs 3,312 hours. These nonconvexities make retirement a discrete and, in practice, irreversible choice, and they determine which workers can respond when the reward for working an additional year changes. In our companion paper (Ndiaye and Yu 2026), we show that these nonconvexities are the main driver of nonparticipation at older ages in this economy, ahead of the effects of health on wages and on mortality.

We use the estimated model to compute the MVPF of 66 reforms in five families: the claiming-age dependence of Social Security benefits (the NRA, the DRC, and symmetric adjustments around the NRA), the progressivity and level of the benefit formula, Medicare coverage after age 65, income taxes after age 62, and the return to saving. For each reform, we compute the ex ante willingness to pay of a worker at age 30, before health and wage histories are realized, the direct cost of the reformed program, the change in the net cost of the other taxes and transfers, and the implied MVPF. The model thus serves as a laboratory in which both parts of the MVPF of a life-cycle policy are computed over the whole life cycle: the net cost includes fiscal effects that materialize decades after behavior changes, and the ex ante willingness to pay includes the value of a reform as insurance against

health risk.

Our central result is that, for the cohorts we study, the MVPF of a claiming-age reform depends on whether the reform rewards later claiming or lowers benefits at every claiming age. Consider first the reforms that reward later claiming. Raising the DRC from the cohort-average 4.5 percent to between 8 and 10 percent, or making the adjustment of benefits around the NRA symmetric and steeper, raises welfare by 0.10 to 0.80 percent of consumption at every age and lowers net government spending by 0.12 to 1.57 percent. These reforms have an infinite MVPF: they pay for themselves. Raising the NRA to 66, 67, or 68 instead lowers welfare by 0.38 to 1.09 percent while saving 0.87 to 2.38 percent of government spending. Its MVPF, 0.16 to 0.18 on the consumption base we use for reforms that begin at 62, is the welfare cost of each dollar saved, and it is lower than the MVPF of every benefit expansion in our grid except the Medicare expansions. Reading those expansions in reverse, raising the NRA is a comparatively inexpensive way to save money, but rewards for later claiming save money while raising welfare. We call an NRA increase a mandate in a loose sense: it does not prevent anyone from claiming at 62, but it lowers the benefit at every claiming age, so that a worker must claim later to receive a given benefit.

The welfare and fiscal sides of this result have different explanations. On the welfare side, a higher DRC weakly enlarges every worker's choice set at given prices and taxes: it raises the benefit of anyone who claims after the NRA and lowers no one's. Its value, assessed at age 30 before health is known, combines a higher reward for working longer when work remains feasible with unchanged terms for workers whose health leads them to claim early. A higher NRA lowers the benefits of all claimants, including those who cannot postpone retirement at low cost; the additional participation it generates at ages 65 to 69, 0.60 to 3.27 percentage points, is consistent with an income effect of lower benefits at every claiming age rather than a response to a higher reward. The fiscal side is the substantive result. Social Security's own outlays fall under every DRC increase, by 2.46 to 5.05 percent, even though a higher credit would raise outlays at unchanged behavior; the behavioral responses within the program—the induced changes in claiming ages, together with the effects of work through the earnings test and the AIME—therefore more than finance the higher credit. A one-year delay lowers the present value of a claimant's benefits whenever the credit is below the rate that would make the delay actuarially fair from the government's perspective, given the claimant's survival prospects and the interest rate, and this rate is higher for claimants with lower survival, as is typical in worse health. Abstracting from the earnings test and the AIME, the fall in outlays implies that the induced delays occur, on balance, where the credit is below that rate, in line with evidence

that delaying beyond the NRA was not actuarially advantageous for cohorts that faced credits well below 8 percent (Coile et al. 2002; Shoven and Slavov 2014). The moderate versions of these reforms also raise participation in every age group we track, and the net cost of the other taxes and transfers falls, consistent with the additional work at ages 65 to 69 that every one of these reforms induces.

Two of our reforms correspond to the end points of the two changes to claiming incentives in the 1983 Amendments: an NRA of 67 and a DRC of 8 percent. Evaluated for the 1920–1940 cohorts, whose schedule we represent with an NRA of 65 and a cohort-average credit of 4.5 percent, the credit increase raises welfare by 0.16 percent and saves 0.45 percent of government spending, whereas the increase in the NRA lowers welfare by 0.74 percent and saves 1.75 percent, an MVPF of 0.16. The two halves of the change sit at opposite ends of the ranking. In our grid, rewarding later claiming also dominates raising the NRA whenever the required fiscal saving is moderate: raising the DRC to 10 percent saves nearly as much as raising the NRA to 66 while raising welfare, and the steepest symmetric adjustment we consider saves more than the increase to 66 and raises welfare by 0.80 percent.

The remaining reform families sharpen the picture. Every expansion of benefit generosity except the smallest raises net spending. The smallest increase in the bottom replacement rate of the benefit formula is approximately budget neutral: its indirect saving, which our decomposition attributes mainly to income taxes paid by additional participants, offsets its direct cost; larger increases in the bottom rate have MVPFs between 0.69 and 1.14, above those of proportional increases in all rates, between 0.30 and 0.61. The bottom rate targets workers whose lifetime earnings were depressed by low ability or poor health. Raising the upper replacement rates on top of the bottom one adds little: from the largest increase in the bottom rate to the largest increase in the second rate, willingness to pay rises by 0.15 percent of consumption while the net cost rises by 0.83 percent of government spending. Expanding Medicare coverage after age 65 by 20 percentage points has the lowest MVPF of any expansion, about 0.15. Cutting income taxes after age 62 generates the largest participation responses of any reform we study, up to 8.9 percentage points at ages 65 to 69, yet its MVPF, 0.23 to 0.26, is lower than that of any expansion of the benefit formula. Subsidies to saving raise welfare at a small net cost, because additional saving raises taxable interest income, and they lower participation at older ages; their MVPFs rest on a different consumption base from the other reforms. Finally, eliminating each instrument in turn shows that Social Security is the most valuable of them: without it, welfare would be 5.89 percent of consumption lower. Because such eliminations do not hold the budget fixed, we rely on the MVPF to compare reforms.

These results suggest an order of operations for closing a financing gap for the cohorts and rules we study. The MVPF ranks the ways of saving a dollar by their welfare cost. Reforms that reward later claiming come first: they save money and raise welfare at the same time. To a first-order approximation that reads our simulated expansions in reverse, raising the NRA costs less welfare per dollar saved than raising income taxes after age 62 or cutting all replacement rates proportionally, but it is dominated by rewards for later claiming up to the largest fiscal saving that a reward-delay reform in our grid achieves.

Three features of the MVPF of a life-cycle policy drive these conclusions and are easy to miss in short-run evaluations. First, willingness to pay is computed ex ante, before health is known, so it values the protection a reform provides, or removes, for workers whose health later deteriorates. Second, part of the fiscal consequences of retirement reforms falls outside the reformed program and accrues over many years after the reform first changes behavior: the smallest expansion of the bottom replacement rate is approximately budget neutral because its indirect saving offsets its direct cost. Third, MVPF levels depend on how the willingness to pay is converted into dollars. We value it over consumption from age 62 onward for reforms that operate through retirement benefits, Medicare, or old-age taxes. This convention leaves every ranking among these reforms and every infinite MVPF unchanged, but the finite levels are about one-fifth of those obtained by valuing the willingness to pay over lifetime consumption, the normalization under which a transfer with no behavioral response has an MVPF of one (Section 7). We report the willingness to pay and the fiscal effects separately and state the formula that maps them into the MVPF; up to rounding, every MVPF computed with the net cost equals the ratio of the willingness to pay to the net cost times a factor that depends only on the consumption base.

Related literature. This paper contributes to four literatures. First, it builds on structural life-cycle models of retirement, which have shown how Social Security, pensions, health, and health insurance shape labor supply and saving at older ages (Gustman and Steinmeier 1986; Rust 1989; Rust and Phelan 1997; French 2005; van der Klaauw and Wolpin 2008; De Nardi et al. 2010; French and Jones 2011; Haan and Prowse 2014; Low and Pistaferri 2015; Blundell et al. 2016; De Nardi et al. 2024; Fan et al. 2024), and which have been used to evaluate changes in the NRA and other Social Security reforms (Conesa and Krueger 1999; Imrohoroğlu and Kitao 2012; Kitao 2014; Laun and Wallenius 2015; Li 2018; Bairoliya and McKiernan 2023). Jones and Li (2023) find that, with mortality that differs by education, the welfare-maximizing reform of the current system weakens the dependence of benefits on lifetime earnings and reduces claiming-age adjustments. Their exercise searches for the best reform of the current system; ours measures the value per dollar of discrete reforms

from the cohort-average credit of 4.5 percent, and we discuss the relation between the two in Section 7. Our model follows French (2005) and French and Jones (2011) in its treatment of preferences, wages, and the Social Security rules. It adds a disability state with disability insurance, which provides an alternative pathway out of the labor force at older ages (Maestas et al. 2013; French and Song 2014; Low and Pistaferri 2015; Yu 2024), fixed costs of work that depend on health, and age-dependent progressive income taxes (Borella et al. 2023), and it targets the population rather than male household heads. Relative to this literature, our contribution is to map a large set of reforms into a common welfare metric that accounts for their fiscal consequences across programs.

Second, the paper contributes to the welfare analysis of government policies with the MVPF (Hendren 2016; Hendren and Sprung-Keyser 2020) and to the sufficient-statistics approach to social insurance (Chetty 2009; Kleven 2021). Closest to our question, Haller (2022) evaluates pension reforms with sufficient statistics in the Baily–Chetty tradition, where the welfare effect of a reform compares a consumption-smoothing benefit with the total fiscal effect of the reform, and Kolsrud et al. (2024) show with Swedish data that workers who retire earlier consume less in retirement, so that reforms pushing workers to retire later carry a substantial consumption-smoothing cost. We show how the components of our MVPF map into the Baily–Chetty comparison. The structural approach measures the fiscal effects of behavioral responses over the life cycle and across programs and computes an ex ante willingness to pay that includes the value of each reform as insurance; sufficient-statistics estimates discipline the local behavioral responses on which such calculations rest. The two approaches are complements.

Third, the paper relates to the evidence on how workers respond to Social Security rules. Program rules generate strong early-retirement incentives across countries (Gruber and Wise 1998; Gruber and Wise, eds 2004), and older Americans have been working longer (Coile 2019). Workers respond to benefit levels, to the marginal return to additional work, and to the early and normal retirement ages (Coile and Gruber 2007; Liebman et al. 2009; Mastrobuoni 2009; Staubli and Zweimüller 2013; Manoli and Weber 2016). Claiming and labor force exit are closely linked (Shoven et al. 2017); an information intervention raised labor force participation without detectable effects on claiming (Liebman and Luttmer 2015); the earnings test affects employment (Gelber et al. 2020a,b); and an increase in the NRA raises disability insurance enrollment (Duggan et al. 2007). Statutory retirement ages also affect behavior beyond the financial incentives they create: they act as reference points for retirement (Behaghel and Blau 2012; Seibold 2021; Lalive et al. 2023), and when the NRA rose from 65 to 66, claiming moved with it while labor force exit remained concentrated at 65 (Deshpande et al. 2024). Our reforms of the claiming-age schedule work only through

financial incentives; we discuss how reference-point effects of statutory ages would change our conclusions.

Fourth, the paper complements the normative analysis of retirement and social insurance (Diamond and Mirrlees 1978; Cremer et al. 2004; Golosov and Tsyvinski 2006; Huggett and Parra 2010; Golosov et al. 2013; Hosseini 2015; Nishiyama and Smetters 2007; Hosseini and Shourideh 2019). In our companion paper (Ndiaye and Yu 2026), we characterize the constrained-efficient allocation in a dynamic Mirrlees economy with endogenous retirement, estimated on the same data, and find that optimal retirement consumption rises with the claiming age more steeply than the statutory delayed retirement credit implies. The present paper takes the opposite route. Instead of characterizing the optimum, it evaluates reforms within the existing system. The two analyses point in the same direction: the reforms that steepen the claiming-age schedule are those that pay for themselves.

Outline. Section 2 presents the model. Section 3 describes the data and the estimation and evaluates the model's fit. Section 4 defines the welfare and fiscal measures, the MVPF, and its decomposition. Section 5 presents the results, and Section 6 draws their implications for closing a financing gap. Section 7 discusses the measurement conventions, general equilibrium, reference-point effects of statutory ages, the relation between the cohorts we study and current law, the role of preferences and disability insurance, and distribution. Section 8 concludes. The appendices contain the details of the model, the data, the estimation, the construction of the reforms, and the derivations.

2 A Life-Cycle Model of Retirement with Health

This section presents the life-cycle model in which we evaluate retirement reforms and the government accounts with which we measure their fiscal effects. In the model, worse health raises the fixed time cost of working, lowers wages, raises out-of-pocket medical expenses, lowers survival, and, in its worst state, opens access to disability insurance. Retirement is therefore a response both to incentives and to health, and health determines who can respond to a reform and at what cost. The institutions are those through which retirement reforms change behavior and the budget: the Social Security benefit formula with its claiming-age adjustments and earnings test, disability insurance, pensions, Medicare, the income and payroll taxes, and a means-tested consumption floor.

2.1 Demographics, Timing, and Choices

The economy is populated by a single cohort of individuals. Time is discrete and annual, and we index periods by age t. Individuals enter the model at age 30 and live at most until

age $T = 85$; between consecutive ages they face a mortality risk that depends on age and health. The institutional environment is that faced by the cohort born between 1920 and 1940, whose life cycle we observe in the data (Section 3). We model individuals rather than households: the budget constraint contains no spousal income, and the model is estimated to match the outcomes of the whole cohort rather than those of male household heads.

At the beginning of age t, an individual observes her health h_t and her ability θ_t, which together determine her hourly wage. She holds assets a_t, a record of past earnings summarized by her annualized average indexed monthly earnings $aime_t$, and two indicators from the previous period: whether she has claimed Social Security retirement benefits, $b_{t-1} \in \{0, 1\}$, and whether she participated in the labor market, $P_{t-1} \in \{0, 1\}$. She then chooses consumption c_t, hours of work $n_t \geq 0$, and, if eligible, whether to claim retirement benefits, b_t. Participation is $P_t = \mathbf{1}\{n_t > 0\}$. At the end of the period she survives to age $t + 1$ with a probability that depends on her current health and age, and next period's health and ability are drawn.

The life cycle has three stages that differ in the choice set. Between ages 30 and 61, individuals choose consumption and hours. Between ages 62 and 69, they also decide whether to claim Social Security retirement benefits. Claiming is a one-time, irreversible decision, $b_t \geq b_{t-1}$, and it is not available before the early retirement age of 62. Claiming must occur by age 70: individuals who have not claimed earlier receive benefits from age 70, so that between ages 70 and 85 every individual receives benefits and chooses consumption and hours. Labor supply remains a choice at every age up to 85. Retirement, the decision to stop working, is therefore distinct from claiming: individuals can claim while working, stop working before claiming, and work past age 70.

2.2 Preferences

Flow utility is nonseparable in consumption and leisure, as in French and Jones (2011):

$$u(c_t, l_t) = \frac{1}{1-\nu} \left(c_t^{\gamma} \, l_t^{1-\gamma} \right)^{1-\nu}, \tag{1}$$

where $\gamma \in (0, 1)$ is the weight on consumption and $\nu > 0$ is the curvature of utility in the consumption–leisure composite. Leisure is the time endowment net of hours worked and of two time costs of working:

$$l_t = L - n_t - \phi_n(h_t) \, \mathbf{1}\{n_t > 0\} - \phi_{RE} \, RE_t, \qquad RE_t = P_t \, (1 - P_{t-1}). \tag{2}$$

Here L is the annual time endowment in hours, $\phi_n(h_t)$ is a fixed time cost incurred whenever hours are positive, and ϕ_{RE} is a reentry cost incurred by an individual who

works at age t after not working at age $t-1$. Individuals discount the future with factor β, so that expected lifetime utility at age 30 is

$$U = \mathbb{E}_{30} \sum_{t=30}^{T} \beta^{t-30} \left(\prod_{k=31}^{t} s_k \right) u(c_t, l_t),$$

where s_k is the probability of surviving from age $k-1$ to age k (Section 2.3). There is no bequest motive: assets held at death are accidental bequests that the individual does not value, and $a_{T+1} = 0$ (Section 2.12 discusses this simplification).

The fixed cost $\phi_n(h_t)$ makes labor supply nonconvex. The first hour of work costs $1 + \phi_n(h_t)$ hours of leisure, so short workweeks are dominated: an individual either works enough hours to justify the fixed cost or does not work at all. As incentives to work weaken with age, hours fall continuously only down to a threshold, below which participation drops discretely to zero. This is the mechanism through which life-cycle models generate retirement as a discrete exit rather than as a gradual reduction in hours (French 2005; Prescott et al. 2009; Rogerson and Wallenius 2013), and it places the participation margin at the center of the response to old-age incentives.

The reentry cost ϕ_{RE} makes an exit costly to reverse: an individual who has stopped working returns only if the gain from working covers ϕ_{RE} in addition to the fixed cost. Reentry is rare in the data, and the estimated reentry cost is large (Section 3). Incentives to work longer therefore act mainly on individuals who are still working when the incentive applies, rather than on those who have already left.

The fixed cost depends on health. Declining health makes work less pleasant (Blundell et al. 2016), and a larger fixed cost raises the wage at which working becomes worthwhile, so that at a given wage unhealthy individuals are more likely to exit. Because the PSID contains few observations of individuals with a disability, we impose $\phi_n(1) = \phi_n(2)$ in estimation: poor health and disability share one participation cost. Disability nevertheless differs from poor health through wages, medical expenses, survival, and eligibility for disability insurance.

Three properties of (1) are used below. First, the coefficient of relative risk aversion over consumption, holding leisure fixed, is $-u_{cc}c/u_c = 1 - \gamma(1-\nu)$, which equals 1.38 at the estimates of Section 3.[1] Second, when $\nu > 1$, as in our estimates, the marginal utility of consumption falls with leisure ($u_{cl} < 0$), so that, other things equal, optimal consumption drops when an individual stops working. Third, utility is homogeneous in consumption: scaling consumption at every age and in every state by e^x multiplies flow utility, and

[1] With $\gamma = 0.5354$ and $\nu = 1.7029$, $1 - \gamma(1-\nu) = 1 + 0.5354 \times 0.7029 \approx 1.38$.

hence expected lifetime utility, by $e^{\gamma(1-\nu)x}$. Section 4.2 uses this property to express welfare changes as consumption equivalents.

2.3 Health, Disability, and Mortality

Health takes three values, $h_t \in \mathcal{H} \equiv \{0, 1, 2\}$: good health ($h_t = 0$), poor health ($h_t = 1$), and disability ($h_t = 2$). In the data, health is measured from self-reported work limitations in the PSID and self-perceived health in the MEPS (Section 3 and Appendix B). Health follows a Markov chain with age-dependent transition probabilities

$$\pi_{j,i,t+1} = \Pr\left(h_{t+1} = j \,|\, h_t = i,\, t+1\right), \qquad i, j \in \mathcal{H}.$$

The probability of surviving to age $t+1$ depends on current health and age, $s_{t+1} = S(h_t, t+1)$, with $s_{T+1} = 0$. Health at age 30 is drawn, jointly with assets, wages, and past participation, from the empirical distribution in the PSID (Appendix B).

Health and survival evolve exogenously: neither depends on past work, consumption, or medical spending. This is the identification assumption of French (2005), under which work decisions do not affect health or wages. It implies that reforms in the model change behavior, welfare, and the budget, but not the distribution of health or longevity in the cohort. Figure 1 in Section 3 shows the estimated transition, survival, medical-expense, and wage profiles by health.

2.4 Wages

The hourly wage has a deterministic component that depends on health and age and a persistent idiosyncratic component, ability:

$$\ln w_t(\theta_t, h_t) = W(h_t, t) + \ln \theta_t, \tag{3}$$

$$\ln \theta_t = \rho \, \ln \theta_{t-1} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \sigma^2), \tag{4}$$

where the innovations ε_t are independent over time and across individuals. The deterministic profiles $W(h, t)$ are estimated from the PSID with a correction for selection into work, and ability follows an AR(1) process with $\rho = 0.981$ and $\sigma^2 = 0.016$ (Section 3). At every age, the estimated wage profile is highest in good health and lowest with a disability, so that a deterioration of health lowers the return to work at the same time as it raises its fixed cost. Wages do not depend on past work, so the model abstracts from human-capital accumulation on the job. Because ability is highly persistent, a wage shock near retirement changes expected wages for the remainder of the working life and, through $aime_t$, future benefits.

2.5 Out-of-Pocket Medical Expenses

Out-of-pocket medical expenses depend on health, age, and participation, $m_t = M(h_t, t, P_t)$. They are total medical expenditure net of the part paid by health insurance before age 65 and by Medicare from age 65. Medical expenses are not a choice: given health, age, and participation they are deterministic, so medical-expense risk in the model is health risk. Out-of-pocket expenses rise with age and are considerably higher in poor health and with a disability (Section 3). The model does not distinguish types of health insurance; Blau and Gilleskie (2008), French and Jones (2011), and Imrohoroğlu and Kitao (2012) model employer-provided coverage explicitly.

From age 65, Medicare pays the part of total medical expenditure m_t^{tot} that is not paid out of pocket, $m_t^G = m_t^{tot} - m_t$. These payments do not enter the individual's budget. They are the fiscal object of the Medicare reforms of Section 5.4, which raise the share of expenditure paid by Medicare and thereby shift expenditure from m_t to m_t^G. These reforms are evaluated against a baseline in which Medicare's coverage rate varies with health and working status as measured in the MEPS (Appendix D).

2.6 Social Security

Social Security retirement benefits depend on lifetime earnings, the age at which benefits are claimed, and earnings after claiming. We model each element following the rules faced by the 1920–1940 cohort; Table A.1 in Appendix A lists the parameters, in 1987 dollars.

Lifetime earnings. Benefits are based on the average indexed monthly earnings (AIME), approximately the average of the highest 35 years of earnings. We track an annualized version, $aime_t$, which is capped at $aime_{max}$ and evolves according to

$$aime_{t+1} = \begin{cases} \min\left\{aime_t + \dfrac{w_t n_t}{35},\ aime_{max}\right\} & \text{if } t < 60, \\ \min\left\{aime_t + \max\left\{0, \dfrac{w_t n_t - aime_t}{35}\right\},\ aime_{max}\right\} & \text{if } t \geq 60. \end{cases} \tag{5}$$

Individuals are assumed to start working at age 25, so that they enter the model with five years of earnings (French 2005); ages 25 to 59 thus span the 35 years of the averaging period. From age 60, an additional year raises $aime_t$ only if its earnings exceed the current average, which approximates the replacement of a low-earnings year in the highest 35. This weakens the incentive to keep working once a full earnings record has been accumulated.

Benefit formula. The primary insurance amount (PIA) is a piecewise-linear function of $aime_t$:

$$PIA_t = 0.90\ \min\{aime_t, aime_0\} + 0.32\ \min\left\{\ \max\{aime_t - aime_0, 0\},\ aime_1 - aime_0\right\} + 0.15\ \max\{aime_t - aime_1, 0\}, \tag{6}$$

with bend points $aime_0$ and $aime_1$. The marginal replacement rate falls from 90 to 32 to 15 percent across the three brackets, so the formula replaces a larger share of lifetime earnings for low earners. The progressivity and level reforms of Section 5.3 raise these three rates.

Claiming age. The benefit to which an individual is entitled before the earnings test is $ssb_t = \alpha(t^c)\, PIA_t$, where t^c is the claiming age and $\alpha(\cdot)$ is the actuarial adjustment. Benefits can be claimed from the early retirement age (ERA) of 62. Claiming before the normal retirement age (NRA) reduces benefits, and delaying a claim past the NRA raises them through the delayed retirement credit (DRC), up to age 70. For our cohort the NRA is 65, and the DRC is set to 4.5 percent, approximately the cohort average: as the 1983 Amendments phased in a credit of 8 percent, the DRC rose with the year of birth from 3 percent for those born before 1925 to 7 percent for those born in 1939–1940. Under these rules, an individual who claims at 62 receives 80 percent of her PIA. Each year of delay raises the benefit by 6.67 percentage points, to 100 percent at the NRA of 65, and each year of delay after the NRA raises it by the DRC of 4.5 percent of the PIA, to 122.5 percent at 70. Table D.1 in Appendix D reports this schedule (row AN0) together with those of the claiming-age reforms. The reduction for early claiming was roughly actuarially fair for an individual with average mortality, whereas the DRCs faced by these cohorts were below the actuarially fair rate: on average, delaying a claim beyond the NRA did not raise the expected present value of benefits (Coile et al. 2002), and the later rise of the DRC to 8 percent is one reason why delay has become more valuable for later cohorts (Shoven and Slavov 2014). In the model, survival depends on health, so the actuarial value of a delay differs across individuals; Section 4.6 derives when a marginal delay saves the government money.

Earnings test. Beneficiaries younger than 70 who work are subject to the retirement earnings test (RET). Each dollar of labor earnings above an exempt amount y_{ret} reduces benefits by τ_{ret} dollars, until the entire benefit is withheld:

$$ss_t = \max\left\{0,\ ssb_t - \tau_{ret}\max\{0,\ w_t n_t - y_{ret}\}\right\} \qquad \text{if } t < 70, \tag{7}$$

and $ss_t = ssb_t$ from age 70. Below the NRA, $\tau_{ret} = 50$ percent and $y_{ret} = \$6{,}000$; between the NRA and age 69, $\tau_{ret} = 33$ percent and $y_{ret} = \$8{,}186$. The model holds these parameters fixed, a stylized version of rules that changed during the cohort's old age (Appendix A.1). Withheld benefits are not lost. Following French and Jones (2011), they are credited as if claiming had been delayed: a year's worth of benefits withheld before the NRA raises future benefits by the early-claiming adjustment for one year, and a year's worth withheld after the NRA raises them by the DRC. The earnings test thus taxes work only insofar as these credits fall short of actuarial fairness for the individual or withholding tightens her liquidity (see Gelber et al. 2020a, for evidence on its employment effects). A higher DRC therefore also lowers the implicit tax on beneficiaries who keep working after the NRA.

2.7 Disability Insurance and Pensions

Disability insurance. Individuals with a disability who are older than 50 and younger than the NRA receive disability insurance (DI) benefits if their labor earnings are below a threshold y_{db}:

$$db_t = \pi^{db}\, PIA_t\, \mathbf{1}\{h_t = 2\}\, \mathbf{1}\{w_t n_t \leq y_{db}\} \qquad \text{if } 50 < t < NRA, \tag{8}$$

and $db_t = 0$ otherwise. The threshold $y_{db} = \$3{,}600$ is the annual equivalent of the monthly substantial gainful activity amount of \$300 for nonblind individuals with a disability in effect in 1987, the price base of our program parameters. The benefit is the PIA without the reduction for early claiming, scaled by π^{db}, the probability that an older individual with severe work limitations receives DI benefits, taken from Low and Pistaferri (2015). DI thus enters as its expected value: application and award are not modeled, and every eligible individual receives the fraction π^{db} of her PIA.

Disability insurance is an alternative pathway out of the labor force before the NRA (Low and Pistaferri 2015; Yu 2024). The income test makes it a notch: earnings above y_{db} forfeit the entire benefit, so that for an individual with a disability after age 50 the return to earning just above y_{db} is sharply negative. Because DI is paid until the NRA, a reform that raises the NRA also lengthens the period during which individuals with a disability are eligible for DI benefits, which are not subject to the reduction for early claiming. The NRA reforms of Section 5.2 therefore act on the budget through DI outlays as well as through retirement benefits.

Pensions. Pension income pb_t captures defined-benefit (DB) plans, which depend on the work history, are illiquid before age 62, and often create strong incentives to leave the labor force at particular ages through their accrual rates (Gustman et al. 2000; Ippolito

1997). Following French (2005), we impute pension income as a function of Social Security benefits, $pb_t = pb(PIA_t) + \xi_t$ for $t \geq 62$, and $pb_t = 0$ before 62. The function $pb(\cdot)$ is based on the estimates of Gustman and Steinmeier (1999), and the age-specific term ξ_t accounts for the difference between pension and Social Security accrual over the life cycle, based on Gustman et al. (2000). Pensions are not a reform instrument here, but they shape the incentive to exit in the early 60s on which the claiming-age reforms act. Appendix A gives details on DI and pensions.

2.8 Taxes

Individuals pay an income tax on total income y_t and a payroll tax on labor earnings:

$$T_t(y_t) = y_t - \lambda_t\, y_t^{1-\tau_t}, \qquad T_t^p(w_t n_t) = \tau_t^p\ \min\{w_t n_t,\, \bar{y}_t^{ss}\}. \tag{9}$$

The income tax takes the form of Bénabou (2002) and Heathcote et al. (2017): after-tax income is $\lambda_t y_t^{1-\tau_t}$, where λ_t governs the level of taxation and τ_t its progressivity. Both parameters depend on age and are estimated from the PSID as in Borella et al. (2023); individuals anticipate their evolution. Total income comprises interest income, labor earnings, and pension income (Section 2.9), so interest income is taxed as it accrues. The payroll tax, which finances Social Security's Old-Age, Survivors, and Disability Insurance (OASDI) and Medicare's Hospital Insurance (HI), applies at rate τ_t^p to earnings up to the taxable maximum $\bar{y}_t^{ss}$; both follow the historical SSA schedules for the cohort, with the combined employee rate ranging from 3 to 7.65 percent and the threshold from \$14,459 to \$64,974 (Table A.1). Two features simplify the rules faced by the cohort: Social Security and DI benefits are not part of taxable income, and the HI tax shares the OASDI taxable maximum (Appendix A.4). A reform that raises participation at some age raises income- and payroll-tax revenue at that age, and a reform that raises saving raises the tax on interest income later. These revenues are part of the indirect fiscal effects of Section 4. The income-tax reforms of Section 5.5 raise λ_t after age 61.

2.9 Budget Constraint, Borrowing, and the Consumption Floor

Assets evolve according to

$$\begin{aligned} a_{t+1} &= a_t + Y_t + b_t\, ss_t + db_t + tr_t - m_t - c_t, \\ y_t &= r\, a_t + w_t n_t + pb_t, \qquad Y_t = y_t - T_t(y_t) - T_t^p(w_t n_t), \\ a_{t+1} &\geq 0, \end{aligned} \tag{10}$$

where y_t is taxable income, Y_t is after-tax income, $b_t ss_t$ are retirement benefits net of the earnings test, db_t are DI benefits, and tr_t are means-tested transfers. The net return on assets is $r = (1 - \tau_k)\, r_f$. The pretax interest rate r_f is 4 percent (French 2005), and the wedge $\tau_k = 20$ percent represents corporate and other direct capital taxes that are not modeled (Stantcheva 2017). The net return is therefore 3.2 percent. Because individuals cannot borrow against future Social Security benefits and can rarely borrow against pension wealth, the borrowing limit is zero (Zeldes 1989; Hubbard et al. 1995). The absence of borrowing matters for claiming: an individual with few assets who stops working before 62, or who faces high medical expenses, values the liquidity of an early claim.

Government transfers guarantee a minimum level of consumption $\underline{c}$, as in Hubbard et al. (1995) and De Nardi et al. (2010):

$$tr_t = \max\Big\{0,\ \underline{c} + m_t - \big(a_t + Y_t + b_t\, ss_t + db_t\big)\Big\}. \tag{11}$$

The floor is set to $\underline{c} = \$3{,}500$, the value that French and Jones (2004) use as an estimate of the average means-tested benefits available to a household without children or members aged 65 or older; French and Jones (2011) estimate a floor of similar magnitude jointly with their preference parameters. It stands for safety-net programs such as the Supplemental Nutrition Assistance Program and Supplemental Security Income (modeled explicitly by Braun et al. 2017), and insures individuals against health and medical-expense risk. Because the transfer falls one for one with the individual's own resources when it is positive, it also taxes earnings, saving, and Social Security benefits implicitly at a 100 percent rate for individuals who receive transfers (Hubbard et al. 1995). An increase in benefits for low earners is therefore partly offset, in the aggregate, by lower transfers, which is one of the fiscal externalities of the generosity reforms of Section 5.3.

2.10 The Individual's Problem

The state of an individual at age t is $X_t = (a_t, \theta_t, h_t, b_{t-1}, P_{t-1}, aime_t)$: assets, ability, health, whether benefits have been claimed, whether she worked in the previous period, and the earnings record.[2] Out-of-pocket medical expenses, pension income, DI benefits, and transfers are functions of the state and current choices and are not separate state variables. The value function satisfies

$$V_t(X_t) = \max_{c_t,\, n_t,\, b_t} \Big\{u(c_t, l_t) + \beta\, s_{t+1}\, \mathbb{E}_t\big[V_{t+1}(X_{t+1})\big]\Big\}, \tag{12}$$

[2]Given health and age, ability and the wage are in one-to-one correspondence through (3), so the state can equivalently be written in terms of the wage w_t, as in the numerical implementation (Appendix A.5).

subject to the leisure constraint (2), the budget constraint and borrowing limit (10), the consumption floor (11), the benefit rules (5)–(8), the tax functions (9), and the claiming constraints $b_t \geq b_{t-1}$, $b_t = 0$ for $t < 62$, and $b_t = 1$ for $t \geq 70$, with $V_{T+1} \equiv 0$. The expectation is taken over next period's health and ability, conditional on survival:

$$\mathbb{E}_t\big[V_{t+1}(X_{t+1})\big] = \sum_{j \in \mathcal{H}} \pi_{j,h_t,t+1} \int V_{t+1}\big(a_{t+1},\, \theta_{t+1},\, j,\, b_t,\, P_t,\, aime_{t+1}\big)\, d\Phi_\sigma(\varepsilon_{t+1}),$$

where $\ln \theta_{t+1} = \rho \ln \theta_t + \varepsilon_{t+1}$, Φ_σ is the normal distribution with mean zero and variance σ^2, and survival enters (12) through $s_{t+1} = S(h_t, t+1)$.

Between 62 and 69, an individual who has not yet claimed trades one more year of benefits now against a permanently higher benefit later. Delay raises the benefit by the actuarial adjustment and, for an individual who keeps working, avoids the withholding of benefits by the earnings test. Its value depends on survival, and hence on health, which governs how long the higher benefit is received, and on assets, which govern how much the individual values liquidity now. The joint choice of hours and claiming links retirement to the claiming-age schedule: a steeper schedule raises the return to delay, which those who can keep working are best placed to collect, whereas a shift of the schedule to later ages lowers the benefit at every claiming age.

The solution consists of decision rules $c_t(X_t)$, $n_t(X_t)$, and $b_t(X_t)$, from which participation P_t and next-period assets a_{t+1} follow. We solve the problem by backward induction from age T and simulate the cohort forward from initial conditions drawn from the PSID (Appendix A.5).

2.11 Government Accounts

The government runs Social Security, DI, and Medicare, provides the consumption floor, and levies the income and payroll taxes. We measure its fiscal position vis-à-vis the cohort with per capita present values at age 62, the early retirement age, discounted at the net interest rate r. For a flow x_t, let $\bar{x}_t$ denote its per capita value at age t in the simulated cohort, and define

$$\mathcal{N}[x] \equiv \sum_{t=30}^{T} (1+r)^{-(t-62)}\, \bar{x}_t.$$

The fiscal aggregates are the present values of Social Security retirement outlays, DI outlays, consumption-floor transfers, income-tax revenue, payroll-tax revenue, and Medicare

payments:

$$SS = \mathcal{N}[b_t\, ss_t], \qquad DI = \mathcal{N}[db_t], \qquad TR = \mathcal{N}[tr_t],$$
$$IT = \mathcal{N}[T_t(y_t)], \qquad PT = \mathcal{N}[T_t^p(w_t n_t)], \qquad MC = \mathcal{N}\left[m_t^G\, \mathbf{1}\{t \geq 65\}\right].$$

Net government spending on the cohort is

$$G \equiv SS + DI + TR - IT - PT,$$

that is, outlays on retirement benefits, disability benefits, and transfers, net of income- and payroll-tax revenue. Section 4 measures the fiscal effect of a reform by its effect on G and splits it into a direct effect on the program that the reform changes—retirement outlays SS for Social Security reforms and income-tax revenue IT for income-tax reforms—and an indirect effect on the other components of G. Medicare payments are not part of G. Except for the two Medicare reforms, whose fiscal accounting adds Medicare's payments to the beneficiaries each reform targets, the net cost of a reform therefore covers retirement benefits, DI, means-tested transfers, and income and payroll taxes, but not Medicare outlays, although payroll-tax revenue includes the HI tax. The savings-subsidy reforms finance a higher net return with an explicit subsidy, whose present value is an additional outlay. Section 4.3 and Appendix D.7 give the exact construction of each fiscal measure.

The analysis is partial equilibrium: the wage profiles, the ability process, and the interest rate are fixed, and reforms do not change prices (Section 7 discusses general-equilibrium effects). The budget need not balance: a reform changes G, and Section 4 values that change through the marginal value of public funds rather than by financing it with a particular instrument. Each reform is a permanent change in the rules faced by the cohort, known from age 30, so individuals adjust saving and labor supply over the whole life cycle in anticipation of rules that apply later. The aggregates thus measure net government spending on one cohort over its lifetime, not the annual balance of a pay-as-you-go system.

2.12 Relation to Existing Life-Cycle Retirement Models

The model belongs to the class of structural life-cycle models of retirement with health reviewed in Section 1. It differs from French (2005) and French and Jones (2011) in seven respects, each of which matters for the valuation of claiming-age and benefit reforms.

First, individuals can work at every age up to 85, rather than having endogenous retirement decisions end at age 70, while claiming must occur by 70. Participation responses to reforms are therefore not truncated at the last claiming age. Second, the model is estimated on the outcomes of the whole cohort rather than on male household heads, which aligns

the behavioral responses with the per capita fiscal aggregates that enter the MVPF. Third, the model has an explicit disability state and DI benefits. DI is an early-exit pathway that is essential for explaining the participation declines of unhealthy individuals in their 50s (Yu 2024), and it is a margin through which NRA reforms affect the budget outside retirement outlays (Section 2.7). Fourth, the fixed cost of participation depends on health, whereas French and Jones (2011) specify a time cost of bad health that does not depend on participation and a time cost of participation that does not depend on health. Making the nonconvexity itself depend on health lets health determine who can respond to a reward for delay: a higher DRC pays off most for those who can delay claiming while still working, whereas a higher NRA lowers benefits also for those for whom continued work is too costly.

Fifth, health has three states. Poor health lowers wages and raises the costs of working and of medical care but leaves DI out of reach; disability opens the DI pathway. Distinguishing the two separates individuals who can respond to old-age incentives, at a higher cost of work, from those who can leave the labor force through DI before the NRA. Sixth, income taxes follow an age-dependent function estimated as in Borella et al. (2023). Together with the payroll tax, it determines how participation responses at each age reach the budget, and it provides the instrument for the old-age income-tax reforms. Seventh, the model serves as a laboratory for the MVPF of a large grid of reforms: for 66 reforms, we compute the ex ante willingness to pay and the fiscal effects on retirement benefits, DI, means-tested transfers, and income and payroll taxes, together with Medicare payments for the two Medicare reforms; for 5 instrument-elimination experiments, we compute the welfare effects.

The model makes three simplifications. It has no bequest motive, which matters for the decumulation of wealth late in life (De Nardi et al. 2010; Lockwood 2018) and for the fit of the asset profiles after the mid-50s (Section 3); it does not distinguish types of health insurance, which French and Jones (2011) show matter for retirement before 65; and health is unaffected by work.

3 Data and Estimation

We estimate the model of Section 2 on US panel data for the cohorts born between 1920 and 1940 and compare the estimated economy with the life-cycle profiles of participation, hours, wealth, and reentry that discipline it.

3.1 Estimation Strategy

We follow the two-step strategy that is standard for structural life-cycle models (Gourinchas and Parker 2002; Cagetti 2003). In the first step, we estimate or calibrate the parameters that can be identified without solving the model: the age profiles of health transitions, survival, out-of-pocket medical expenses, and wages by health status and the stochastic process for ability, all estimated directly from the data; the interest rate and the consumption floor, taken from the literature; and the rules of Social Security, disability insurance, pensions, and taxes. We collect these first-step parameters in a vector χ. In the second step, we estimate the preference parameters

$$\mathcal{P} = \left(\gamma,\ \nu,\ \beta,\ L,\ \phi_n(0),\ \phi_n(1) = \phi_n(2),\ \phi_{RE}\right)$$

by the method of simulated moments (MSM), taking χ as given.

The division separates the environment from behavior. The first-step processes describe the health, mortality, wage, and medical-expense risks that individuals face; apart from the Medicare reforms, which change the coverage of medical expenses after age 65, the reforms of Section 5 leave them unchanged. The second-step parameters govern how individuals respond when program rules change: the fixed costs of participation determine how many workers are close to the margin between working and retiring, the reentry cost determines whether retirement can be undone, and γ, ν, and β determine the value of insurance and the strength of income effects. Because the willingness to pay for a reform depends on the value of insurance and its fiscal externalities depend on behavioral responses (Section 4), we discipline the second step with the profiles that reflect these parameters most directly: participation and hours by health status, wealth, and labor market reentry.

Two of the differences from French (2005) and French and Jones (2011) listed in Section 2.12 concern the estimation directly. The target profiles describe household heads and spouses of the 1920–1940 cohorts rather than male household heads, so that the fiscal accounts used to compute MVPFs refer to the population the reforms reach. And the fixed cost of participation depends on health, so that health can change the cost of working in addition to its effects on wages, mortality, medical expenses, and access to disability insurance. In addition, we estimate wages outside the model with a selection correction, whereas these studies estimate the wage profiles within the structural model.

3.2 Data

Samples. Our main data source is the Panel Study of Income Dynamics (PSID), a longitudinal survey of a representative sample of the US population that records labor market

outcomes, income, health, and wealth. We use the 1968–2015 waves of its nationally representative component and keep household heads and spouses with nonmissing labor and health information. We focus on individuals born between 1920 and 1940 for two reasons. First, these cohorts were subject to similar Social Security rules: an early retirement age (ERA) of 62, a normal retirement age (NRA) of 65, and a delayed retirement credit (DRC) that rose with the year of birth and whose cohort average, 4.5 percent, we use in the model. A single policy environment therefore describes the incentives behind the behavior we observe. Second, the PSID observes these cohorts over much of their working lives and through the ages at which they retire. The final sample has 2,248 individuals and 44,404 person–year observations. We measure medical spending in the 1999–2012 waves of the Medical Expenditure Panel Survey (MEPS), a nationally representative survey in which spending is reported at the individual level and cross-checked with providers and insurers; the sample has 120,731 household heads aged 20 and older and 211,709 person–year observations. Appendix B describes both samples.

Health. Health must be measured consistently over the entire life cycle. In the PSID, we use self-reported work-limitation questions asked in every wave and classify individuals, following Low and Pistaferri (2015), as in good health ($h = 0$) if they report no condition that limits the amount of work they can do, in poor health ($h = 1$) if a condition limits it somewhat or a little, and as having a disability ($h = 2$) if a condition prevents them from doing any work, which corresponds to the Social Security Administration (SSA) criterion for disability insurance (DI). In the MEPS, we follow Pashchenko and Porapakkarm (2019) and classify individuals by self-rated health. The two classifications give similar shares of people with a disability between ages 50 and 70 and a declining share in good health with age, although the MEPS classifies more people in poor health (Table B.3); this consistency allows us to combine transition, survival, and wage profiles from the PSID with medical-expense profiles from the MEPS.

Target profiles. The second step targets four PSID profiles over ages 30 to 69: labor force participation, defined as working more than 800 hours in the year; annual hours of participants; nonpension assets, which exclude pension and Social Security wealth; and reentry, a transition from nonparticipation into participation. Participation and hours are measured separately for healthy individuals ($h = 0$) and unhealthy individuals ($h \in \{1, 2\}$), whom we pool because the PSID contains few observations of people with a disability at each age. We purge the profiles of family-size, business-cycle, and individual fixed effects with the procedure of French (2005) (Appendix C). Monetary values are in 2016 dollars.

3.3 First-Step Estimates

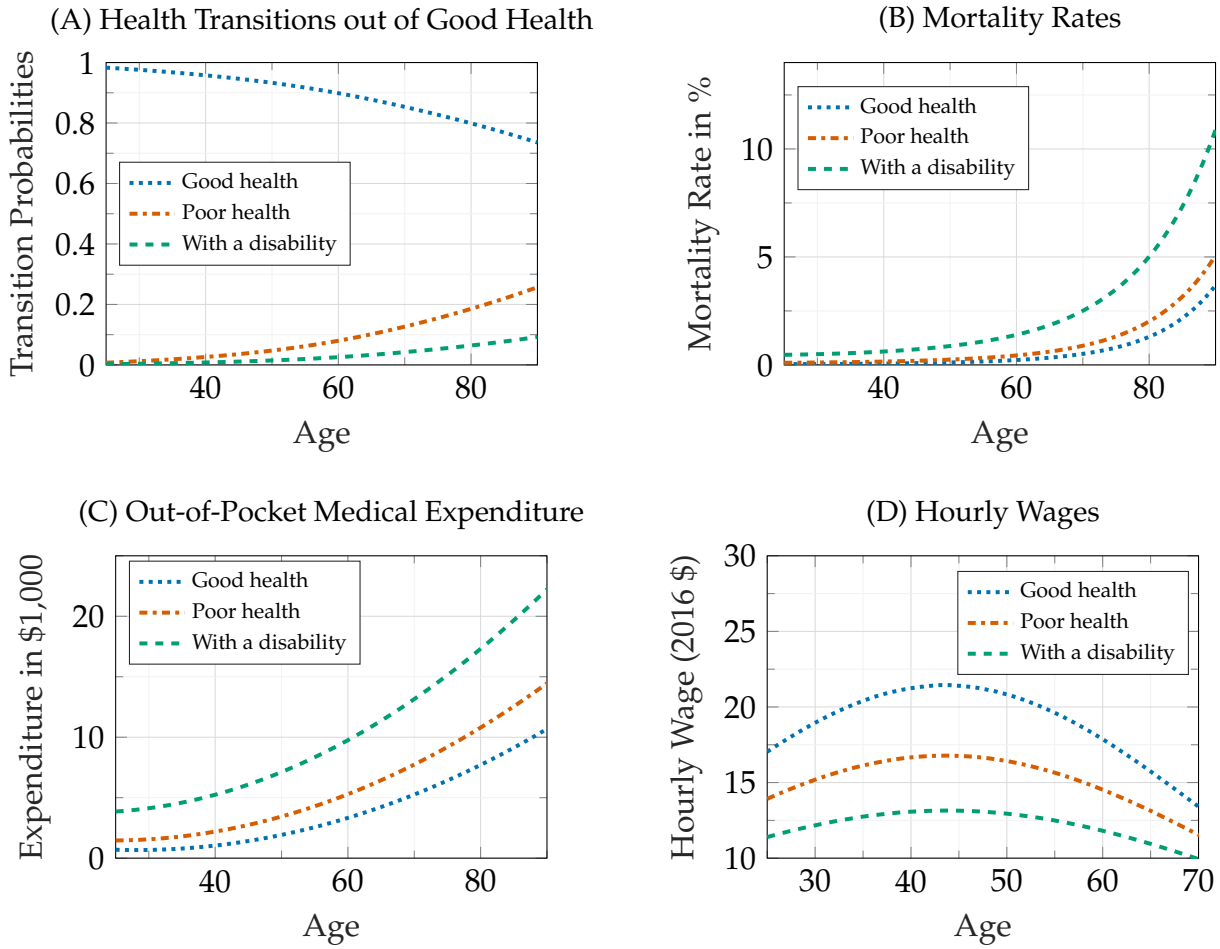
Health, survival, and medical expenses. We estimate the health transition probabilities $\pi_{j,i,t+1}$ from probit regressions on age dummies for each origin state, and survival probabilities $S(h,t)$ from a logistic regression of survival on a polynomial in age, lagged health, and their interactions. Out-of-pocket medical expenses $M(\cdot)$ are total medical expenditures in the MEPS net of the share covered by insurance before age 65 and by Medicare from age 65, by age and health status. Panels A to C of Figure 1 display the profiles. The probability of remaining in good health falls steadily with age, mostly through transitions into poor health; mortality rises with age in every health state and much faster for people with a disability; and out-of-pocket expenses rise with age and are considerably higher in poor health and with a disability. Through these profiles, worse health shortens the expected horizon over which a delayed claim pays off and raises the medical risk against which retirees self-insure.

Wages. Wages are observed only for participants, and those who keep working when old or unhealthy are likely to be those with high offered wages, so observed wages overstate offered wages most for older and unhealthy individuals. We correct for this selection with the two-step estimator of Heckman (1976), as in Guner et al. (2012): a participation probit generates an inverse Mills ratio that enters a regression of log hourly wages on an age polynomial and its interactions with health, with family size and year of birth excluded from the wage equation. The coefficient on the inverse Mills ratio is positive and significant (Table B.5), so participants are positively selected. Healthy individuals earn more than individuals in poor health, who earn more than individuals with a disability, at every age (panel D of Figure 1). The ability process (4), estimated from the wage residuals by maximum likelihood, is highly persistent, with $(\rho, \sigma^2) = (0.981, 0.016)$. Wage shocks are therefore close to permanent: an adverse shock in a worker's 50s lowers her expected wages, and hence the return to continued work, throughout the ages at which she decides when to retire.

Calibrated parameters and program rules. The gross interest rate is $r_f = 4$ percent (French 2005), and a 20 percent wedge between the gross and the net return captures corporate and other direct capital taxes not represented in the income tax function (Stantcheva 2017), so the net return is $r = 3.2$ percent. The consumption floor, $\underline{c} = \$3{,}500$, follows French and Jones (2004) (Section 2.9). The age-dependent income tax function is estimated from the PSID as in Borella et al. (2023); Social Security rules are those that applied to the 1920–1940 cohorts (Table A.1); DI benefits are scaled by the probability of a successful application estimated by Low and Pistaferri (2015); and pension benefits follow French

(2005), with parameters from Gustman and Steinmeier (1999) and Gustman et al. (2000). Initial conditions at age 30 are drawn from the joint distribution of assets, wages, health, and participation in the PSID. Table 1 summarizes the first-step inputs, and Appendix B.3 details each estimation.

Figure 1. First-Step Estimates: Profiles by Health Status



Notes: Panel A shows the probability of being in good health, in poor health, or with a disability at age $t+1$ for an individual in good health at age t. Panel B shows annual mortality rates (percent), panel C annual out-of-pocket medical expenditures (thousands of dollars), and panel D offered hourly wages corrected for selection (dollars), by age and health status. Blue dotted, red dash-dotted, and green dashed lines refer to good health, poor health, and disability. Monetary values are in 2016 dollars. *Data sources:* Panel Study of Income Dynamics (panels A, B, and D) and Medical Expenditure Panel Survey (panel C).

3.4 Second-Step Estimation by the Method of Simulated Moments

For each age from 30 to 69, we match seven moments: the participation rates of healthy and unhealthy individuals, the mean annual hours of healthy and unhealthy participants, mean and median nonpension assets, and the mean reentry rate. This yields $7 \times 40 = 280$

Table 1. First-Step and Calibrated Parameters

Parameter	Description	Value	Source
A. Estimated from data outside the model			
$\pi_{j,i,t+1}$	Health transition probabilities	Fig. 1A	PSID
$S(h,t)$	Survival probabilities	Fig. 1B	PSID
$M(h,t,P)$	Out-of-pocket medical expenses	Fig. 1C	MEPS
$W(h,t)$	Deterministic wage profiles	Fig. 1D	PSID
ρ	Autocorrelation of log ability	0.981	PSID
σ^2	Variance of ability innovations	0.016	PSID
λ_t, τ_t	Income tax function by age	Age-specific	Borella et al. (2023)
B. Taken from the literature			
r_f	Gross interest rate	4%	French (2005)
τ_k	Wedge between gross and net return	20%	Stantcheva (2017)
$\underline{c}$	Consumption floor	\$3,500	French and Jones (2004)
C. Program rules, cohorts born 1920–1940			
ERA	Early retirement age	62	SSA
NRA	Normal retirement age	65	SSA
DRC	Delayed retirement credit	4.5%	SSA
	Other Social Security rules	Table A.1	SSA

Notes: Parameters fixed before the second-step estimation. Panel A lists objects estimated directly from the data (Appendix B.3); ρ and σ^2 are the autocorrelation and innovation variance of log ability $\ln \theta_t$; the income tax function $T_t(y) = y - \lambda_t y^{1-\tau_t}$ is estimated by age from the PSID following Borella et al. (2023). Panel B lists values taken from earlier studies; the net return on savings is $r = (1-\tau_k)r_f = 3.2\%$. Panel C lists Social Security rules for the 1920–1940 birth cohorts; the DRC is the cohort average of credits that varied with the year of birth. Abbreviations: ERA, early retirement age; NRA, normal retirement age; DRC, delayed retirement credit; PSID, Panel Study of Income Dynamics; MEPS, Medical Expenditure Panel Survey; SSA, Social Security Administration.

moment conditions. Because the labor-supply moments pool poor health and disability, the two states share a fixed cost of participation. Let $\hat{\varphi}(\mathcal{P};\chi)$ denote the 280-element vector of differences between the data moments and the moments simulated from the model's decision rules (Appendix A.5) for individuals who start from the empirical initial conditions and draw shocks from the first-step processes. The MSM estimator is

$$\hat{\mathcal{P}} = \arg\min_{\mathcal{P}} \; \frac{N}{1 + N/N_s} \, \hat{\varphi}(\mathcal{P};\chi)' \, \hat{\mathbf{W}} \, \hat{\varphi}(\mathcal{P};\chi),$$

where N and N_s are the numbers of observations in the data and in the simulation, and the factor $1 + N/N_s$ accounts for simulation error. The weighting matrix $\hat{\mathbf{W}}$ is the inverse of a diagonal matrix that holds the sampling variances of the data moments, so each moment is weighted by its precision. Appendix C gives the moment conditions, the algorithm, and the standard errors.

Identification. As in French (2005), identification rests on two assumptions: work decisions do not affect health or wages, and age shifts the incentives to work and save—through wages, health, mortality, medical expenses, and the rules of Social Security and pensions—but not preferences. The first step then measures how incentives change with age and health, and the behavior these incentives leave unexplained identifies preferences. The discount factor β is identified by intertemporal substitution, and hence by the asset and labor-supply profiles, especially at the ages at which individuals accumulate wealth for retirement. The time endowment L and the fixed costs $\phi_n(h)$ are identified by participation and hours by health status: with a fixed cost, individuals either work substantial hours or do not work at all, so the participation rate and the hours of those who work carry separate information, and the gap between the participation of healthy and unhealthy individuals, given their first-step wages and medical costs, pins down the health gradient in the fixed cost. The reentry cost ϕ_{RE} is identified by the low reentry rates in the data. The curvature parameters ν and γ are identified by the asset and labor-supply profiles, through different features of them. In the within-period trade-off between consumption and leisure, γ determines the share of full income devoted to consumption, so the hours of participants relative to the time endowment, given their wages and assets, inform γ. Given γ, ν governs the willingness to substitute consumption over time and across states, and hence the strength of the precautionary motive, so it is informed mainly by the slope of the asset profile during the years of accumulation: higher risk aversion raises precautionary saving and encourages more work earlier in life.

3.5 Parameter Estimates

Table 2 reports the estimates. The discount factor is 0.98. The annual time endowment is 4,491 hours, and the fixed cost of working is 791 hours for healthy individuals and 868 hours for individuals in poor health or with a disability. A participant thus gives up about 18 percent of her time endowment before working the first hour if healthy and about 19 percent if unhealthy. This nonconvexity makes retirement a discrete decision: workers choose between substantial hours and full exit. The health gradient in the fixed cost adds to the lower wages and the access to DI of unhealthy individuals, both of which also pull their exit forward. The time endowment and the fixed costs fall within the range of estimates in French and Jones (2011) and Borella et al. (2023). The reentry cost is 3,312 hours, about three-quarters of the annual time endowment. Leaving the labor force is therefore close to irreversible, and a reform that encourages later retirement acts on workers before they exit rather than drawing retirees back into work.

The curvature of the consumption–leisure composite is $\nu = 1.70$ and the consumption

weight is $\gamma = 0.54$, which imply a coefficient of relative risk aversion over consumption, $-u_{cc}c/u_c = 1 - \gamma(1-\nu)$, of approximately 1.38. Because $\nu > 1$, the marginal utility of consumption falls as leisure rises, so consumption optimally declines at retirement (Section 2.2), and the product $\gamma(1-\nu)$ converts changes in expected utility into the consumption-equivalent welfare measure of Section 4.2. The curvature also governs the value of insurance: at a relative risk aversion of about 1.38, individuals value protection against health, medical-expense, and longevity risk less than they would at higher curvature, so the willingness to pay for reforms whose value lies mainly in insurance, such as wider Medicare coverage, is lower than it would be under more curvature.

The overidentification statistic is 1463 with $280 - 7 = 273$ degrees of freedom, so the test formally rejects the model, as is common for life-cycle retirement models estimated on many precisely measured moments (French 2005; French and Jones 2011). The test requires the model to match all 280 moments within sampling error; the next subsection shows which patterns the model reproduces and where it departs from the data.

Table 2. Estimated Preference Parameters

Parameter	Description	Estimate	Std. error
Preferences			
γ	Consumption weight	0.5354	0.0047
ν	Curvature of the consumption–leisure composite	1.7029	0.0681
β	Time discount factor	0.9797	0.0023
Time endowment and time costs (hours per year)			
L	Time endowment	4490.5	21.65
$\phi_n(0)$	Fixed cost of work, good health	791.07	15.65
$\phi_n(1) = \phi_n(2)$	Fixed cost of work, poor health or disability	867.54	22.68
ϕ_{RE}	Labor market reentry cost	3311.5	82.02
Overidentification statistic, χ^2 (273 degrees of freedom)		1463	

Notes: Method-of-simulated-moments estimates of the preference parameters and their asymptotic standard errors (Appendix C). γ is the weight on consumption and ν the curvature in $u(c,l) = (c^{\gamma}l^{1-\gamma})^{1-\nu}/(1-\nu)$; the implied relative risk aversion over consumption is $1-\gamma(1-\nu) \approx 1.38$. L, $\phi_n(h)$, and ϕ_{RE} are measured in hours per year; poor health ($h = 1$) and disability ($h = 2$) share one fixed cost of work. The overidentification statistic tests the 280 moment conditions against 7 parameters. *Data source:* Panel Study of Income Dynamics.

3.6 Match of Moments

Figure 2 compares the simulated profiles of participation and hours with their PSID counterparts. In the data, participation is high and stable in the 30s and 40s, declines from the late 40s and through the 50s for individuals in poor health or with a disability, and drops sharply in the 60s in both health groups, as wages fall and the retirement

incentives of pensions and Social Security grow. The model reproduces these patterns. As emphasized by Yu (2024), the disability state and DI benefits are essential for the decline in the participation of unhealthy individuals from their 50s: DI provides an alternative pathway out of the labor force (Maestas et al. 2013; French and Song 2014; Low and Pistaferri 2015; Li 2018), and modeling it yields a closer fit for this group than in earlier work such as French (2005).

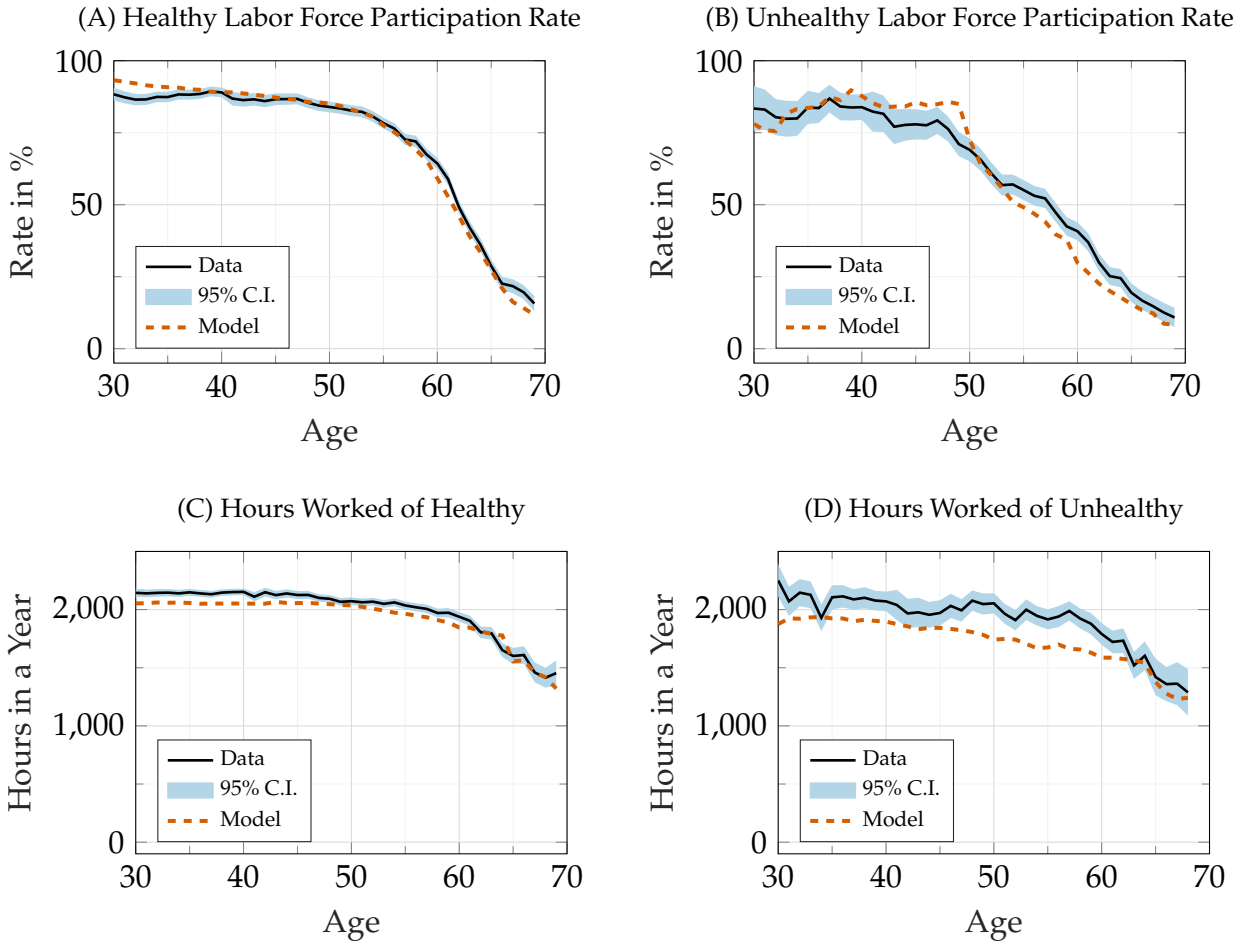
The model departs from the data in four respects. For unhealthy individuals, the model overstates participation in the 40s and concentrates the decline around age 50, when DI becomes available in the model, whereas in the data the decline is spread over the late 40s and early 50s. In both health groups, simulated exit is somewhat early: from the mid-50s to the mid-60s, simulated participation of the unhealthy lies below the confidence band, and that of the healthy lies below it between ages 58 and 64 and again from 67 to 69. For healthy individuals, the model overstates participation in the 30s. Hours of participants decline in the 60s in the model as in the data, but the model places the hours of unhealthy workers, and to a lesser extent those of healthy workers, below the data before age 60.

The fit of unhealthy participation between 55 and 69 matters for the reform analysis because at these ages three exit routes compete for the same individuals: DI, whose benefit is based on the primary insurance amount without an early-claiming reduction and is paid until the NRA; early claiming of reduced retirement benefits from 62; and continued work. Claiming-age reforms shift the relative value of these routes. Raising the NRA lowers retirement benefits at every claiming age while extending the ages at which DI is paid; raising the DRC increases the reward for delay, which individuals with higher mortality value less. The resulting reallocation across programs is what the indirect fiscal effects of Section 4 measure.

Claiming ages are not targeted in the estimation, and we do not compare the model's claiming responses with quasi-experimental estimates of the responses to the increase in the NRA for later cohorts (Mastrobuoni 2009; Behaghel and Blau 2012). The claiming responses on which the results of Section 5 rest are disciplined only indirectly: by the targeted profiles of participation and exit by health, which determine who is still working when benefits become available; by the asset profiles, which determine who can finance a delayed claim after leaving work; and by the statutory rules that tie benefits to work, namely the earnings test, which withholds and later credits the benefits of claimants who keep working before age 70, and the recomputation of the AIME, which raises the benefits of those whose additional earnings exceed their current average (Section 2.6). The misfits at ages 55 to 69 bear on these responses. Because simulated participation of the unhealthy lies below the data from the mid-50s to the mid-60s, the model has fewer unhealthy workers

at the claiming ages than the data, and hence fewer whose exit and claiming decisions a reform can move. It therefore likely understates the responses of unhealthy individuals to claiming-age reforms, and with them the tax components of the indirect effects. Lower simulated wealth in old age and the absence of reentry likewise limit the responses that a reform can induce among those who have already left work: the former leaves fewer of them able to finance a delayed claim, and the latter rules out a return to work. We do not attempt to sign the net effect of these misfits on the MVPFs.

Figure 2. Data and Simulated Profiles: Participation and Hours



Notes: Panels A and B show labor force participation rates (percent) of healthy individuals ($h = 0$) and unhealthy individuals (poor health or disability, $h \in \{1, 2\}$); panels C and D show annual hours worked conditional on participation for the same groups. In the data, an individual participates if she works more than 800 hours in the year. Solid lines are data profiles estimated from the Panel Study of Income Dynamics (Appendix C), shaded areas are 95% confidence intervals, and dashed lines are profiles simulated from the model at the estimates of Table 2. All profiles are targeted in the estimation.

Figure 3 shows the wealth and reentry profiles. The model tracks mean assets until the early 50s but overstates median assets in the 30s and 40s, when the data median lies

far below the data mean; the model generates less dispersion in wealth at young ages than the data. From the 50s on, simulated wealth falls increasingly short of the data, from the early 50s for the mean and from the mid-50s for the median. The divergence likely reflects the absence of a bequest motive: the model has accidental bequests only, whereas bequest motives raise saving at older ages (Lockwood 2018). Late-life medical expenses, which De Nardi et al. (2010) find to be a major reason why the elderly save, enter the model only up to its terminal age of 85. We abstract from bequest motives to keep the model comparable with the literature on life-cycle taxation (Farhi and Werning 2013; Golosov et al. 2016; Stantcheva 2017). Simulated households therefore enter old age with less self-insurance than their counterparts in the data, which raises the weight of Social Security income in their old-age resources. A specification with separable logarithmic utility over consumption, estimated in Appendix C.5, fits old-age mean assets better but participation worse; the reform analysis uses the baseline specification.

Reentry rates are low in the data, below 5 percent at every age, and the model generates no reentry: at the estimated reentry cost, exit from the labor force is effectively absorbing. The model thus reproduces the participation and hours profiles through which retirement reforms operate; it understates wealth after the early 50s and generates no reentry. Ndiaye and Yu (2026) use the same estimated economy to quantify the channels—fixed costs of work, health-related productivity losses, and differential mortality—through which health shapes retirement.

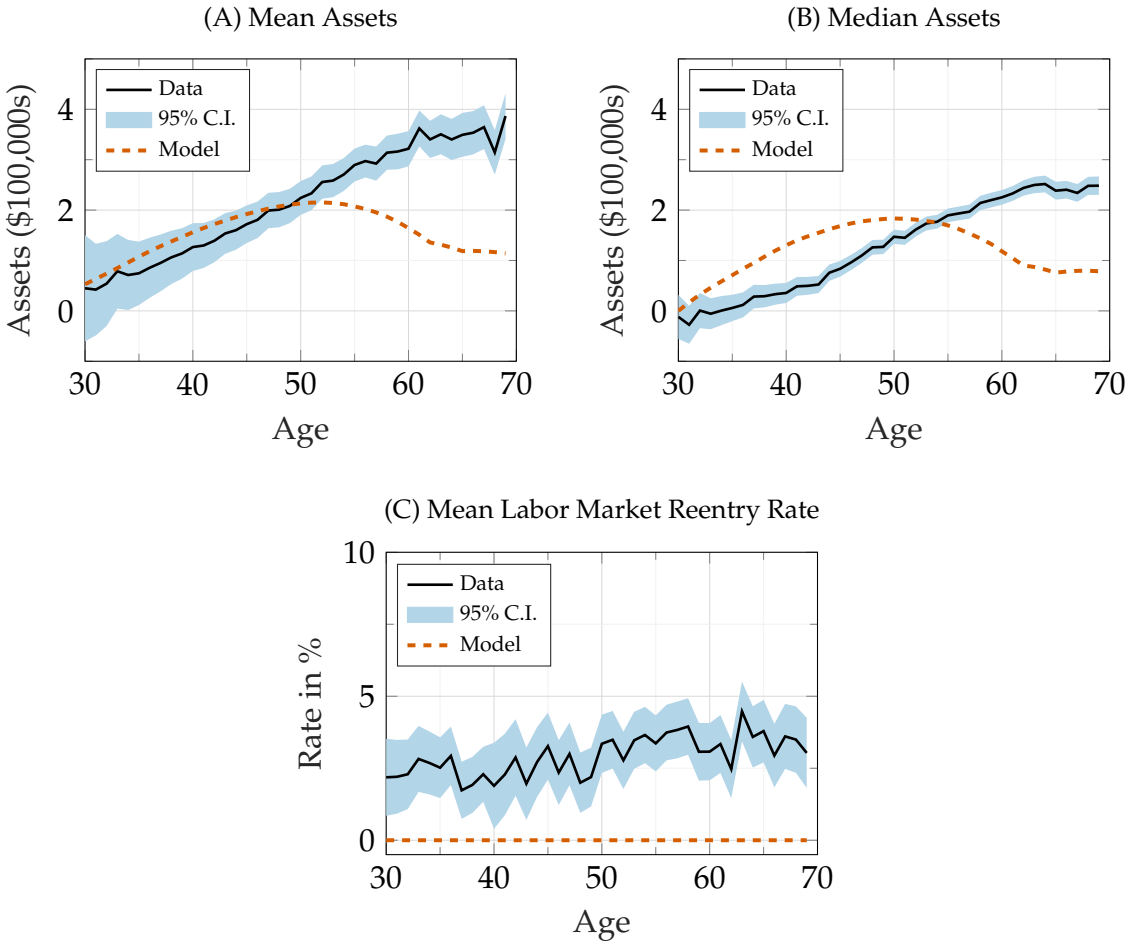
4 Measuring the Value of Retirement Reforms

The marginal value of public funds (MVPF) of a policy is the willingness to pay of its beneficiaries divided by its net cost to the government, including the fiscal externalities of the behavioral responses it induces (Hendren 2016; Hendren and Sprung-Keyser 2020). For a life-cycle policy, both objects must be measured over the whole life cycle: the willingness to pay before individuals know the health and earnings histories that determine what they receive, and the net cost to Social Security, DI, means-tested transfers, and the income and payroll taxes, including responses that occur decades before or after the reformed rule applies. This section defines both objects in our model, relates their ratio to the sufficient-statistics analysis of pension reform, and derives the accounting identity that governs the fiscal effects of claiming-age reforms.

4.1 Reforms

A reform is a permanent change in the policy rules of Section 2, announced before the cohort enters the model at age 30 and in force over its entire life; the status quo is the

Figure 3. Data and Simulated Profiles: Assets and Reentry



Notes: Panels A and B show mean and median nonpension assets (hundreds of thousands of 2016 dollars); panel C shows the mean labor market reentry rate (percent). Solid lines are data profiles estimated from the Panel Study of Income Dynamics (Appendix C), shaded areas are 95% confidence intervals, and dashed lines are profiles simulated from the model at the estimates of Table 2. All profiles are targeted in the estimation. Asset and reentry profiles are not conditioned on health.

estimated economy of Section 3. For each reform, we solve the individual's problem (12) under the new rules and simulate the cohort from the same initial distribution. Because the reform is known from age 30, its effects include responses long before the reformed rule applies: a worker who expects a more generous benefit formula adjusts her hours at 35, when her earnings enter her AIME, as well as her exit at 63.

We study 66 reforms in five families, constructed in Appendix D. The first family (17 reforms) changes the claiming-age adjustment $\alpha(\cdot)$ of Section 2.6. Reforms AN1–AN3 raise the NRA from 65 to 66, 67, and 68, which shifts the entire schedule of benefits by claiming age to the right. Reforms AD1–AD5 raise the DRC from 4.5 percent to 8, 8.5, 9, 9.5, and 10 percent and leave the reductions for early claiming unchanged. Reforms

AL1–AL9 make the adjustment symmetric around the NRA of 65: the benefit changes by x percent of the PIA for each year by which the claim precedes or follows the NRA, with x from 7 to 15 percent. Figure 4 and Table D.1 display the schedules. The second family (39 reforms) raises the replacement rates of the benefit formula (6). Reforms P1.1–P1.9 raise the first rate from 0.90 to 0.91, ..., 0.99; with the first rate at 0.99, reforms P2.0–P2.9 raise the second rate from 0.32 in ten steps of 1 percent of its initial value, to 0.3232, ..., 0.352; and with the second rate at 0.352, reforms P3.0–P3.9 raise the third rate from 0.15 in the same proportional steps, to 0.1515, ..., 0.165. Reforms L0–L9 raise all three rates in proportion, by 1 to 10 percent.

The third family (two reforms) raises the share of medical expenditures after age 65 that Medicare covers by 20 percentage points, for all individuals (Part A) or for labor market nonparticipants (Part B), relative to a baseline in which coverage rates vary with health and working status as in the MEPS. The fourth family (five reforms) lowers income taxes in old age: reforms I1–I5 raise the level parameter of the tax function (9) from λ_t to $\lambda_t + x(t-61)$ at every age $t \geq 62$, with $x = 0.01, 0.02, 0.03, 0.04$, and 0.05, which raises after-tax income at age t by the proportion $x(t-61)/\lambda_t$ at every income level. The fifth family (three reforms) raises the net return on assets at every age by 0.2, 0.4, and 0.8 percentage points; the largest subsidy sets the net return equal to the pretax return r_f. The reforms of the first four families operate through retirement benefits, Medicare coverage, or old-age taxes, which apply from age 62, the earliest claiming age, or later (from 65 for Medicare); we refer to them as the reforms that start at 62. The savings subsidies change the return to saving from age 30. This distinction motivates the conversion of welfare into dollars (Section 4.4). The progressivity and level reforms also raise DI benefits, which are proportional to the PIA and paid to eligible individuals with a disability from age 51 (8).

We also remove the existing instruments one at a time: D1 sets Social Security benefits to zero, D2 compares the status quo with full coverage of medical expenditures after age 65, D3 sets the net return equal to the pretax return, D4 eliminates the income tax, and D5 eliminates DI. These experiments are not financed, and we report only their welfare effects (Section 5.1).

4.2 Welfare

Let $\bar{V}^0$ be the average over simulated individuals of the value at age 30, $V_{30}(X_{30})$, in the status quo, and $\bar{V}^1$ the same average under a reform. Flow utility (1) is homogeneous of degree $\gamma(1-\nu)$ in consumption: multiplying consumption by e^x at every age and in every state, with leisure unchanged, multiplies flow utility by $e^{\gamma(1-\nu)x}$, and therefore multiplies each individual's expected lifetime utility, and their average $\bar{V}^0$, by the same

factor. The consumption-equivalent welfare change (CE) of a reform is the x that solves $\bar{V}^1 = e^{\gamma(1-\nu)x}\,\bar{V}^0$:

$$CE = \frac{\ln\left(\bar{V}^1/\bar{V}^0\right)}{\gamma(1-\nu)}, \qquad LE = \frac{\ln\left(\bar{V}^1/\bar{V}^0\right)}{(1-\gamma)(1-\nu)}. \tag{13}$$

The ratio $\bar{V}^1/\bar{V}^0$ is positive because, with $\nu > 1$, lifetime utility is negative under every policy. CE is the proportional increase in status-quo consumption, at every age and in every state, that the cohort values as much as the reform. The leisure equivalent LE, obtained by scaling leisure instead, equals $CE\,\gamma/(1-\gamma)$ and ranks reforms identically. We report CE, in percent, as the willingness to pay for the reform, WTP. As a log ratio, it is antisymmetric: the CE of a move from one set of rules to another is minus the CE of the reverse move.

The willingness to pay is ex ante. It is evaluated at age 30, before health, ability, and survival histories are realized, and it averages expected utility with equal weights across individuals who differ in initial assets, wages, and health. It therefore includes the value of a reform as insurance. The welfare effect of a change in resources in a given state is weighted by the marginal utility of consumption in that state, so a reform that raises resources in states where marginal utility is high is worth more ex ante than its expected transfer, and a reform that lowers them in such states costs more than its expected saving. Health shocks, low ability, and medical expenses determine which states these are; the claiming-age schedule and the benefit formula determine the resources available in them.

Many applications of the MVPF instead sum the ex post willingness to pay of the individuals a policy affects, each in her own dollars, which for small policies equals the mechanical transfer each receives; distributional and insurance considerations then enter through separate welfare weights (Hendren and Sprung-Keyser 2020; Finkelstein and Hendren 2020; Saez and Stantcheva 2016). Our measure embeds the weights of a utilitarian evaluation of one cohort behind the veil of ignorance. It does not weigh the welfare of different cohorts (Section 6).

4.3 Fiscal Accounting

The fiscal effects of a reform are changes in the government accounts of Section 2.11: the per capita present values at age 62, discounted at the net return r, of Social Security retirement outlays SS, DI outlays DI, consumption-floor transfers TR, income-tax revenue IT, payroll-tax revenue PT, and Medicare payments. Net government spending, $G = SS + DI + TR - IT - PT$, is outlays on retirement benefits, DI, and transfers net of income- and payroll-tax revenue. Medicare payments are not part of G: except for the two Medicare reforms, whose cost includes Medicare's payments to the beneficiaries they target, the net cost of a reform excludes Medicare outlays. Flows before age 62, such as the taxes paid

during the working life, are compounded forward to 62. We denote by Δ the difference between the reform and the status quo. No reform is required to balance the budget: the MVPF values the change in net cost without specifying how it is financed or how a saving is used.

We split the net cost of a reform into a direct cost on the program that the reform changes and an indirect cost on the other programs in the accounts. The direct cost includes the behavioral responses within the reformed program: for a claiming-age reform, it is the change in retirement outlays, which combines the change in benefits at given claiming ages with the effect of the induced changes in claiming ages. The indirect cost collects the fiscal externalities on the other components of G: income and payroll taxes, DI, means-tested transfers, and, for reforms outside Social Security, Social Security itself. In dollars per capita, the direct cost DC, the indirect cost IC, and the net cost $NC = DC + IC$ are

$$\begin{array}{lll}
\text{Social Security:} & DC = \Delta SS, \quad IC = \Delta(DI + TR - IT - PT), & NC = \Delta G, \\
\text{Medicare:} & DC = \Delta MC^{tg}, IC = \Delta G, & NC = \Delta G + \Delta MC^{tg}, \\
\text{Income tax:} & DC = -\Delta IT, \quad IC = \Delta(SS + DI + TR - PT), & NC = \Delta G, \\
\text{Savings subsidy:} & DC = K, \quad IC = \Delta G, & NC = \Delta G + K,
\end{array}$$

where the Social Security family comprises the claiming-age, progressivity, and level reforms, MC^{tg} is the present value of Medicare's payments to the beneficiaries that the reform targets (all individuals aged 65 and older for Part A, nonparticipants aged 65 and older for Part B), and K is the present value of the savings subsidy, the additional return the government pays on assets from age 30.

The results tables report the three fiscal effects in percent, as defined in Appendix D.7; a positive entry is a cost to the government and a negative entry a saving. Net is the percentage change in the present value of net government spending; for the Medicare reforms it adds the change in Medicare's payments to the targeted beneficiaries, and for the savings subsidies the subsidy, both as shares of baseline government spending. Direct is the percentage change in the present value of retirement outlays for Social Security reforms and of the income-tax revenue forgone for income-tax reforms. For Medicare reforms, it is the change in the present value of Medicare's payments to the targeted beneficiaries as a share of baseline payroll-tax revenue, so that it is the proportional increase in payroll-tax revenue that would finance the additional payments at unchanged behavior; for savings subsidies, it is the present value of the subsidy as a share of baseline government spending. Indirect is the percentage change in the present value of the aggregate it measures. The three columns thus have different bases, so Net is in general not the sum of Direct and

Indirect; the dollar amounts add up.

4.4 The Marginal Value of Public Funds

We convert the willingness to pay into dollars with a present value of baseline per capita consumption $\bar{c}_t^0$:

$$MVPF = \frac{WTP^{\$}}{NC}, \qquad WTP^{\$} = \frac{WTP}{100}\,\bar{C}_s^0, \qquad \bar{C}_s^0 = \sum_{t=s}^{T}(1+r)^{-(t-62)}\,\bar{c}_t^0, \tag{14}$$

with $s = 62$ for the reforms of Social Security, Medicare, and income taxes and $s = 30$ for the savings subsidies. Numerator and denominator are per capita present values at age 62, in dollars.

We follow the conventions of Hendren and Sprung-Keyser (2020). When the net cost is positive, the MVPF is the welfare delivered per dollar of government spending. When the WTP is positive and the net cost is zero or negative, the MVPF is infinite: the reform pays for itself, since the government could implement it, return the saving to the cohort, and keep the entire welfare gain. When the WTP and the net cost are both negative—a cut that lowers welfare and saves money—the MVPF is the welfare cost per dollar of revenue raised, the shadow price of raising revenue from the beneficiaries of a policy by reducing spending on it. By the antisymmetry of the CE and of the dollar fiscal effects, it equals the MVPF of the reverse reform up to the baseline used for the dollar conversion. Financing a reform with MVPF $MVPF_1$ by a cut with MVPF $MVPF_2$ raises welfare when $MVPF_1 > MVPF_2$, the comparison Section 6 uses to order the ways of closing a financing gap.

For some reforms, we also report the MVPF computed with the direct cost alone:

$$\underline{MVPF} = \frac{WTP^{\$}}{DC}. \tag{15}$$

For a reform with positive WTP, positive direct cost, and negative net cost, the savings on other programs are what make the MVPF infinite. $\underline{MVPF}$ is the MVPF the reform would have if none of those savings materialized; it shows how much of the evaluation rests on the reformed program alone. When the direct cost is negative as well, the MVPF is infinite under either cost. For the savings subsidies, the model taxes interest income as it accrues, as if all capital income were realized and taxed every period, whereas saving for retirement, much of it in retirement accounts, is long-term investment whose returns are taxed later or not at all. The model therefore overstates the revenue from taxing the additional interest income, and $WTP^{\$}/NC$ is an upper bound on the MVPF of a subsidy.

We report it together with $WTP^{\$}/DC$, the MVPF that counts the subsidy alone. The latter is a lower bound only if the fiscal externalities other than the taxation of interest are savings, which is not assured: the subsidies lower participation after 62 (Section 5.6), and with it labor-income and payroll-tax revenue.

The dollar conversion is a convention, and its base matters for the levels of finite MVPFs. The CE scales consumption at every age from 30 to T, whereas (14) values it with baseline consumption from age 62 for the reforms that operate through retirement benefits, Medicare, or old-age taxes, and with consumption over the whole life cycle for the savings subsidies, which change the return to saving from age 30. For the progressivity and level reforms, which also raise DI benefits from age 51, the post-62 base slightly understates the ages at which the reform operates. The base does not affect whether an MVPF is finite or infinite, which depends only on the signs of WTP and net cost, nor the ranking of reforms within and across the four families that start at 62, which share the base. It does affect the levels of finite MVPFs. On the lifetime base $\bar{C}^0_{30}$, the normalization of external MVPF estimates, under which a transfer that induces no behavioral response has an MVPF of about one, every finite MVPF of a reform that starts at 62 would be larger by the common factor $\bar{C}^0_{30}/\bar{C}^0_{62}$, about five: the table entries imply a ratio between 4.7 and 5.2. We therefore compare the MVPF levels of the reforms that start at 62 with one another. Comparisons of these levels with one, with external estimates such as the MVPF of the introduction of Medicare, or with the MVPFs of the savings subsidies require the conversion of Section 7.1.

The construction implies a regularity in the results tables. The dollar WTP is the WTP entry times a fixed baseline aggregate, $\bar{C}^0_s$, and the dollar net cost is the Net entry times another fixed baseline aggregate, so the ratio $MVPF \times Net/WTP$ depends on the baseline but not on the reform. Computed from the rounded table entries, it lies between 0.38 and 0.40 for all 48 reforms with a finite MVPF that start at 62 and between 1.87 and 1.99 for the three savings subsidies (with the MVPF computed with the net cost), a dispersion of the order of the rounding error in the entries. Every MVPF computed with the net cost can thus be recovered, up to rounding, from the WTP and Net columns.

4.5 Decomposition and Sufficient Statistics

The net cost of a reform also splits into a mechanical and a behavioral component. Let ME be the change in net cost that the reform would cause if every individual kept her status-quo hours, claiming age, and saving, and let the fiscal externality FE be the remainder, due to the behavioral responses. Then

$$NC = ME + FE = DC + IC. \tag{16}$$

The direct cost contains the mechanical cost on the reformed program together with the behavioral responses within it; the indirect cost contains the fiscal externalities on the other programs in the accounts together with mechanical interactions between programs, such as the fall in consumption-floor transfers when benefits rise for individuals at the floor (Appendix E).

For a small reform, the envelope theorem implies that behavioral responses have no first-order effect on welfare: since individuals were optimizing, the WTP equals the mechanical change in each individual's resources weighted by the marginal utility of consumption in the state in which it occurs (Hendren 2016). In our units, the CE of a small reform is the ratio of this marginal-utility-weighted transfer to the marginal-utility-weighted consumption stream of the status quo (Appendix E); the result holds with discrete participation and claiming choices because the value function is the upper envelope of the values of the alternatives (Milgrom and Segal 2002). Behavioral responses then affect the MVPF only through the budget. We do not rely on this approximation: we compute WTP exactly from the value functions and the net cost from the simulated responses. We do not compute the mechanical effect separately. The decomposition relates our measures to the sufficient-statistics approach, and it signs the fiscal externality when the sign of the mechanical effect is known, as for the claiming-age reforms (Section 4.6).

The fiscal externality has three sources. The first is participation and hours: a worker who enters the labor force or postpones her exit pays income and payroll taxes, may forgo DI and transfers through their income and means tests, may have benefits withheld by the earnings test, and may raise her AIME and future benefits. Each such response lowers the net cost by the participation tax on her earnings: the taxes paid, benefits withheld, and DI and transfers forgone, net of the increase in future benefits (Kleven and Kreiner 2006). The second is the claiming age, whose effect on retirement outlays Section 4.6 characterizes. The third is saving, which raises taxable interest income and lowers means-tested transfers. The fiscal columns are organized by program rather than by margin, but the margins map into programs: for a Social Security reform, claiming responses enter Direct, and participation and saving responses enter Indirect, except for their effects on retirement outlays through the earnings test and the AIME.

Haller (2022) evaluates pension reforms with a dynamic version of the Baily–Chetty formula, which compares the value of the consumption smoothing a reform provides, per dollar of mechanical transfer, with a fiscal multiplier, the ratio of the reform's total fiscal cost to its mechanical cost; a reform is desirable when the former exceeds the latter. With

the multiplier $FM \equiv NC/ME$,

$$MVPF = \frac{WTP^{\$}/ME}{FM} :$$

the MVPF is the value of consumption smoothing per mechanical dollar divided by the fiscal multiplier. A reform is desirable when its MVPF exceeds that of the instrument that finances it, and the Baily–Chetty comparison is the case in which that instrument has an MVPF of one (Appendix E). For a reform with a positive mechanical cost, an infinite MVPF corresponds to a negative multiplier: the behavioral responses more than finance the mechanical transfer.

The structural model measures the fiscal effects of behavioral responses over the whole life cycle, across Social Security, DI, means-tested transfers, and the income and payroll taxes, and by health, and it computes the ex ante WTP of reforms outside the range of historical variation. Quasi-experimental evidence provides credible local responses without a full model: of retirement to financial incentives (Liebman et al. 2009; Manoli and Weber 2016), of employment to the earnings test (Gelber et al. 2020a), of disability enrollment to the NRA (Duggan et al. 2007), and of retirement to statutory ages (Seibold 2021). Such responses can discipline and test those of a structural model, which in turn extends them to the life cycle and across programs; the two approaches are complements (Chetty 2009; Kleven 2021).

4.6 The Fiscal Logic of Claiming-Age Reforms

The fiscal effects of claiming-age reforms rest on an accounting identity. Consider an individual alive at age 62, with a fixed primary insurance amount *PIA*, who claims at age a at or after the NRA and receives $b(a) = PIA\,[1 + \delta(a - NRA)]$ every year from a until death, where δ is the DRC. We abstract from the earnings test. Let $\mathcal{S}(t)$ be the probability, as of age 62, that she survives to age t, with $\mathcal{S}(62) = 1$, and let the government discount at the net return r.

Lemma 1. *Let* $A(a) \equiv \sum_{t=a}^{T}(1+r)^{-(t-62)}\,\mathcal{S}(t)$. *The present value at 62 of the individual's benefits when she claims at age* $a \in \{NRA, \ldots, 69\}$ *is* $B(a) = b(a)\,A(a)$, *and*

$$B(a+1) - B(a) = \delta\,PIA\,A(a+1) \;-\; b(a)\,(1+r)^{-(a-62)}\,\mathcal{S}(a). \tag{17}$$

A one-year delay therefore lowers the present value of her benefits if and only if $\delta < \delta^*(a)$, *where*

$$\delta^*(a) \equiv \frac{b(a)\,(1+r)^{-(a-62)}\,\mathcal{S}(a)}{PIA\,A(a+1)} = \frac{b(a)/PIA}{\sum_{t=a+1}^{T}(1+r)^{-(t-a)}\,\mathcal{S}(t)/\mathcal{S}(a)}. \tag{18}$$

The rate $\delta^*(a)$ is strictly decreasing in the conditional survival probability $\mathcal{S}(t)/\mathcal{S}(a)$ at every age $t > a$ and nonincreasing in the terminal age T.

Appendix E gives the proof. The first term of (17) is the cost of the higher benefit, paid for the rest of the individual's life; the second is the saving on the year of benefits she forgoes. The rate $\delta^*(a)$ is the credit that is actuarially fair from the government's perspective for this individual: it makes the program indifferent to her delaying by one year, given the government's discount rate and her survival prospects. At the NRA, where $b(a) = PIA$, it is the inverse of the present value of an annuity of one dollar a year starting one year later.

The fair rate is individual-specific, and in the model it varies through health. Survival is lower in poor health and with a disability at every age (panel B of Figure 1), so an individual in worse health typically has lower conditional survival and a higher $\delta^*(a)$ (Appendix E gives a sufficient condition). At a common credit, a one-year delay saves the program more per dollar of PIA when the claimant's conditional survival is lower, and the same delay can save money for one claimant and cost money for another. The fair rate also depends on the length of the horizon over which the higher benefit is paid, which in the model ends at $T = 85$. Whether a delay is privately valuable is a separate question: the individual discounts with β and her marginal utility of consumption, values the longevity insurance of a larger annuity, and, if she keeps working, avoids the withholding of benefits by the earnings test.

Consider an increase in the DRC from δ to $\delta + d\delta$, with the reductions for early claiming unchanged. Its welfare and fiscal effects have different sources. At given taxes and other rules, the reform raises the benefit for every claim after the NRA, including the credit on benefits withheld by the earnings test, and lowers none. It therefore weakly enlarges every individual's choice set, and its WTP is positive by construction as long as some individuals claim after the NRA. Its fiscal effect has no such sign. Its direct cost has a mechanical part, the higher benefit paid to inframarginal late claimers, which is nonnegative for the same reason, and a behavioral part, the change in the present value of benefits of individuals whose claiming age changes. Each such change is a sequence of one-year steps valued by (17) or its analog before the NRA, so the behavioral part is negative when the induced delays occur, on balance, at ages and for individuals for whom the increment of the new schedule is below its actuarially fair value, $\delta^*(a)$ after the NRA (Appendix E); by Lemma 1, this value is higher for claimants with lower survival. The direct cost is negative if and only if the behavioral part is negative and exceeds the mechanical cost in absolute value. The indirect cost adds the taxes paid during longer careers and the resulting changes in DI and transfers. The earnings test adds an interaction: benefits withheld after the NRA are credited at the DRC, so a higher DRC lowers the implicit tax on work after the NRA and

raises the future benefits of those whose benefits are withheld. The symmetric reforms AL1–AL9 also raise the reduction for early claiming. They trade a lower benefit for early claiming against a higher one for late claiming, so neither their WTP nor their mechanical effect is signed in general: the mechanical effect is a saving on early claimers and a cost on late claimers, and the same identity values the induced changes in claiming ages.

An increase in the NRA works differently. It shifts the schedule to the right, so that the benefit falls at every claiming age from 62 to 70 (Table D.1). Its mechanical effect is a cut for every claimant, a saving whatever the claiming ages, to which behavioral responses add or from which they subtract. It does not raise the credit for delay beyond the new NRA; it changes the year-to-year increments of the schedule only by moving the boundary between the early-claiming reductions and the DRC. Its welfare cost falls on all claimants, including those in poor health, whose higher fixed cost of work makes it costlier for them to offset the cut by working longer. Because DI is paid until the NRA without the reduction for early claiming, an NRA increase also lengthens the period over which individuals with a disability can receive DI. A higher DRC rewards delay and raises only the benefits of those who claim after the NRA; a higher NRA mandates delay by lowering the benefits of everyone, including those who do not delay. Section 5.2 shows what this difference implies for welfare and the budget.

5 Results

This section reports the welfare and fiscal effects of the five instrument eliminations and of the 66 reforms. Each entry comes from simulating the estimated model for the cohort under the reform, announced before age 30 and in force over the whole life cycle, with wages and the interest rate held fixed. Each reform table reports the WTP, the three fiscal columns Direct, Indirect, and Net, the MVPF, and the changes in participation at ages 30–49, 62–64, and 65–69, as defined in Section 4. The MVPFs of the reforms that start at 62 use the post-62 consumption base (Section 4.4). We compare their levels with one another; comparisons with one, with external estimates, and with the savings subsidies require the conversion of Section 7.1. Small non-monotonicities in the tables, such as a WTP of 0.98 percent for L9 against 1.02 percent for L8 although L9 sets a higher rate in every bracket, reflect simulation and approximation error, and we do not interpret differences of a few hundredths of a percentage point.

5.1 The Value of Existing Instruments

We first measure what each existing instrument contributes to welfare by removing it, one at a time, from the estimated economy. Table 3 reports the results. Eliminating

Social Security benefits (D1) lowers welfare by 5.89 percent of consumption, the largest change among the five experiments. Eliminating only the retirement benefits, and keeping disability benefits, lowers welfare by 5.34 percent in consumption-equivalent and 6.15 percent in leisure-equivalent terms. Social Security benefits are an annuity whose level depends on lifetime earnings through a progressive formula. They insure longevity risk, which individuals can otherwise self-insure only by holding wealth that is lost at death, and they insure part of the consequences of low ability and poor health for lifetime earnings.

Table 3. Welfare Value of Existing Policy Instruments

	Experiment	CE (%)	LE (%)
(D1)	No Social Security benefits	−5.89	−6.79
(D2)	Incomplete Medicare coverage after 65 (loss relative to full coverage)	−4.28	−4.93
(D3)	Full savings subsidy ($r = r_f$)	3.60	4.15
(D4)	No income tax	3.16	3.64
(D5)	No disability benefits	−0.34	−0.40

Notes: Welfare change, in percent, from eliminating one policy instrument from the estimated economy, all other rules unchanged. CE is the ex ante consumption-equivalent change at age 30: the proportional change in consumption at every age that gives the same expected lifetime utility (equation (13)). LE is the analogous leisure-equivalent change, equal to CE $\times\ \gamma/(1-\gamma)$ (Section 4.2). D1 sets Social Security benefits to zero. D2 compares the status quo with an economy in which all medical expenditures after age 65 are covered for participants and nonparticipants; it is reported as the welfare loss from the incomplete coverage of the status quo, so full coverage would raise welfare by 4.28 percent (CE). D3 sets the net return on assets r equal to the pretax return r_f. D4 sets the income tax to zero. D5 sets disability insurance benefits to zero. The experiments are not budget balanced. Each entry comes from simulating the estimated model for the cohort under the counterfactual, with prices held fixed. Participation profiles under each experiment are in Figure F.1 and Figure F.2.

The next largest change concerns Medicare. Covering all medical expenditures after age 65, for participants and nonparticipants alike, would raise welfare by 4.28 percent (reported in Table 3 as the loss from the incomplete coverage of the status quo). Out-of-pocket medical expenses rise with age and are higher in poor health and with a disability (Section 2.5), so their uninsured part is a large old-age risk (De Nardi et al. 2010). A full subsidy to saving that raises the net return to the pretax return (D3) raises welfare by 3.60 percent, and abolishing the income tax (D4) raises it by 3.16 percent. Eliminating disability insurance (D5) lowers welfare by only 0.34 percent. The loss falls on individuals with a disability after age 50, a small share of the cohort, and the ex ante measure weights it by the probability of reaching that state. The leisure-equivalent changes are proportional to the consumption-equivalent ones and preserve the ordering.

The elimination of Social Security also moves participation the most (Figure F.1 and Figure F.2 in Appendix F). Without benefits, participation falls before age 50, when work no longer builds future benefits: by up to about 4 percentage points in the aggregate, and

by more in poor health and with a disability. It rises from the 50s on in every health state, as individuals must finance old-age consumption from their own earnings and savings, starting at 50 for individuals with a disability and in the mid-50s for those in good health. Eliminating DI raises participation after age 50 in every health state, by far the most for individuals with a disability, for whom DI is a pathway out of the labor force before the NRA. The three experiments that raise welfare—full Medicare coverage, the full savings subsidy, and the abolition of the income tax—lower aggregate participation at older ages, consistent with an income effect of the additional old-age resources that each provides.

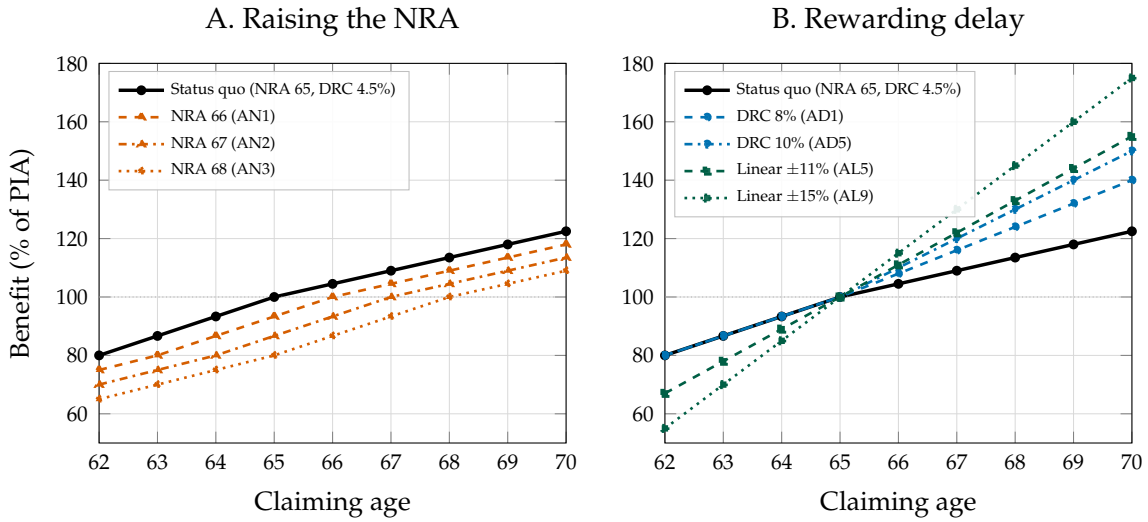
In welfare terms, in our model, which does not tie health insurance to employment (Section 2.5), these rankings qualify the emphasis that French and Jones (2011) place on Medicare in retirement decisions: completing Medicare coverage after 65 would be worth 4.28 percent of consumption, less than the 5.89 percent that Social Security benefits are worth.

None of these experiments holds the government budget fixed. D3 and D4 transfer resources to households without financing them, and the other experiments change government spending without any offsetting change elsewhere in the budget. Their welfare effects measure the value of the instruments as they stand, not the desirability of changing them: the gain from D4 says nothing about who would pay for abolishing the income tax. The MVPF puts the welfare effect of each reform against its net cost to the government, and we use it for the rest of the section.

5.2 Rewarding versus Mandating Delay

Reforms that reward later claiming raise welfare and save money; reforms that mandate it, by raising the NRA, save money and lower welfare. Figure 4 shows how each family changes the benefit, in percent of the PIA, that an individual receives for each claiming age between 62 and 70. Raising the NRA (Panel A) shifts the whole schedule to the right, so that the benefit falls at every claiming age: under an NRA of 66 (AN1), a claim at 66 pays what a claim at 65 pays in the status quo. The DRC reforms AD1–AD5 (Panel B) raise the benefit at every claiming age above the NRA and leave it unchanged at and below the NRA. The symmetric reforms AL1–AL9 set the reduction for early claiming and the credit for delay to a common rate and rotate the schedule around the NRA, which stays at 65: under AL9, benefits range from 55 percent of the PIA at 62 to 175 percent at 70, against 80 and 122.5 percent in the status quo. At unchanged claiming ages, an NRA increase lowers the benefit of every claimant, whereas an AD reform raises the benefit of every claimant who claims after 65 and lowers no one's. An NRA increase mandates later retirement; a higher credit rewards it.

Figure 4. Claiming-Age Schedules under the Status Quo and Selected Reforms



Notes: Retirement benefit received by an individual who claims at each age from 62 to 70, in percent of the primary insurance amount (PIA). Status quo (black): normal retirement age (NRA) of 65, reduction of 6.67 percent of the PIA per year of claiming before the NRA, and delayed retirement credit (DRC) of 4.5 percent of the PIA per year of claiming after the NRA, approximately the average credit faced by the 1920–1940 birth cohorts. Panel A: AN1–AN3 raise the NRA to 66, 67, and 68. Panel B: AD1 and AD5 raise the DRC to 8 and 10 percent; AL5 and AL9 set both the early-claiming reduction and the DRC to 11 and 15 percent per year. Table D.1 reports the schedules of all reforms; Appendix D describes their construction.

Table 4 reports the results. An NRA of 66, 67, or 68 lowers welfare by 0.38, 0.74, or 1.09 percent and lowers net government spending by 0.87, 1.75, or 2.38 percent. The saving comes from Social Security itself: retirement outlays fall by 6.13, 11.82, and 16.75 percent, whereas the indirect effects are small, between -0.02 and -0.12 percent. Participation rises at 65–69, by 0.60, 1.98, and 3.27 points, consistent with an income effect of lower benefits at every claiming age. It falls slightly at 30–49, consistent with the lower return to the earnings that build the AIME when each dollar of PIA pays less, and, for AN2 and AN3, at 62–64. In the model, the small indirect effects combine the taxes lost through these declines, the taxes paid by those who work longer at 65–69, and longer DI receipt: because DI is paid until the NRA, a higher NRA lengthens DI receipt for individuals with a disability (Section 2.7). The Indirect column does not separate these components.

The MVPFs of the NRA increases, 0.17, 0.16, and 0.18 on the post-62 base, are welfare costs per dollar saved. Up to the baseline used for the dollar conversion, they are also the MVPFs of the reverse reforms, which would return the NRA to 65 for a cohort that faces a higher NRA: the shadow price of raising revenue by cutting the policy (Hendren and Sprung-Keyser 2020). The MVPF is nearly constant across the three increases because WTP and the net saving rise nearly in proportion with the NRA.

Table 4. Reforms of the Claiming-Age Schedule

		Welfare		Fiscal cost (% change)			Δ Participation (pp)		
	Reform	MVPF	WTP (%)	Net	Direct	Indirect	30–49	62–64	65–69
A. Raising the normal retirement age (NRA)									
(AN1)	NRA at 66	0.17	−0.38	−0.87	−6.13	−0.02	−0.04	0.38	0.60
(AN2)	NRA at 67	0.16	−0.74	−1.75	−11.82	−0.12	−0.07	−0.04	1.98
(AN3)	NRA at 68	0.18	−1.09	−2.38	−16.75	−0.08	−0.14	−0.43	3.27
B. Raising the delayed retirement credit (DRC)									
(AD1)	DRC at 8%	∞	0.16	−0.45	−2.46	−0.10	0.11	0.15	0.74
(AD2)	DRC at 8.5%	∞	0.19	−0.44	−2.87	−0.04	0.10	0.18	0.57
(AD3)	DRC at 9%	∞	0.23	−0.66	−4.22	−0.07	0.09	−0.36	1.20
(AD4)	DRC at 9.5%	∞	0.26	−0.73	−4.56	−0.09	0.12	−0.49	1.37
(AD5)	DRC at 10%	∞	0.30	−0.78	−5.05	−0.08	0.14	−0.97	1.39
C. Symmetric actuarial adjustment around the NRA									
(AL1)	Linear ±7%	∞	0.10	−0.12	−0.74	−0.01	0.05	0.24	0.22
(AL2)	Linear ±8%	∞	0.15	−0.46	−2.81	−0.07	0.11	0.05	0.61
(AL3)	Linear ±9%	∞	0.21	−0.72	−4.74	−0.07	0.11	−0.31	1.21
(AL4)	Linear ±10%	∞	0.29	−0.86	−5.54	−0.09	0.14	−0.93	1.42
(AL5)	Linear ±11%	∞	0.37	−1.17	−7.02	−0.19	0.19	−1.15	2.01
(AL6)	Linear ±12%	∞	0.47	−1.37	−7.72	−0.28	0.30	−1.46	2.61
(AL7)	Linear ±13%	∞	0.57	−1.38	−8.21	−0.23	0.32	−2.90	2.50
(AL8)	Linear ±14%	∞	0.69	−1.47	−8.22	−0.31	0.34	−3.30	2.42
(AL9)	Linear ±15%	∞	0.80	−1.57	−8.45	−0.37	0.36	−3.49	2.26

Notes: Each row reports the outcomes of simulating the estimated model for the 1920–1940 birth cohorts under the reform, announced before age 30 and in force over the life cycle, with prices held fixed (partial equilibrium). Status quo: NRA of 65, early-claiming reduction of 6.67 percent of the PIA per year, DRC of 4.5 percent per year (approximately the cohort average). AN1–AN3 raise the NRA to 66, 67, and 68; AD1–AD5 set the DRC at 8 to 10 percent, in steps of 0.5 percentage points; AL1–AL9 set the early-claiming reduction and the DRC to a common rate of 7 to 15 percent per year (Figure 4 and Table D.1). WTP: ex ante consumption-equivalent welfare change at age 30, in percent of consumption at every age. Direct: percentage change in the present value of Social Security retirement benefits, including the effect of claiming responses. Indirect: percentage change in the present value of all other taxes and transfers (DI benefits and consumption-floor transfers, net of income- and payroll-tax revenue). Net: percentage change in the present value of government spending. Present values are per capita and discounted to age 62 at the net interest rate (Section 4.3); because the three fiscal columns have different bases, Net is not the sum of Direct and Indirect. MVPF: WTP valued with the present value of per capita baseline consumption from age 62, divided by the dollar change in the present value of per capita government spending (equation (14)). ∞: WTP > 0 and net cost < 0; here the direct cost is also negative, so the lower bound of equation (15) is infinite as well. For AN1–AN3, which lower welfare and save money, the MVPF is the welfare cost per dollar saved. Δ Participation: change in the labor force participation rate at ages 30–49, 62–64, and 65–69, in percentage points. Abbreviations: AIME, average indexed monthly earnings; DI, disability insurance; DRC, delayed retirement credit; NRA, normal retirement age; PIA, primary insurance amount.

Reforms that reward delay raise welfare and save money. Every AD and AL reform has a positive WTP, from 0.10 percent for AL1 to 0.80 percent for AL9, and lowers net, direct, and indirect spending. Their MVPF is therefore infinite whether it is computed with the net cost or with the direct cost alone (equation (15)). For the AD reforms, the sign of the WTP holds by construction: at fixed prices and taxes, an AD reform raises the benefit of every claim after the NRA and lowers none, so it weakly enlarges every individual's choice set. The AL reforms trade a lower benefit for early claiming against a higher one for late claiming, and the sign of their WTP is a result of the simulation.

The direct effect is the most informative column. Because an AD reform raises the benefit at every claiming age above the NRA and lowers none, its mechanical effect on retirement outlays, evaluated at unchanged behavior, is nonnegative. Yet retirement outlays fall, by 2.46 percent under AD1 and by 5.05 percent under AD5. The behavioral responses within Social Security—the induced changes in claiming ages, together with the effects of work on the earnings test and the AIME—therefore more than pay for the higher credit, before any effect on taxes. Lemma 1 gives the accounting (Section 4.6). A one-year delay lowers the present value of a claimant's benefits when the credit is below the rate δ^* that makes the delay actuarially fair for that claimant from the government's perspective (equation (18)), and this rate is higher for claimants with lower survival. Abstracting from the earnings test and the AIME, the fall in direct outlays thus means that, at credits of 8 to 10 percent, the induced delays occur on balance among claimants for whom the new credit is still below δ^*.[3] The status quo leaves room for this: at the credits these cohorts faced, delaying a claim past 65 was, on average, not actuarially advantageous (Coile et al. 2002; Shoven and Slavov 2014). In terms of the decomposition of Section 4.5, the behavioral fiscal externality of an AD reform is negative and larger in absolute value than its mechanical cost, so its fiscal multiplier, the ratio of net to mechanical cost, is negative. The Indirect column adds to the saving: the net cost of income and payroll taxes, DI, and means-tested transfers falls by 0.01 to 0.37 percent across the AD and AL reforms, consistent with the additional work at 65–69 that every one of them induces.

The moderate reforms raise participation in every age group we track. AD1, AD2, AL1, and AL2 raise participation by 0.05 to 0.11 points at 30–49, by 0.05 to 0.24 points at 62–64, and by 0.22 to 0.74 points at 65–69. The response at young ages is consistent with the higher value of a dollar of PIA for those who expect to claim late. Steeper schedules move participation at 62–64 and at 65–69 in opposite directions. From AD3 and AL3 on,

[3] By Lemma 1, two features of the model raise the fair rate: the horizon ends at age 85, which truncates the stream of benefits that a delay raises, and the government discounts at the net return r. Survival beyond 85 or a lower discount rate would lower it (Section 7.4).

participation at 62–64 falls, by up to 3.49 points under AL9, while participation at 65–69 rises by 1.20 to 2.61 points. We interpret this pattern as consistent with selection by health. In the model, a steeper schedule widens the gap between the benefits of early and late claimers and sorts individuals more sharply by the cost of continued work: those in good health, with low costs of work, have the most to gain from staying longer to collect the larger credit, while less healthy workers have the least. The tables report participation by age, not by health, so they do not test this interpretation directly. The larger credit may also lower participation before 65 through an income effect on those who would claim late in any case, for whom it is a windfall at unchanged behavior. The net saving nonetheless rises with the slope of the schedule, from 0.72 percent under AL3 to 1.57 percent under AL9, even as participation at 62–64 falls, and the indirect saving is largest for the steepest reforms (0.23 to 0.37 percent for AL6–AL9, against 0.07 percent for AL3).

The welfare contrast between the two families reflects the difference between a reward and a mandate (Section 4.6). A credit is paid only to those who delay claiming, and in the model delay is cheapest for individuals in good health, whose fixed cost of work is lower and whose wages are higher. Under the AD reforms, those who claim early, in the model often because poor health makes work costly, keep the benefits of the status quo. Because WTP is computed at age 30, before health histories are realized, it values the higher reward in the states in which work remains feasible together with the unchanged terms of an early claim in the states in which it does not; the tables do not separate the two. An NRA increase lowers the benefit at every claiming age. Its cost falls also on those who cannot delay cheaply: individuals in poor health face a higher fixed cost of work and lower wages (Section 3) and receive less whatever their claiming age.

The symmetric reforms combine a larger credit with a larger reduction for early claiming, and comparing reforms with the same credit isolates the role of the reduction. AL2 and AD1 both pay a credit of 8 percent; AL4 and AD5 both pay 10 percent. The additional reduction for early claiming changes the saving (-0.46 against -0.45 percent; -0.86 against -0.78 percent) and the WTP (0.15 against 0.16 percent; 0.29 against 0.30 percent) by amounts at or near the precision of the simulations. The reward does most of the work. The steepest AL reforms, with credits of 11 to 15 percent, raise welfare by more than any AD reform, whose credits stop at 10 percent. Within the AL family, WTP and fiscal savings both rise with the slope of the schedule up to the steepest reform we consider, so our grid does not reach the slope at which further steepening would stop paying for itself. This is in line with the finding of Ndiaye and Yu (2026) that, in the constrained-efficient allocation of the same economy, consumption in retirement rises with the claiming age more steeply than the statutory credit implies.

The Social Security Amendments of 1983 raised the DRC gradually to 8 percent per year, in a phase-in that began with workers turning 65 in 1990 and therefore partly covers the cohorts we study, and raised the NRA toward 67, beginning with workers born in 1938 (Svahn and Ross 1983). AN2, an NRA of 67, and AD1, a DRC of 8 percent, correspond to the end points of these two changes, evaluated for our cohort relative to its cohort-average schedule. In the model, the credit increase raises welfare by 0.16 percent and lowers government spending by 0.45 percent, an infinite MVPF; the NRA increase lowers welfare by 0.74 percent and saves 1.75 percent, an MVPF of 0.16. The two halves of the change in claiming incentives sit at opposite ends of the ranking. This is a comparison within the model, for cohorts born between 1920 and 1940. It is not an evaluation of the 1983 law, which changed other provisions as well and phased in its changes over many cohorts (Section 7.4).

Section 6 compares the fiscal savings of the two families. In the model, the NRA affects behavior only through the benefit, DI, and earnings-test rules it determines; Section 7.3 discusses evidence that statutory ages also act as reference points (Seibold 2021).

5.3 Benefit Progressivity and Level

The second family changes the benefit formula itself. P1.1–P1.9 raise the replacement rate of the first bracket of the PIA formula from 0.90 to between 0.91 and 0.99. The level reforms L0–L9 raise all three rates proportionally, by 1 to 10 percent. Table F.1 in Appendix F reports reforms that raise the second rate (P2) and then the third rate (P3) on top of a first rate of 0.99. These reforms include the increase of the first rate to 0.99 of P1.9, so their entries, MVPFs included, measure the full change from the status quo. At unchanged behavior, each of these reforms raises the benefit of every retiree with a given AIME, so its mechanical effect is a transfer to retirees. All of them raise Social Security outlays, and all but the smallest raise net government spending (Table 5).

The smallest expansion of the bottom bracket, P1.1, is the exception. It raises retirement outlays by 0.57 percent and changes net government spending by -0.03 percent, so its MVPF is infinite as computed; counting only its direct cost gives a lower bound of 0.52. A net saving of this size is within the precision of the simulations, and we read P1.1 as approximately budget neutral. Its indirect saving offsets its direct cost: P1.1 raises participation in all three age groups, and in our decomposition of this reform the income taxes paid by the additional participants account for 57 percent of its indirect fiscal effect. The larger expansions of the first rate, P1.2–P1.9, have MVPFs between 0.69 and 1.14, the lowest for the largest reform. Their net costs are small, 0.09 to 0.44 percent, so errors of a few hundredths of a percentage point move these MVPFs noticeably, and we do not

Table 5. Reforms of Benefit Progressivity and Level

		Welfare		Fiscal cost (% change)			Δ Participation (pp)		
	PIA rates	MVPF	WTP (%)	Net	Direct	Indirect	30–49	62–64	65–69
A. Raising the first PIA rate (bottom bracket)									
(P1.1)	(0.91, 0.32, 0.15)	$[0.52, +\infty)$	0.10	−0.03	0.57	−0.10	0.08	0.20	0.06
(P1.2)	(0.92, 0.32, 0.15)	0.97	0.22	0.09	1.23	−0.07	0.14	−0.07	0.00
(P1.3)	(0.93, 0.32, 0.15)	0.92	0.31	0.13	1.66	−0.09	0.14	0.15	0.01
(P1.4)	(0.94, 0.32, 0.15)	0.84	0.40	0.19	2.16	−0.10	0.14	−0.17	−0.23
(P1.5)	(0.95, 0.32, 0.15)	0.91	0.50	0.21	2.32	−0.09	0.16	−0.34	−0.31
(P1.6)	(0.96, 0.32, 0.15)	1.14	0.59	0.20	2.80	−0.16	0.19	−0.27	−0.15
(P1.7)	(0.97, 0.32, 0.15)	1.01	0.67	0.26	3.22	−0.16	0.21	−0.57	−0.50
(P1.8)	(0.98, 0.32, 0.15)	0.86	0.75	0.34	3.72	−0.15	0.24	−0.71	−0.49
(P1.9)	(0.99, 0.32, 0.15)	0.69	0.80	0.44	4.12	−0.11	0.25	−0.85	−0.60
B. Raising all PIA rates proportionally (level)									
(L0)	(0.909, 0.3232, 0.1515)	0.51	0.16	0.12	1.34	−0.05	0.11	−0.05	0.01
(L1)	(0.918, 0.3264, 0.153)	0.59	0.32	0.21	2.47	−0.11	0.14	−0.15	−0.13
(L2)	(0.927, 0.3296, 0.1545)	0.49	0.46	0.36	3.48	−0.10	0.13	−0.49	−0.08
(L3)	(0.936, 0.3328, 0.156)	0.61	0.60	0.38	4.25	−0.18	0.18	−0.29	−0.30
(L4)	(0.945, 0.336, 0.1575)	0.59	0.74	0.49	5.28	−0.21	0.21	−0.48	−0.43
(L5)	(0.954, 0.3392, 0.159)	0.52	0.87	0.64	6.31	−0.19	0.23	−0.72	−0.68
(L6)	(0.963, 0.3424, 0.1605)	0.47	0.95	0.79	7.41	−0.20	0.24	−1.00	−1.22
(L7)	(0.972, 0.3456, 0.162)	0.42	1.00	0.92	8.66	−0.23	0.26	−1.40	−1.43
(L8)	(0.981, 0.3488, 0.1635)	0.35	1.02	1.14	9.80	−0.18	0.27	−1.76	−1.86
(L9)	(0.99, 0.352, 0.165)	0.30	0.98	1.24	10.70	−0.20	0.28	−2.37	−1.99

Notes: Each row reports the outcomes of simulating the estimated model for the cohort under the reform, announced before age 30 and in force over the life cycle, with prices held fixed (partial equilibrium). The second column gives the three replacement rates of the PIA formula (first, second, and third bracket of the AIME); the status quo is (0.90, 0.32, 0.15). P1.1–P1.9 raise the first rate by 1 to 9 percentage points; L0–L9 raise all three rates proportionally by 1 to 10 percent. L9 sets the same rates as P3.9 in Table F.1. WTP: ex ante consumption-equivalent welfare change at age 30, in percent. Direct: percentage change in the present value of Social Security retirement benefits. Indirect: percentage change in the present value of all other taxes and transfers (DI benefits and consumption-floor transfers, net of income- and payroll-tax revenue). Net: percentage change in the present value of government spending. Present values are per capita and discounted to age 62 (Section 4.3); because the three fiscal columns have different bases, Net is not the sum of Direct and Indirect. MVPF: WTP valued with the present value of per capita baseline consumption from age 62, divided by the dollar change in government spending (equation (14)). For P1.1, the net cost is negative and the MVPF infinite; the interval reports as lower bound the MVPF computed with the direct cost only (equation (15)). Δ Participation: change in the labor force participation rate at the indicated ages, in percentage points. Abbreviations: AIME, average indexed monthly earnings; DI, disability insurance; PIA, primary insurance amount.

interpret their non-monotone pattern across P1.2–P1.9. The level reforms have MVPFs between 0.30 and 0.61, and the P2 and P3 reforms, measured from the status quo, between 0.29 and 0.56.

The generosity reforms move participation in opposite directions over the life cycle. At ages 30–49 participation rises slightly, by 0.08 to 0.28 points across all P and L reforms, consistent with a higher return to the earnings that build the AIME in the brackets whose rates rise. At older ages, the larger reforms lower participation through an income effect; under L9, participation falls by 2.37 points at 62–64 and by 1.99 points at 65–69. The indirect effects are small and negative for every reform. In the model, they net out opposing changes that the reported columns do not separate: more work at 30–49 raises tax revenue and higher benefits lower consumption-floor transfers, whereas less work at older ages lowers tax revenue and DI benefits, which are proportional to the PIA, rise.

The bottom bracket delivers the most welfare per dollar because of whom it pays. Comparing reforms with similar costs isolates this. P1.9 and L3 have similar direct costs (4.12 and 4.25 percent) and net costs (0.44 and 0.38 percent), but WTP of 0.80 and 0.60 percent. Similarly, P1.6 and L1 have net costs of 0.20 and 0.21 percent, but WTP of 0.59 and 0.32 percent. The difference lies in welfare, not in fiscal externalities. An increase in the first rate adds the same amount to the PIA of every retiree whose AIME exceeds the first bend point, and less to those below it; a proportional increase adds to each PIA in proportion to its level. Per dollar spent, the first-bracket reform therefore pays relatively more to retirees with a low AIME. In the model, a low AIME is the cumulative result of low ability and of health shocks that lowered wages, raised the cost of work, and shortened careers; retirees with such histories are likely to have few assets and a high marginal utility of consumption. Because WTP is measured before these histories are realized, shifting the transfer toward them is insurance. Raising the upper rates on top of a first rate of 0.99 adds little welfare. Because the P2 and P3 reforms include P1.9, their marginal effect is the difference between rows (Table F.1): from P1.9 to P2.9, WTP rises by 0.15 percent (from 0.80 to 0.95) while the net cost rises by 0.83 percent (from 0.44 to 1.27); from P2.9 to P3.9, WTP rises by 0.03 percent and the net cost falls by 0.03 percent (from 1.27 to 1.24), differences within the precision of the simulations. The marginal value of raising the upper rates is thus far below the cumulative MVPFs of the P2 and P3 reforms.

5.4 Medicare

Panel A of Table 6 reports two expansions of Medicare. For these reforms, the baseline coverage rates of medical expenditures after age 65 are re-estimated from the MEPS by health and working status, and the model with these rates serves as the status quo

(Appendix D), so the Medicare reforms are evaluated against a different baseline from the other reforms. The first reform, MA (labeled Part A), raises the coverage rate of all individuals aged 65 and over by 20 percentage points; the second, MB (labeled Part B), raises it by 20 points for nonparticipants only.[4]

Both expansions raise welfare substantially, by 1.36 and 1.30 percent, but at a high cost. Medicare's payments to the targeted beneficiaries rise by an amount equal to 45.17 percent of baseline payroll-tax revenue for MA and 41.61 percent for MB, and net government spending, including these payments, rises by 3.53 and 3.19 percent. The cost is driven mainly by large medical expenditures after age 70. The MVPFs are 0.146 and 0.155, the lowest finite MVPFs in our grid. Coverage after 65 lowers participation at older ages: by 1.70 and 0.38 points at 62–64 and 65–69 under MA, and by 1.95 and 3.44 points under MB. Better coverage reduces the need to work and save for medical expenses in old age, including in the years before 65. Under MB, the additional coverage is also conditional on not working, so that a worker over 65 who keeps working forgoes it; the expansion acts as an implicit tax on work, consistent with its much larger participation response at 65–69. The indirect effects are positive, 0.10 and 0.03 percent, consistent with the lower tax revenue from lower participation.

Using the estimates of Finkelstein and McKnight (2008), Hendren and Sprung-Keyser (2020) put the MVPF of the introduction of Medicare in 1965 at about 1.6, with a 95 percent confidence interval of $[0.48, 3.55]$. That estimate uses the normalization under which a transfer with no behavioral response has an MVPF of about one, so our post-62 MVPFs must be converted before the two are compared (Section 7.1). Converted, our MVPFs for expansions of an established program lie below that point estimate, as diminishing returns to coverage would suggest, but inside its confidence interval.[5]

5.5 Old-Age Income Taxes

Panel B of Table 6 reports cuts in income taxes from age 62. Reforms I1–I5 raise the level parameter λ_t of the after-tax income schedule $\lambda_t y^{1-\tau_t}$ at each age $t \geq 62$ to $\lambda_t + x(t-61)$, with $x = 0.01, \ldots, 0.05$, so that the cut grows with age. These reforms produce the largest increases in old-age participation of any reform we study. Participation rises by 0.76 to 4.11 points at 62–64 and by 1.98 to 8.92 points at 65–69. The cut raises the after-tax return to

[4]Part A covers mainly inpatient services and is premium-free for most beneficiaries. Part B covers physician and outpatient services, which most employer-provided plans also cover, and requires a premium; we therefore target the Part B expansion at nonparticipants.

[5]Estimates for subsidized health insurance for low-income adults find a willingness to pay well below the cost of coverage (Finkelstein et al. 2019); they concern a different population and program and are not directly comparable to ours.

Table 6. Reforms of Medicare, Old-Age Income Taxes, and Savings Returns

		Welfare		Fiscal cost (% change)			Δ Participation (pp)		
	Reform	MVPF	WTP (%)	Net	Direct	Indirect	30–49	62–64	65–69
A. Medicare: coverage of medical expenditures after age 65 raised by 20 percentage points									
(MA)	Part A: all individuals	0.146	1.36	3.53	45.17	0.10	0.22	−1.70	−0.38
(MB)	Part B: nonparticipants	0.155	1.30	3.19	41.61	0.03	0.21	−1.95	−3.44
B. Old-age income tax: level parameter λ_t raised by $x(t-61)$ at ages $t \geq 62$									
(I1)	$x = 1\%$	0.25	0.44	0.69	0.70	−0.81	0.06	0.76	1.98
(I2)	$x = 2\%$	0.26	0.87	1.28	1.31	−1.51	0.15	1.58	4.12
(I3)	$x = 3\%$	0.25	1.29	2.01	2.03	−2.18	0.18	2.76	6.07
(I4)	$x = 4\%$	0.23	1.69	2.80	2.80	−2.78	0.20	3.19	7.71
(I5)	$x = 5\%$	0.23	2.09	3.46	3.47	−3.58	0.26	4.11	8.92
C. Savings subsidy: net return on assets raised at all ages									
(K1)	+0.2 pp	[0.96, 20.37]	0.87	0.08	1.77	−1.69	0.15	−1.27	−0.85
(K2)	+0.4 pp	[0.92, 19.38]	1.75	0.18	3.77	−3.59	0.36	−2.58	−1.75
(K3)	+0.8 pp	[0.84, 5.29]	3.60	1.34	8.44	−7.10	0.38	−5.98	−4.11

Notes: Each row reports the outcomes of simulating the estimated model for the cohort under the reform, announced before age 30 and in force over the life cycle, with prices held fixed (partial equilibrium). WTP: ex ante consumption-equivalent welfare change at age 30, in percent. Fiscal columns (Section 4.3; present values per capita, discounted to age 62; because the three columns have different bases, Net is not the sum of Direct and Indirect): *Panel A.* Coverage rates after 65 by health and working status re-estimated from the Medical Expenditure Panel Survey (MEPS) form the baseline; MA raises coverage of all individuals aged 65 and over, MB coverage of nonparticipants aged 65 and over. Direct: change in the present value of Medicare's payments to the targeted beneficiaries, in percent of baseline payroll-tax revenue. Indirect: percentage change in the present value of all taxes and transfers, including Social Security benefits. Net: change in government spending plus the change in Medicare's payments to the targeted beneficiaries, in percent of baseline government spending. MVPFs in Panel A are reported to three decimals. *Panel B.* I1–I5 replace the level parameter λ_t of the after-tax income schedule $\lambda_t y^{1-\tau_t}$ at each age $t \geq 62$ by $\lambda_t + x(t-61)$, with $x = 0.01, \ldots, 0.05$ (rows $x = 1\%$ to $x = 5\%$; Appendix D). Direct: income-tax revenue forgone, in percent of its baseline present value. Indirect: percentage change in the present value of all other taxes and transfers, including Social Security benefits, net of payroll-tax revenue. Net: percentage change in government spending. *Panel C.* K1–K3 raise the net return on assets at all ages; K3 sets it equal to the pretax return, as in D3 of Table 3. Direct: present value of the subsidy, in percent of baseline government spending. Indirect: percentage change in government spending excluding the subsidy. Net: percentage change in government spending including the subsidy. MVPF (equation (14)): WTP valued with the present value of per capita baseline consumption from age 62 in Panels A and B and from age 30 in Panel C, divided by the dollar change in government spending. In Panel C, the interval reports as lower bound the MVPF with the subsidy as the only cost (equation (15)) and as upper bound the MVPF with the net cost, which treats capital income as realized and taxed every period and therefore overstates the indirect saving. Because of the different consumption base, the levels of the MVPFs in Panel C are not comparable with those in Panels A and B (Section 7.1). Δ Participation: change in the labor force participation rate at the indicated ages, in percentage points.

work at older ages, and the substitution effect dominates the income effect. Participation also rises slightly at 30–49, by 0.06 to 0.26 points.

Despite these responses, the tax cuts do not pay for themselves. The net cost of all other taxes and transfers, including Social Security, falls by 0.81 to 3.58 percent, consistent with higher payroll-tax revenue from the induced work. The revenue loss dominates: net spending rises by 0.69 to 3.46 percent, and the MVPFs lie between 0.23 and 0.26. The welfare gain per dollar is lower than for any expansion of the benefit formula because of whom the cut pays. A higher λ_t raises after-tax income by $x(t-61)\,y^{1-\tau_t}$, which rises with taxable income. In old age, taxable income consists of earnings, pensions, and interest, while Social Security and DI benefits are not taxed (Section 2.8). The cut therefore pays the most to those who keep working, typically in good health, and to those with large pensions and assets, whose marginal utility is low, and little to retirees who live mainly on Social Security. Ex ante, it is a transfer toward good states rather than insurance against bad ones.

The comparison with the claiming-age reforms shows that a large participation response is neither necessary nor sufficient for a high MVPF. The larger tax cuts raise participation at 65–69 by far more than any reward-delay reform, by up to 8.92 points against at most 2.61 points, and cost money; AD1 raises it by 0.74 points and saves money. What matters is the fiscal saving that a reform recovers from the behavior it induces relative to the transfer it makes to individuals who would have behaved the same way without it. A credit for delay pays nothing to those who claim at or before the NRA; a tax cut on old-age income pays every individual with taxable income from age 62.

5.6 Savings Subsidies

Panel C of Table 6 reports reforms that raise the net return on saving at all ages by 0.2, 0.4, and 0.8 percentage points. K3 raises the net return to the pretax return r_f and is therefore the full subsidy of experiment D3; its WTP of 3.60 percent equals the welfare gain in Table 3. The subsidies raise welfare by 0.87 to 3.60 percent. Their direct cost, the present value of the subsidy, is 1.77 to 8.44 percent of baseline government spending, but most of it returns to the budget: higher saving raises taxable interest income, and government spending excluding the subsidy falls by 1.69 to 7.10 percent, which leaves net costs of 0.08 to 1.34 percent. This recovery is an upper bound, because the model taxes interest income every period, as if capital income were realized and taxed as it accrues. We therefore report each MVPF as an interval. The lower bound counts only the subsidy as a cost and lies between 0.84 and 0.96; the upper bound uses the net cost and lies between 5.29 and 20.37. Because a subsidy changes returns from age 30, its WTP is converted into dollars with the

lifetime consumption base (Section 4.4), so the levels of the savings MVPFs, including their lower bounds, are not comparable with those of the reforms that start at 62; Section 7.1 places the two on a common base.

The subsidies have a strong income effect at older ages. Participation falls by 1.27 to 5.98 points at 62–64 and by 0.85 to 4.11 points at 65–69, as individuals accumulate more wealth and retire earlier, while it rises slightly at 30–49. A higher return makes saving a better substitute for continued work as a source of old-age income.

5.7 Synthesis: The MVPF Map

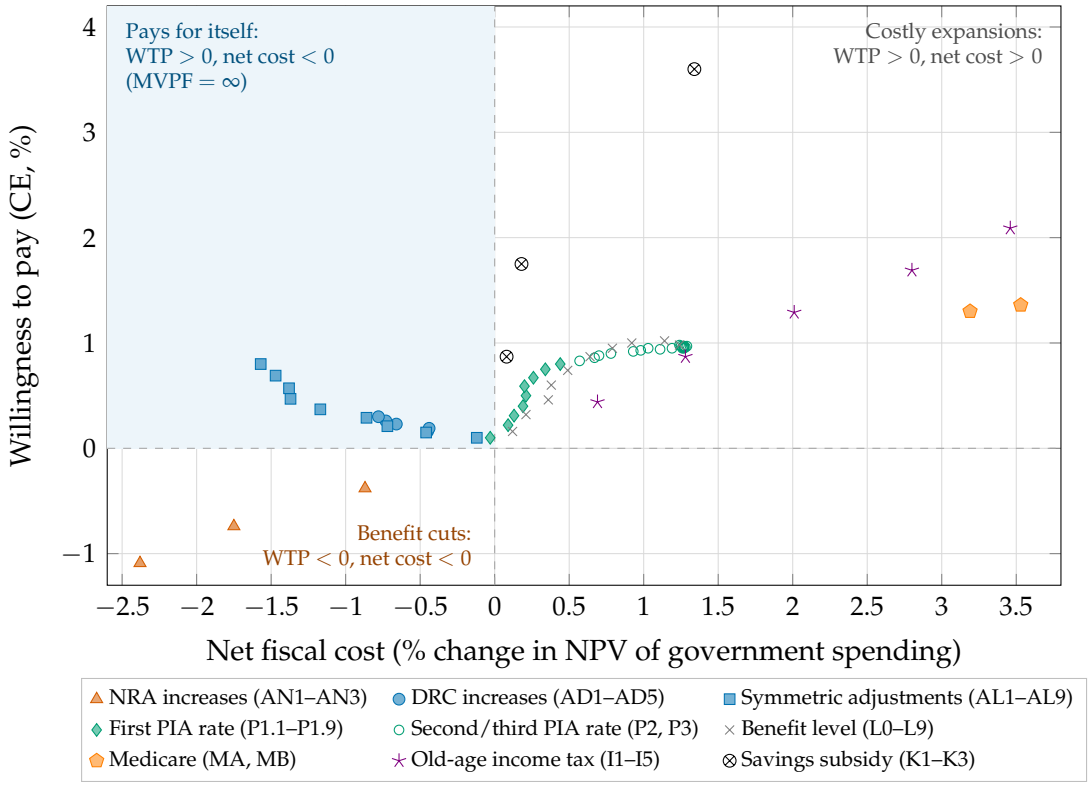
Figure 5 places every reform in the plane of net fiscal cost and WTP. Both axes are measured in the same units for every reform, so the slope of the ray from the origin to a reform, WTP divided by net cost, is proportional to its MVPF on a common consumption base. For the reforms that start at 62, the MVPF equals, up to rounding, this slope times a common factor (Section 4.4).

The map has three regions. The shaded upper-left region contains the fourteen reforms that reward delay, which pay for themselves, and, at its edge, the smallest expansion of the bottom bracket, which is approximately budget neutral. The lower-left quadrant contains the NRA increases, which lower welfare and save money, on rays about as flat as those of the Medicare expansions. The upper-right quadrant contains the costly expansions. Within it, the first-rate reforms lie above the level reforms at the same net cost; the P2 and P3 reforms extend to the right of P1.9 at a nearly constant WTP, adding cost with little additional welfare; the income-tax cuts lie along a ray of nearly constant slope, as their nearly constant MVPFs imply; and the Medicare expansions, at net costs similar to the largest tax cuts but with lower WTP, lie on the flattest rays among the expansions.

Because the reforms that start at 62 share a common consumption base, their MVPFs can be ranked directly. Reforms that reward delay come first, with infinite MVPFs. Expansions of the bottom bracket follow, with MVPFs between 0.69 and 1.14; the smallest is approximately budget neutral. Level reforms and the P2 and P3 reforms follow at 0.29 to 0.61, then cuts in old-age income taxes at 0.23 to 0.26. A reduction of the NRA, the reverse of AN1–AN3, at 0.16 to 0.18 up to the baseline of the dollar conversion, and the Medicare expansions, at 0.146 to 0.155 against their own baseline, come last. The savings subsidies appear on the same axes, but their net costs rest on the upper-bound treatment of capital-income taxes, and Section 7.1 places their MVPFs on the common base.

The ranking reflects two properties of a reform: whom it pays at unchanged behavior, and which behavior it rewards. A credit for delay pays only those who claim after the NRA, and the delays it induces lower the program's outlays. The bottom bracket shifts

Figure 5. The MVPF Map: Welfare and Net Fiscal Cost of All Reforms



Notes: Each marker is one reform of Tables 4–6 and F.1; the markers of P2.9, P3.0–P3.9, and L9 overlap near a net cost of 1.25 and a WTP of 0.97. Horizontal axis: net fiscal cost, the percentage change in the present value of per capita government spending discounted to age 62 (Net). Vertical axis: willingness to pay, the ex ante consumption-equivalent welfare change at age 30, in percent (WTP). Shaded region: WTP > 0 and net cost < 0, where the MVPF is infinite. Lower-left quadrant: reforms that lower welfare and save money, for which the MVPF is the welfare cost per dollar saved. Upper-right quadrant: reforms that raise welfare at a positive net cost. For the reforms that start at age 62, the MVPF equals, up to rounding, the slope of the ray from the origin times a common factor (Section 4.4). The savings subsidies use a different consumption base for their MVPFs, and their net costs treat capital income as taxed every period (Table 6). All entries come from simulations of the estimated model with prices held fixed. Abbreviations: CE, consumption equivalent; DRC, delayed retirement credit; NPV, net present value; NRA, normal retirement age; PIA, primary insurance amount.

benefits toward retirees whose lifetime earnings were depressed by low ability or poor health. A lower NRA pays every claimant, and an old-age tax cut pays more to those with more taxable income in old age. A Medicare expansion pays for medical expenditures concentrated at the oldest ages, which drive its large cost.

6 Closing a Financing Gap

Population aging puts the financing of Social Security under pressure (Board of Trustees, OASDI 2024; Congressional Budget Office 2024), and closing a financing gap requires raising revenue or cutting outlays. The MVPF ranks the instruments available for this

purpose by the welfare cost of each dollar they save (Hendren 2016; Hendren and Sprung-Keyser 2020). All statements below refer to the reforms in our grid and to the cohort and rules we study.

Reforms that save money and raise welfare. A reform with an infinite MVPF relaxes the budget at no welfare cost and therefore comes first in any consolidation. In our grid, apart from P1.1, which is approximately budget neutral (Section 5.3), these are the reforms that reward later claiming (Table 4), and their savings are not small. The DRC increases lower the present value of government spending by 0.44 to 0.78 percent, and the symmetric adjustments by 0.12 percent (AL1) to 1.57 percent (AL9), while raising welfare by 0.10 to 0.80 percent. Within the symmetric family, fiscal saving and welfare rise together with the slope of the schedule, so the reform that saves the most, AL9, also raises welfare the most. For these reforms there is no trade-off between the budget and welfare to resolve: the government obtains the saving by paying a larger reward to those who delay, the induced delays more than finance the reward within Social Security, and the net cost of income and payroll taxes, DI, and means-tested transfers falls as well (Section 5.2).

Rewarding delay and raising the NRA. The natural comparison is with increases in the NRA, the instrument the 1983 Amendments used alongside a higher credit. For a fiscal saving up to the size of AN1, 0.87 percent of government spending, several reward-delay reforms deliver a similar or larger saving with a welfare gain instead of a welfare loss. AD5 saves 0.78 percent and AL4 saves 0.86 percent; AL5, AL6, AL7, AL8, and AL9 save 1.17, 1.37, 1.38, 1.47, and 1.57 percent. AN1 lowers welfare by 0.38 percent; each of these reforms raises it, AD5 by 0.30 percent and AL9 by 0.80 percent. AL9 also raises participation at 65–69 by more than AN1, by 2.26 against 0.60 points. Larger consolidations are a different matter. AN2 and AN3 save 1.75 and 2.38 percent, more than any single reward-delay reform in our grid. We have not computed combinations of reforms, such as a steeper claiming schedule together with a higher NRA, so the grid does not say how a saving of that size would best be obtained. It does show that, for the cohort and rules we study, savings of up to 1.57 percent of government spending can be obtained while raising welfare.

The welfare cost of a dollar saved. Once the reforms with infinite MVPF are exhausted, further savings require reforms that lower welfare, and the MVPF of such a reform measures its welfare cost per dollar saved. The NRA increases have MVPFs of 0.16 to 0.18. Our grid contains three other families against which this cost can be compared—the level of benefits, Medicare coverage after 65, and income taxes from age 62—but we simulate

them as expansions, with MVPFs of 0.30 to 0.61, 0.146 to 0.155, and 0.23 to 0.26. To a first order, the MVPF of a cut equals the MVPF of the corresponding expansion. For a small change $d\zeta$ in a policy parameter ζ, both the WTP and the net cost are proportional to $d\zeta$; reversing the sign of $d\zeta$ reverses the sign of both and leaves their ratio unchanged. Read this way, raising the NRA costs about as much welfare per dollar saved as reversing an expansion of Medicare coverage after 65, whose MVPFs are measured against a baseline with re-estimated coverage rates (Section 5.4). It costs less welfare per dollar than raising income taxes on older individuals or cutting all replacement rates of the benefit formula proportionally, whose MVPFs as cuts would be, to a first order, 0.23 to 0.26 and 0.30 to 0.61. All of these MVPFs share the post-62 consumption base, so the comparison does not depend on the dollar conversion (Section 4.4). Among the welfare-reducing instruments we consider, NRA increases are thus, to a first order, a comparatively inexpensive source of revenue, even though they are dominated by rewards for delay for moderate savings.

This symmetry is a first-order argument, and our reforms are not marginal. If the WTP is concave in the policy parameter, as diminishing returns imply, the welfare lost by moving the parameter down exceeds the welfare gained by moving it up by the same amount, and the reversed MVPFs understate the welfare cost of a cut. The grid speaks to this curvature only within each family of expansions. The MVPFs of I1–I5 hardly change with the size of the reform, which suggests that the curvature is mild for old-age income taxes over the range we consider. The MVPFs of the level reforms fall from 0.61 for L3 to 0.30 for L9; for a cut in replacement rates, the smaller reforms L0–L4, with MVPFs of 0.49 to 0.61, are the more relevant comparison. The two Medicare expansions have the same size and are silent on curvature. Behavioral responses to cuts and expansions also need not be symmetric when participation involves fixed costs of work and a cost of reentry. We have not simulated the reverse reforms, and the comparison between NRA increases and tax increases, Medicare cuts, or proportional benefit cuts is an approximation rather than a computed result. The comparison between NRA increases and reward-delay reforms involves no such approximation.

Who bears the gap. Our analysis concerns a single cohort, facing reforms announced before age 30 and in force over its whole life. A financing gap also raises the question of how a legacy debt is shared across cohorts, and a cohort analysis does not determine who should bear it. The MVPF ranks the instruments for raising a dollar from a given cohort; it does not say which cohorts should contribute that dollar. The transition generations, cohorts already in the middle of their careers or near retirement when a reform is announced, are not modeled. For them, both the welfare cost and the fiscal saving of a reform depend on

how much of their working life remains when it is announced: an NRA increase announced late in a career leaves little room to adjust saving and labor supply, and a higher credit can change the behavior only of those who have not yet claimed. Prices are held fixed throughout, so the analysis also abstracts from the effects of a consolidation on wages and interest rates (Section 7.2).

With these qualifications, the MVPFs suggest an order for closing a financing gap for the cohort and rules we study. Reforms that reward later claiming come first, because they save money and raise welfare. Beyond the savings they can deliver, the choice among reforms that lower welfare should follow their welfare cost per dollar saved, on which increases in the NRA compare favorably, to a first order, with higher taxes on older individuals and with proportional cuts in benefits.

7 Discussion

This section examines which of our results depend on measurement conventions and modeling choices.

7.1 Converting Welfare into Dollars

The WTP for a reform is the proportional change in consumption, at every age from 30 to T and in every state, that the cohort values as much as the reform (equation (13)). Equation (14) converts it into dollars with the present value of baseline per capita consumption from age 62, $\bar{C}^0_{62}$, for the reforms that start at 62, and from age 30, $\bar{C}^0_{30}$, for the savings subsidies. Only the lifetime base values the consumption change that defines the CE: on it, a transfer that raises consumption in the same proportion at every age and in every state, and induces no behavioral response, has an MVPF of one, whereas on the post-62 base its MVPF is $\bar{C}^0_{62}/\bar{C}^0_{30}$. The tables pin down this ratio. Every MVPF computed with the net cost equals the ratio of the WTP and Net entries times a factor that depends on baseline aggregates but not on the reform; computed from the rounded entries, the factor lies between 0.38 and 0.40 for the reforms that start at 62 and between 1.87 and 1.99 for the savings subsidies (Section 4.4). Because every Net entry is a percentage change of the same baseline aggregate, government spending in the status quo, the two factors differ only through the consumption base. The lifetime base, which adds the consumption of the 32 years from age 30 to 61, is therefore about five times the post-62 base: the ratio lies between 4.7 and 5.2.

Three kinds of statements do not depend on the base: whether an MVPF is infinite, which depends only on the signs of WTP and net cost; the ranking of reforms within a family, which share a base; and the ranking across the families that start at 62 (Section 5.7),

which share the post-62 base. The central results of the paper are of these kinds. The base matters for the levels of finite MVPFs and for two comparisons. First, on the post-62 base, the lower bounds of the savings MVPFs, 0.84 to 0.96, would be about one-fifth as large, roughly 0.16 to 0.20: close in level to the MVPFs of the NRA increases and below those of the old-age income-tax cuts. Their upper bounds would be about 4 for K1 and K2 and about 1 for K3. Where the subsidies rank among the reforms that start at 62 thus depends on how much of the indirect saving from taxing interest income, an upper bound in the model, materializes.

Second, on the lifetime base, every finite MVPF of the reforms that start at 62 would be about five times larger. The MVPFs of the old-age income-tax cuts and of every expansion of the benefit formula would exceed one. The MVPFs of the Medicare expansions, about 0.7 to 0.8, would lie below the point estimate of about 1.6 for the introduction of Medicare but inside its 95 percent confidence interval of 0.48 to 3.55; on the post-62 base, they lie below that interval (Section 5.4). Each dollar saved by raising the NRA would cost about 75 to 94 cents of willingness to pay, rather than the 16 to 18 cents measured on the post-62 base. A comparison of an MVPF with one is therefore meaningful only on the lifetime base.

7.2 General Equilibrium

Every reform is evaluated at a fixed wage per efficiency unit and a fixed interest rate, without balancing the budget. Fixed prices are exact in a small open economy, in which, with constant returns to scale, the world return to capital also fixes the wage. In a closed economy, a reform applied to every cohort would also move the wage, through aggregate labor input, the return to capital, through aggregate saving, and the taxes or spending that absorb its effect on the budget. Overlapping-generations analyses of Social Security reform incorporate these channels (Conesa and Krueger 1999; Nishiyama and Smetters 2007; Imrohoroğlu and Kitao 2012; Kitao 2014).

The price channels are pecuniary. By the envelope theorem, individuals' responses to a small price change have no first-order effect on their welfare, which changes only through the direct effect of prices on their income. With constant returns to scale, the fall in the wage and the rise in the return that follow an increase in labor input leave aggregate pretax income unchanged at the initial factor supplies. They redistribute it from workers to asset holders, and they move the WTP of a reform only through differences in marginal utility between those who gain and those who lose. Their effect on the budget differs: additional labor raises output and the tax base in general equilibrium as in partial equilibrium, and the shift of income from labor to capital changes revenue by the difference between the tax

rates on the two.

The sign of the fiscal saving of the reward-delay reforms is likely to survive these feedbacks. Its first source, the fall in retirement outlays, comes from claiming responses whose fiscal sign depends on prices only through the discount rate of Lemma 1: both terms of equation (17) are proportional to the PIA, so the wage affects their size but not their sign. Its second source is additional work. These reforms raise participation at 65–69 and, in their moderate versions, in every age group we track; a lower wage would reduce the revenue from each additional hour but, for a small reform, would not reverse the response that generates it. Returned through any instrument with a positive MVPF, the saving raises welfare further while balancing the budget (Appendix E). The saving of the NRA increases is mechanical, so its sign does not depend on prices either. Correcting the size of these effects would require, beyond the cohort's responses that the model delivers, the elasticities of the wage and the return to capital with respect to aggregate labor and saving, each reform's effect on these aggregates summed over the cohorts alive in a steady state, the tax rates on labor and capital income, and the instrument that balances the budget.

7.3 Statutory Ages as Reference Points

In our model, the NRA matters only through the retirement benefits, DI eligibility, and earnings-test rules it determines. Evidence suggests that statutory ages matter beyond these incentives. Seibold (2021) documents bunching of retirements at statutory ages in Germany, including ages at which the financial incentives to retire change little or not at all, and interprets it as evidence that statutory ages act as reference points. In Switzerland, an increase in women's full retirement age delayed both claiming and labor market exit by more than a standard life-cycle model predicts, whereas changes in the penalties for early claiming with the full retirement age held fixed had much smaller effects (Lalive et al. 2023). In the United States, the spike in benefit claiming at the full retirement age moved with it as the 1983 Amendments raised it (Behaghel and Blau 2012); when it rose from 65 to 66, however, labor force exit remained concentrated at 65 (Deshpande et al. 2024).

Reference dependence would change the fiscal effect of an NRA increase in ways our model does not capture, and it would make its welfare cost harder to measure. If claims and exits both follow the NRA, as in the Swiss evidence, an increase raises participation below the new NRA by more than our model predicts; for a given WTP, the larger fiscal saving would lower its MVPF further. If claims follow the NRA but exits do not, as in the US evidence, the rise in participation at 65–69 that our model predicts, 0.60 to 3.27 points, may overstate the response. Because the indirect effects of the NRA increases are small (Section 5.2), their saving would then change mainly through the shift of claims

toward the new NRA, whose effect on outlays depends on the increments of the schedule relative to claimants' actuarially fair rates (Section 4.6). On the welfare side, if the reference point is part of preferences, moving it changes utility directly; if it is a norm followed without optimizing, the envelope argument under which behavioral responses enter the MVPF only through the budget (Section 4.5) fails. In both cases, a WTP computed from preferences over consumption and leisure misses part of the welfare effect of the reform, and the behavioral evidence does not determine its sign.

The reward-delay reforms leave the NRA, and any reference point at it, at 65. Reference dependence would bear on them through the response to the credit: if the NRA anchors claiming, fewer individuals would delay beyond it when the credit rises. The Swiss finding that early-claiming penalties move behavior much less than the full retirement age itself points in this direction (Lalive et al. 2023). Because the mechanical effect of a DRC increase on retirement outlays is a cost, its infinite MVPF rests on these responses (Section 5.2), and strong anchoring would reduce its saving. Our model matches the concentration of exits in the 60s (Figure 2) but attributes it to incentives, health, and the fixed costs of work; to the extent that anchoring at 65 accounts for part of it, the model may overstate the response to the credit.

7.4 Cohort Rules and Current Law

Our status quo represents the rules faced by the cohorts born between 1920 and 1940 with an NRA of 65, a reduction for early claiming of $6\frac{2}{3}$ percent of the PIA per year, and a DRC of 4.5 percent, approximately the cohort average of credits that rose with the year of birth from 3 to 7 percent. AN2 and AD1 correspond to the end points of the two changes to claiming incentives in the 1983 Amendments, which raised the NRA toward 67 and the DRC to 8 percent, both phased in by birth cohort (Section 5.2; Svahn and Ross 1983). Combined, the two end points give the current-law schedule, which pays 70 percent of the PIA at 62 and 124 percent at 70. Our grid does not contain this combination, whose effects need not equal the sum of those of AN2 and AD1, because the response to a higher credit depends on the age from which it accrues.

The grid still speaks to the direction of further reforms. Because the CE is a log ratio and the Net entries of a table share their baseline aggregate, the difference between two rows evaluates the move from one reform to the other. Relative to the status quo, every credit from 8 to 10 percent in our grid pays for itself. Relative to an 8 percent credit at an NRA of 65 (AD1), a credit of 9, 9.5, or 10 percent (AD3–AD5) raises WTP by 0.07 to 0.14 percent and lowers net government spending by 0.21 to 0.33 percent of its status-quo value, whereas a credit of 8.5 percent (AD2) changes WTP and net spending by amounts

within the precision of the simulations. The reforms AD1–AD5 and AL2–AL9 are at least as steep, at every claiming age, as the current-law schedule, which rises by 5 percent of the PIA per year from 62 to 64, by $6\frac{2}{3}$ percent from 64 to 67, and by 8 percent after 67 (Appendix D). Within the AL family, welfare gains and fiscal savings both rise with the slope of the schedule (Section 5.2), so, for the cohorts we study, the grid points toward schedules steeper than the current-law one.

The grid does not say whether such reforms pay for themselves from current law, and four features of our setting favor the finding that rewards for delay pay for themselves. First, the baseline credit is 4.5 percent, at which delay past 65 was, on average, not actuarially advantageous (Section 5.2); under current law the credit is already 8 percent and accrues from 67, when fewer individuals still work, and who would respond to a further increase can be determined only by solving the model under current law. Second, the horizon ends at 85, which truncates the stream of benefits that a delay raises. Third, the government discounts at the net return r. Fourth, the individuals in our model receive no spousal or survivor benefits. By equation (18), survival beyond 85 and a lower discount rate each lower the actuarially fair credit, and survivor benefits would do the same by lengthening the stream of benefits that a primary earner's delay raises; each lowers the saving from a given delay. Later cohorts also live longer, by more at higher incomes (Chetty et al. 2016). Shoven and Slavov (2014) find that the value of delaying has risen with the credit, with lower real interest rates, and with longer life expectancy, and that at the low real interest rates of recent years delaying is actuarially advantageous for many individuals, most of all for primary earners in married couples. When the claimant and the government discount at the same rate, such a delay raises the program's outlays (Lemma 1). We leave a re-evaluation at current-law parameters and current survival to future work.

For the current system, Jones and Li (2023) reach the opposite conclusion about claiming adjustments. In a life-cycle model of claiming, labor supply, and saving in which mortality differs by education, they find that welfare is maximized when baseline benefits do not depend on lifetime earnings and the claiming adjustments are reduced, and that eliminating the earnings test and the income taxation of benefits adds further gains. Three differences between the two exercises bear on the claiming adjustments. First, they search for the welfare-maximizing system starting from current law, under which the credit is 8 percent; we measure the value per dollar of changes from a credit of 4.5 percent. Second, with mortality that differs by education, a larger credit pays more to long-lived high earners, whose delays save the program less (Lemma 1); in our model, mortality varies with lifetime earnings only through health. Third, in our model part of the effect of a higher credit runs through the earnings test, because benefits withheld after the NRA are

credited at the DRC (Section 4.6); eliminating the earnings test acts on the same margin directly. Our results therefore concern a different starting point and a different mortality gradient, and they do not settle whether rewards for delay would pay for themselves under current law.

7.5 Preferences, Health, and Disability Insurance

The reform results rest on the baseline preferences. The logarithmic specification of Appendix C.5, with fixed utility costs of participation and reentry, also implies a higher cost of work in poor health than in good health, 0.59 against 0.53 utils, and a large reentry cost. It fits old-age assets better but participation worse, and we do not compute reforms under it.

How much the results depend on preferences differs across families. The sign of the WTP of the DRC increases does not depend on preferences at all: at fixed prices and taxes, an AD reform raises the benefit of every claim after the NRA, including the credit on benefits withheld by the earnings test, and lowers no benefit, so it weakly enlarges every individual's set of feasible choices. Its infinite MVPF depends on preferences only through the responses that determine the sign of its net cost. The structural feature behind these responses is the fixed cost of participation, 791 hours a year in good health and 868 hours in poor health or with a disability (Table 2). In the same estimated economy, this nonconvexity is the main driver of nonparticipation at older ages, ahead of the effects of health on wages and on mortality (Ndiaye and Yu 2026). What matters for these responses is primarily the level of this cost, which makes retirement a discrete choice and determines how many workers are close to the margin between work and exit at 62 to 69; its health gradient is modest, about 10 percent. Health sorts workers at this margin through wages, the fixed cost, and access to DI, and, through survival, it determines whether the induced delays occur among claimants for whom the credit is below their actuarially fair rate (Lemma 1). For the other families, the curvature of utility also governs the WTP, through the value of insurance across health and ability states. Simulated households also hold less wealth at old ages than in the data, likely because the model has no bequest motive (Section 3), which raises the weight of Social Security in their old-age resources and is likely to raise the value of an additional dollar of benefits.

Disability insurance is an alternative route out of the labor force before the NRA (Yu 2024). Because DI is paid until the NRA without the reduction for early claiming, an NRA increase lengthens DI receipt and returns part of the saving on retirement outlays as DI outlays (Section 5.2). DI enters the model as an expected benefit, without an application process (Section 2.7), so the model captures this extension but not additional applications.

Duggan et al. (2007) find that the increase in the NRA raised DI enrollment (see also Li 2018); with an application margin, more of the saving from an NRA increase would reappear as DI outlays. The reward-delay reforms leave the NRA, and hence DI eligibility, unchanged.

7.6 Distribution

Our WTP is ex ante and utilitarian for one cohort: the change in average expected utility at age 30, before health, ability, and survival are realized (Section 4.2). It values the insurance a reform provides, but we do not report who gains and who loses by realized health or lifetime earnings. The MVPF framework accommodates such concerns through welfare weights on beneficiaries (Hendren 2016; Hendren and Sprung-Keyser 2020; Saez and Stantcheva 2016), and the rules of each reform indicate how weights would bear on it.

The DRC increases are robust to any weights. Because an AD reform weakly enlarges every individual's set of feasible choices at the prices and taxes of the status quo (Section 7.5), no individual is worse off in any state. Under any welfare weights that are positive for all individuals, its WTP is therefore positive and its MVPF infinite. The symmetric reforms lower the benefit of every claim before 65 and raise it after 65, so at unchanged behavior they transfer resources from early to late claimers; in the model, poor health makes early claiming more likely (Section 5.2). In Swedish data, workers who retire earlier consume less in retirement and experience larger drops in consumption at retirement, so that reforms that push workers to retire later carry a substantial consumption-smoothing cost (Kolsrud et al. 2024). Our ex ante WTP values this cost through the marginal utility of consumption in the states in which individuals claim early. The positive WTP of the symmetric reforms is an average over these and other states, and with sufficiently large weights on individuals in poor health it could turn negative. The NRA increases lower the benefit at every claiming age, including for individuals in poor health who cannot offset the cut by working longer; weights that favor them would tend to raise the welfare cost of each dollar saved and to reinforce the advantage of the DRC increases.

Among the other families, weights that favor low lifetime earners would raise the MVPFs of the bottom-bracket expansions relative to the level reforms and lower those of the old-age income-tax cuts, whose gains rise with taxable income in old age, which excludes Social Security benefits (Section 2.8). Mortality in the model varies with lifetime earnings only through health, whereas in the data life expectancy rises with income (Chetty et al. 2016); with such a gradient, a larger credit would also pay more to long-lived high earners (Section 7.4). Section 6 discusses the distribution across cohorts.

8 Conclusion

Retirement policies change behavior over the entire working life, their effects depend on health, and their fiscal consequences spread across the tax and transfer system. This paper measures the marginal value of public funds of retirement reforms in an estimated life-cycle model with these features. For each reform, it computes the ex ante willingness to pay of workers who do not yet know their health and wage histories and the effect of the reform on Social Security outlays and on the net cost of income and payroll taxes, disability insurance, means-tested transfers, and, for the Medicare reforms, Medicare's payments to the targeted beneficiaries.

For the cohorts we study, one distinction organizes the ranking of reforms: whether a reform rewards later claiming or lowers benefits at every claiming age. Raising the delayed retirement credit, or steepening the adjustment of benefits around the normal retirement age, pays for itself. These reforms raise welfare and lower Social Security outlays; for the increases in the credit, outlays fall even though they would rise at unchanged behavior, so the behavioral responses within Social Security, through claiming ages, the earnings test, and the AIME, more than finance the higher credit. Raising the normal retirement age saves money by lowering benefits for every claimant and reduces welfare. Among expansions of benefits, an increase in the bottom replacement rate of the benefit formula delivers the most welfare per dollar; the smallest such increase is approximately budget neutral. Expansions of Medicare coverage after age 65 deliver the least, and cuts in old-age income taxes less than any expansion of the benefit formula.

These results suggest an order of operations for closing a financing gap for the cohorts and rules we study. Reforms that reward delay come first, because they save money and raise welfare together. To a first-order approximation, an increase in the normal retirement age costs less welfare per dollar saved than higher income taxes after age 62 or proportional cuts in all replacement rates, but in our grid it is dominated by rewards for delay whenever the required saving is moderate. The two changes to claiming incentives in the 1983 Amendments illustrate the distinction: evaluated for the 1920–1940 cohorts, the end point of the increase in the delayed retirement credit pays for itself, while the end point of the increase in the normal retirement age saves money at a welfare cost, with an MVPF of 0.16 on the post-62 consumption base.

Four extensions would show how far these conclusions extend beyond the cohorts we study. The first is to evaluate rewards for delay at current-law parameters, with a delayed retirement credit of 8 percent and a normal retirement age of 67, and with survival beyond age 85. The second is to weight welfare gains by health and lifetime earnings rather than

averaging them ex ante. The third is to allow wages and the government budget to respond in general equilibrium. The fourth is to let statutory retirement ages act as reference points. The logic behind our results is more general than the model: when the cost of continued work differs across workers, a reward for later claiming pays only the workers who choose to delay, so it can lower outlays when delays are cheap for the program, whereas an increase in the normal retirement age lowers the benefits of all claimants, including those for whom working longer is most costly.

References

Bairoliya, Neha and Kathleen McKiernan, "Revisiting Retirement and Social Security Claiming Decisions," 2023. Working paper.

Behaghel, Luc and David M. Blau, "Framing Social Security Reform: Behavioral Responses to Changes in the Full Retirement Age," *American Economic Journal: Economic Policy*, 2012, *4* (4), 41–67.

Bénabou, Roland, "Tax and education policy in a heterogeneous-agent economy: What levels of redistribution maximize growth and efficiency?," *Econometrica*, March 2002, *70* (2), 481–517.

Blau, David M. and Donna B. Gilleskie, "The Role of Retiree Health Insurance in the Employment Behavior of Older Men," *International Economic Review*, May 2008, *49* (2), 475–514.

Blundell, Richard, Eric French, and Gemma Tetlow, "Retirement incentives and labor supply," in "Handbook of the economics of population aging," Vol. 1, Elsevier, 2016, pp. 457–566.

Board of Trustees, OASDI, "The 2024 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds," Technical Report, Social Security Administration, Washington, DC 2024.

Borella, Margherita, Mariacristina De Nardi, and Fang Yang, "The aggregate implications of gender and marriage," *The Journal of the Economics of Ageing*, 2018, *11*, 6–26.

_ , _ , and _ , "Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labor Supply?," *The Review of Economic Studies*, 2023, *90* (1), 102–131.

Braun, R. Anton, Karen A. Kopecky, and Tatyana Koreshkova, "Old, Sick, Alone, and Poor: A Welfare Analysis of Old-Age Social Insurance Programmes," *Review of Economic Studies*, 2017, *84* (2), 580–612.

Cagetti, Marco, "Wealth accumulation over the life cycle and precautionary savings," *Journal of Business and Economic Statistics*, 2003, *21* (3), 339–353.

Chetty, Raj, "Sufficient Statistics for Welfare Analysis: A Bridge Between Structural and Reduced-Form Methods," *Annual Review of Economics*, 2009, *1*, 451–488.

_ , Michael Stepner, Sarah Abraham, Shelby Lin, Benjamin Scuderi, Nicholas Turner, Augustin Bergeron, and David Cutler, "The association between income and life expectancy in the United States, 2001-2014," *Jama*, 2016, *315* (16), 1750–1766.

Coile, Courtney and Jonathan Gruber, "Future Social Security Entitlements and the Retirement Decision," *Review of Economics and Statistics*, 2007, *89* (2), 234–246.

Coile, Courtney C., "Working Longer in the United States: Trends and Explanations," in Courtney C. Coile, Kevin Milligan, and David A. Wise, eds., *Social Security Programs and Retirement around the World: Working Longer*, University of Chicago Press, 2019.

Coile, Courtney, Peter Diamond, Jonathan Gruber, and Alain Jousten, "Delays in Claiming Social Security Benefits," *Journal of Public Economics*, 2002, *84* (3), 357–385. Earlier version: NBER Working Paper 7318 (1999).

Conesa, Juan Carlos and Dirk Krueger, "Social Security Reform with Heterogeneous Agents," *Review of Economic Dynamics*, 1999, *2* (4), 757–795.

Congressional Budget Office, "CBO's 2024 Long-Term Projections for Social Security," Report 60392, Congressional Budget Office August 2024.

Cremer, Helmuth, Jean-Marie Lozachmeur, and Pierre Pestieau, "Social security, retirement age and optimal income taxation," *Journal of Public Economics*, 2004, *88* (11), 2259–2281.

De Nardi, Mariacristina, Eric French, and John Bailey Jones, "Why do the Elderly Save? The role of Medical Expenses," *Journal of Political Economy*, February 2010, *118* (1), 39–75.

_ , Svetlana Pashchenko, and Ponpoje Porapakkarm, "The Lifetime Costs of Bad Health," *Review of Economic Studies*, 2024.

Deshpande, Manasi, Itzik Fadlon, and Colin Gray, "How Sticky Is Retirement Behavior in the United States?," *Review of Economics and Statistics*, 2024, *106* (2), 370–383. Earlier version: NBER Working Paper 27190 (2020), "How Sticky is Retirement Behavior in the U.S.? Responses to Changes in the Full Retirement Age".

Diamond, Peter and Jonathan Gruber, "Social security and retirement in the United States," in "Social security and retirement around the world," University of Chicago Press, 1999, pp. 437–473.

Diamond, Peter Arthur and James A Mirrlees, "A model of social insurance with variable retirement," *Journal of Public Economics*, 1978, *10* (3), 295–336.

Duggan, Mark, Perry Singleton, and Jae Song, "Aching to Retire? The Rise in the Full Retirement Age and Its Impact on the Social Security Disability Rolls," *Journal of Public Economics*, 2007, *91* (7–8), 1327–1350.

Fan, Xiaodong, Ananth Seshadri, and Christopher Taber, "Estimation of a life-cycle model with human capital, labor supply, and retirement," *Journal of Political Economy*, 2024, *132* (1), 48–95.

Farhi, Emmanuel and Iván Werning, "Insurance and taxation over the life cycle," *The Review of Economic Studies*, 2013, *80* (2), 596–635.

Finkelstein, Amy and Nathaniel Hendren, "Welfare Analysis Meets Causal Inference," *Journal of Economic Perspectives*, 2020, *34* (4), 146–167.

_ **and Robin McKnight**, "What did Medicare do? The initial impact of Medicare on mortality and out of pocket medical spending," *Journal of public economics*, 2008, *92* (7), 1644–1668.

_ **, Nathaniel Hendren, and Mark Shepard**, "Subsidizing health insurance for low-income adults: Evidence from Massachusetts," *American Economic Review*, 2019, *109* (4), 1530–1567.

French, Eric, "The effects of health, wealth, and wages on labour supply and retirement behaviour," *The Review of Economic Studies*, 2005, *72* (2), 395–427.

_ **and Jae Song**, "The Effect of Disability Insurance Receipt on Labor Supply," *American Economic Journal: Economic Policy*, 2014, *6* (2).

_ **and John Bailey Jones**, "The Effects of Health Insurance and Self-Insurance on Retirement Behavior," Working Paper CRR 2004-12, Center for Retirement Research at Boston College 2004.

_ **and** _ , "The effects of health insurance and self-insurance on retirement behavior," *Econometrica*, 2011, *79* (3), 693–732.

Gelber, Alexander, Damon Jones, Daniel W Sacks, and Jae Song, "The employment effects of the social security earnings test," *Journal of Human Resources*, 2020.

Gelber, Alexander M., Damon Jones, and Daniel W. Sacks, "Estimating Adjustment Frictions Using Nonlinear Budget Sets: Method and Evidence from the Earnings Test," *American Economic Journal: Applied Economics*, 2020, *12* (1), 1–31.

Golosov, Mikhail, Ali Shourideh, Maxim Troshkin, and Aleh Tsyvinski, "Optimal Pension Systems with Simple Instruments," *American Economic Review*, 2013, *103* (3), 502–507. Papers and Proceedings.

_ **and Aleh Tsyvinski**, "Designing optimal disability insurance: A case for asset testing," *Journal of Political Economy*, 2006, *114* (2), 257–279.

_ **, Maxim Troshkin, and Aleh Tsyvinski**, "Redistribution and social insurance," *The American Economic Review*, 2016, *106* (2), 359–386.

Gourinchas, Pierre-Olivier and Jonathan A. Parker, "Consumption Over the Life Cycle," *Econometrica*, January 2002, *70* (1), 47–89.

Gruber, Jonathan and David A. Wise, eds, *Social Security Programs and Retirement Around the World: Micro-Estimation* International Social Security, Chicago: University of Chicago Press, 2004.

_ **and** _ **, eds**, *Social Security Programs and Retirement around the World: Fiscal Implications of Reform*, Chicago: University of Chicago Press, 2007.

_ **and David Wise**, "Social security and retirement: An international comparison," *The American Economic Review*, 1998, *88* (2), 158–163.

Guner, Nezih, Remzi Kaygusuz, and Gustavo Ventura, "Taxation and Household Labour Supply," *The Review of Economic Studies*, July 2012, *79* (3), 1113–1149.

Gustman, Alan L. and Thomas L. Steinmeier, "A Structural Retirement Model," *Econometrica*, 1986, *54* (3), 555–584.

_ **and** _ , "Effects of pensions on savings: analysis with data from the health and retirement study," *Carnegie-Rochester Conference Series on Public Policy*, June 1999, *50*, 271–324.

_ **, Olivia S. Mitchell, Andrew A. Samwick, and Thomas L. Steinmeier**, "Evaluating Pension Entitlements," in Olivia S. Mitchell, P. Brett Hammond, and Anna M. Rappaport, eds., *Forecasting Retirement Needs and Retirement Wealth*, University of Pennsylvania Press, 2000.

Haan, Peter and Victoria Prowse, "Longevity, life-cycle behavior and pension reform," *Journal of Econometrics*, 2014, *178*, 582–601.

Haller, Andreas, "Welfare Effects of Pension Reforms," 2022. Working Paper, Norwegian School of Economics.

Heathcote, Jonathan, Kjetil Storesletten, and Giovanni L. Violante, "Optimal Tax Progressivity: An Analytical Framework," *The Quarterly Journal of Economics*, 2017, *132* (4), 1693–1754.

Heckman, James J., "The Common Structure of Statistical Models of Truncation, Sample Selection and Limited Dependent Variables and a Simple Estimator for Such Models," *Annals of Economic and Social Measurement*, 1976, *5* (4), 475–492.

Hendren, Nathaniel, "The policy elasticity," *Tax Policy and the Economy*, 2016, *30* (1), 51–89.

_ **and Ben Sprung-Keyser**, "A unified welfare analysis of government policies," *The Quarterly journal of economics*, 2020, *135* (3), 1209–1318.

Hosseini, Roozbeh, "Adverse selection in the annuity market and the role for social security," *Journal of Political Economy*, 2015, *123* (4), 941–984.

_ **and Ali Shourideh**, "Retirement financing: An optimal reform approach," *Econometrica*, 2019, *87* (4), 1205–1265.

_ **, Karen A. Kopecky, and Kai Zhao**, "The Evolution of Health over the Life Cycle," *Review of Economic Dynamics*, 2022.

Hubbard, R. Glenn, Jonathan Skinner, and Stephen P. Zeldes, "Precautionary Saving and Social Insurance," *Journal of Political Economy*, April 1995, *103* (2), 360–399.

Huggett, Mark and Juan Carlos Parra, "How Well Does the US Social Insurance System Provide Social Insurance?," *Journal of Political Economy*, 2010, *118* (1), 76–112.

Imrohoroğlu, Selahattin and Sagiri Kitao, "Social Security Reforms: Benefit Claiming, Labor Force Participation, and Long-Run Sustainability," *American Economic Journal: Macroeconomics*, July 2012, *4* (3), 96–127.

Ippolito, Richard A., *Pension Plans and Employee Performance*, University of Chicago Press, 1997.

Jones, John Bailey and Yue Li, "Social Security Reform with Heterogeneous Mortality," *Review of Economic Dynamics*, April 2023, *48*, 320–344.

Judd, Kenneth L., *Numerical Methods in Economics*, 1 ed., Vol. 1, The MIT Press, September 1998.

Kitao, Sagiri, "Sustainable Social Security: Four Options," *Review of Economic Dynamics*, 2014, *17* (4), 756–779.

Kleven, Henrik Jacobsen, "Sufficient Statistics Revisited," *Annual Review of Economics*, 2021, *13*, 515–538.

_ **and Claus Thustrup Kreiner**, "The marginal cost of public funds: Hours of work versus labor force participation," *Journal of Public Economics*, 2006, *90* (10-11), 1955–1973.

Kolsrud, Jonas, Camille Landais, Daniel Reck, and Johannes Spinnewijn, "Retirement Consumption and Pension Design," *American Economic Review*, 2024, *114* (1), 89–133.

Lalive, Rafael, Arvind Magesan, and Stefan Staubli, "How Social Security Reform Affects Retirement and Pension Claiming," *American Economic Journal: Economic Policy*, 2023, *15* (3), 115–150.

Laun, Tobias and Johanna Wallenius, "A Life Cycle Model of Health and Retirement: The Case of Swedish Pension Reform," *Journal of Public Economics*, 2015, *127*, 127–136.

Li, Yue, "Paradoxical effects of increasing the normal retirement age: A prospective evaluation," *European Economic Review*, January 2018, *101*, 512–527.

Liebman, Jeffrey B and Erzo FP Luttmer, "Would people behave differently if they better understood social security? Evidence from a field experiment," *American Economic Journal: Economic Policy*, 2015, *7* (1), 275–299.

Liebman, Jeffrey B., Erzo F. P. Luttmer, and David G. Seif, "Labor Supply Responses to Marginal Social Security Benefits: Evidence from Discontinuities," *Journal of Public Economics*, 2009, *93* (11–12), 1208–1223.

Lockwood, Lee M, "Incidental bequests and the choice to self-insure late-life risks," *American Economic Review*, 2018, *108* (9), 2513–2550.

Low, Hamish and Luigi Pistaferri, "Disability insurance and the dynamics of the incentive insurance trade-off," *American Economic Review*, 2015, *105* (10), 2986–3029.

Maestas, Nicole, Kathleen J. Mullen, and Alexander Strand, "Does Disability Insurance Receipt Discourage Work? Using Examiner Assignment to Estimate Causal Effects of SSDI Receipt," *American Economic Review*, 2013, *103* (5).

Manoli, Day and Andrea Weber, "Nonparametric Evidence on the Effects of Financial Incentives on Retirement Decisions," *American Economic Journal: Economic Policy*, 2016, *8* (4), 160–182.

Mastrobuoni, Giovanni, "Labor Supply Effects of the Recent Social Security Benefit Cuts: Empirical Estimates Using Cohort Discontinuities," *Journal of Public Economics*, 2009, *93* (11–12), 1224–1233.

Milgrom, Paul and Ilya Segal, "Envelope theorems for arbitrary choice sets," *Econometrica*, 2002, *70* (2), 583–601.

Ndiaye, Abdoulaye and Zhixiu Yu, "Flexible Retirement and Optimal Taxation," Working Paper, New York University and Louisiana State University 2026.

Nishiyama, Shinichi and Kent Smetters, "Does social security privatization produce efficiency gains?," *The Quarterly Journal of Economics*, 2007, *122* (4), 1677–1719.

Pashchenko, Svetlana and Ponpoje Porapakkarm, "Reducing Medical Spending of the Publicly Insured: The Case for a Cash-out Option," *American Economic Journal: Economic Policy*, August 2019, *11* (3), 390–426.

Prescott, Edward C, Richard Rogerson, and Johanna Wallenius, "Lifetime aggregate labor supply with endogenous workweek length," *Review of Economic Dynamics*, 2009, *12* (1), 23–36.

Rogerson, Richard and Johanna Wallenius, "Nonconvexities, retirement, and the elasticity of labor supply," *The American Economic Review*, 2013, *103* (4), 1445–1462.

Rust, John and Christopher Phelan, "How Social Security and Medicare Affect Retirement Behavior in a World of Incomplete Markets," *Econometrica*, 1997, *65* (4), 781–831.

Rust, John P, "A dynamic programming model of retirement behavior," in "The economics of aging," University of Chicago Press, 1989, pp. 359–404.

Saez, Emmanuel and Stefanie Stantcheva, "Generalized social marginal welfare weights for optimal tax theory," *The American Economic Review*, 2016, *106* (1), 24–45.

Seibold, Arthur, "Reference Points for Retirement Behavior: Evidence from German Pension Discontinuities," *American Economic Review*, 2021, *111* (4), 1126–1165.

Shoven, John B. and Sita Nataraj Slavov, "Does It Pay to Delay Social Security?," *Journal of Pension Economics and Finance*, 2014, *13* (2), 121–144.

Shoven, John B, Sita Nataraj Slavov, and David A Wise, "Social Security Claiming Decisions: Survey Evidence," Working Paper 23729, National Bureau of Economic Research August 2017.

Stantcheva, Stefanie, "Optimal Taxation and Human Capital Policies over the Life Cycle," *Journal of Political Economy*, 2017, *125* (6), 1931–1990.

Staubli, Stefan and Josef Zweimüller, "Does Raising the Early Retirement Age Increase Employment of Older Workers?," *Journal of Public Economics*, 2013, *108*, 17–32.

Svahn, John A. and Mary Ross, "Social Security Amendments of 1983: Legislative History and Summary of Provisions," *Social Security Bulletin*, July 1983, *46* (7), 3–48.

van der Klaauw, Wilbert and Kenneth I. Wolpin, "Social Security and the Retirement and Savings Behavior of Low-Income Households," *Journal of Econometrics*, 2008, *145* (1–2), 21–42.

Yu, Zhixiu, "Why are older men working more? The role of social security," *Journal of Public Economics*, 2024, *231.*

Zeldes, Stephen P., "Consumption and Liquidity Constraints: An Empirical Investigation," *Journal of Political Economy*, April 1989, *97* (2), 305–346.

Appendix

A Model Details and Numerical Solution

This appendix documents the institutional parameters of the model of Section 2 and the numerical method used to solve and simulate it.

A.1 Social Security

Program parameters. According to the Social Security Administration (SSA), individuals born between 1920 and 1940 faced an early retirement age (ERA) of 62 and, on average, a normal retirement age (NRA) of 65. Their delayed retirement credit (DRC) rose with the year of birth from 3 percent per year for those born before 1925 to 7 percent for those born in 1939–1940, as the 1983 Amendments phased in a credit of 8 percent; we use 4.5 percent, approximately the cohort average. The PIA formula had bend points at \$3,720 and \$22,392, and the AIME was capped at \$43,800. In the model, the retirement earnings test imposes different rates and thresholds by age: beneficiaries younger than the NRA of 65 lose \$1 of benefits for every \$2 of earnings above \$6,000, and those from the NRA to age 69 lose \$1 for every \$3 of earnings above \$8,186. These parameters are held fixed over the life cycle, which simplifies the rules the cohort faced: the reduction of \$1 for every \$3 above the NRA applied from 1990, and in 2000 the test was eliminated for beneficiaries who had reached the NRA, which removed it for the later cohorts in our sample. Table A.1 summarizes these parameters together with the threshold for disability insurance and the payroll-tax schedule. All monetary values are in 1987 dollars and, like $aime_t$ in (5), at annual rates.

Earnings record. Following French (2005), individuals enter the labor market at age 25, so that by age 30, the first age in the model, they have accumulated five years of earnings. We impute the initial value of $aime_t$ by assuming that each individual worked 2,000 hours per year in these five years at her hourly wage at age 30. Under the SSA rules, earnings up to two years before the age of eligibility, that is, up to age 60, are indexed to average wage growth to account for the rise in living standards over the working life. In the model, $aime_t$ is adjusted for inflation by indexing wages to the personal consumption expenditure price index. Under the law of motion (5), each year of earnings from age 25 to 59, a span of 35 years, adds one thirty-fifth of annual earnings to $aime_t$. From age 60, earnings raise $aime_t$ only if they exceed its current value, which mimics the replacement of a low year among the highest 35.

Claiming schedule. Under the status-quo rules, the benefit as a percentage of the PIA is 80, 86.67, 93.33, 100, 104.5, 109, 113.5, 118, and 122.5 for claiming ages 62 through 70.

Table A.1. Social Security Parameters

Parameter	Description	Value
A. Claiming ages and actuarial adjustment		
ERA	Early retirement age	62
NRA	Normal retirement age	65
DRC	Delayed retirement credit, per year after the NRA	4.5%
B. Benefit formula		
$aime_0$	First bend point of the PIA formula	\$3,720
$aime_1$	Second bend point of the PIA formula	\$22,392
$aime_{max}$	Maximum AIME	\$43,800
C. Retirement earnings test		
τ_{ret}	Benefit reduction rate, below the NRA	50%
τ_{ret}	Benefit reduction rate, from the NRA to age 69	33%
y_{ret}	Exempt amount, below the NRA	\$6,000
y_{ret}	Exempt amount, from the NRA to age 69	\$8,186
D. Disability insurance		
y_{db}	Earnings threshold for DI eligibility	\$3,600
E. Payroll tax		
τ_t^p	Employee payroll tax rate (OASDI and HI)	3%–7.65%
$\bar{y}_t^{ss}$	Payroll tax threshold	\$14,459–\$64,974

Notes: Parameters of the Social Security, disability insurance, and payroll-tax rules faced by the cohort born between 1920 and 1940, as used in the model of Section 2. The NRA is the cohort average, and the DRC approximates the cohort average of the credits, which rose from 3 to 7 percent with the year of birth. The bend points and the maximum of the average indexed monthly earnings (AIME) enter the primary insurance amount (PIA) formula (6) and the law of motion (5). The benefit reduction rate is the dollar reduction in benefits per dollar of labor earnings above the exempt amount (7). The DI threshold is the annual equivalent of the monthly substantial gainful activity amount for nonblind individuals in effect in 1987, \$300 (8). Employee payroll-tax rates and taxable maxima vary over the cohort's working life; the table reports their ranges. Abbreviations: ERA = early retirement age; NRA = normal retirement age; DRC = delayed retirement credit; RET = retirement earnings test; DI = disability insurance; OASDI = Old-Age, Survivors, and Disability Insurance; HI = Hospital Insurance (Medicare). Monetary values in 1987 dollars. *Source:* Social Security Administration.

The schedule rises by 6.67 percentage points per year of delay up to the NRA and by the DRC of 4.5 percentage points per year after it. Figure 4 and Table D.1 report the schedules under the claiming-age reforms. Claiming is irreversible and must occur by age 70.

Earnings test and crediting. Beneficiaries younger than 70 face the earnings test (7). Following French and Jones (2011), benefits withheld by the test are credited through higher future benefits at the rates of the claiming schedule: if a year's worth of benefits is withheld before the NRA, future benefits rise by one year's early-claiming increment of the schedule (6.67 percent of the PIA under the status quo); if a year's worth of benefits is withheld after the NRA, future benefits rise by the DRC. From age 70, benefits are not

subject to the test.

A.2 Disability Insurance

Individuals with a disability are eligible for DI benefits before the NRA if their labor earnings fall below a threshold. We do not model the DI application decision. In models that treat applications explicitly, such as Low and Pistaferri (2015), the probability that an application succeeds depends on age and health. To keep the model tractable, we grant DI benefits exclusively to individuals who have a disability ($h_t = 2$), are older than 50, and earn less than $y_{db} = \$3{,}600$ per year, the annual equivalent of the monthly substantial gainful activity amount of \$300 for nonblind individuals with a disability in effect in 1987. The benefit, given by (8), equals the individual's PIA, without the reduction for early claiming, multiplied by π^{db}, the probability of receiving DI benefits for older individuals with severe work limitations estimated by Low and Pistaferri (2015). Because not all individuals with a disability are awarded benefits, scaling by π^{db} captures the average DI benefit received by individuals with a disability. DI benefits end at the NRA.

A.3 Pensions

Pension benefits capture the main features of defined-benefit (DB) plans. Like Social Security benefits, they depend on the individual's work history and are illiquid before a specified age, which we set to 62. Unlike Social Security, DB plans often create strong incentives to leave the labor force at particular ages through the structure of their accrual rates (Gustman et al. 2000; Ippolito 1997). To capture these features, we impute pension income as a function of Social Security benefits, following French (2005):

$$pb_t = pb(PIA_t) + \xi_t \qquad \text{for } t \geq 62,$$

and $pb_t = 0$ before age 62. The function $pb(\cdot)$ maps lifetime earnings, summarized by the PIA, into pension income, with parameters based on the estimates of Gustman and Steinmeier (1999). The term ξ_t is an age-specific adjustment, based on the pension accrual rates of Gustman et al. (2000), that accounts for the difference between the accrual of pension wealth and the accrual of Social Security wealth over the life cycle; this difference tends to be positive at older ages. French (2005) describes the procedure in detail. Pension income is part of taxable income y_t in (10).

A.4 Taxes and the Return to Saving

Income tax. Individuals face the age-dependent tax function (9) on total income, with parameters (λ_t, τ_t) estimated from PSID data as in Borella et al. (2023). For tractability,

individuals anticipate the changes in these parameters with age. Total income includes interest income, so that interest is taxed as it accrues.

Payroll tax. Payroll-tax rates and taxable maxima are based on the historical data of the SSA for the Old-Age, Survivors, and Disability Insurance (OASDI) and Medicare Hospital Insurance (HI) programs. Since 1960, the employee tax rate for OASDI and, from 1966, HI combined has ranged from 3 to 7.65 percent; the employer pays the same rate. The taxable maximum ranges from \$14,459 to \$64,974 in 1987 dollars (Table A.1). In the model, the HI tax applies up to the same maximum as the OASDI tax, although from 1991 its maximum was higher and from 1994 it applied to all earnings.

Taxation of benefits. In the model, Social Security and DI benefits are not part of taxable income. Under the 1983 Amendments, part of the benefits of beneficiaries whose income exceeds statutory thresholds has been subject to the income tax since 1984, so the model overstates after-tax benefits for higher-income beneficiaries.

Return to saving. The pretax interest rate is $r_f = 4$ percent (French 2005). We set the wedge between the gross and the net return to $\tau_k = 20$ percent, as in Stantcheva (2017). Even though interest income is taxed as part of total income by (9), this wedge captures corporate and other direct capital taxes that the model does not represent. The analysis is therefore equivalent to one with a baseline net interest rate of $r = 3.2$ percent. The savings-subsidy reforms of Section 5.6 raise the net return toward r_f.

A.5 Solution and Simulation

Parameters. An individual's decisions at age t depend on the state X_t, on the preference parameters $\mathcal{P} = (\gamma, \nu, \phi_n(0), \phi_n(1), \phi_{RE}, L, \beta)$, with $\phi_n(2) = \phi_n(1)$, which are estimated in the second step (Section 3), and on the parameters that govern the data-generating process for the state variables and the policy environment,

$$\chi = \Big(r,\, \sigma^2,\, \rho,\, \{W(h,t)\}_t,\, \{\pi_{j,i,t+1}\}_t,\, \{s_{t+1}\}_t,\, \{m_t\}_t,\, \{pb_t\}_t,\, \{ss_t\}_t,\, \{db_t\}_t,\, \Upsilon_t(\cdot)\Big),$$

which are estimated or calibrated in the first step. Out-of-pocket medical expenses and pension income depend on the other state variables and are not separate state variables.

Backward induction. The life cycle is divided into three stages. From 30 to 61, individuals choose consumption and hours; from 62 to 69, they also choose whether to claim; from 70 to 85, all have claimed and choose consumption and hours. We solve the problem (12) by backward induction. At the terminal age $T = 85$, $V_{T+1} \equiv 0$ and $a_{T+1} = 0$, so the individual

consumes her remaining resources and chooses hours. Given the value function at T, we solve for the decision rules and the value function at $T-1$, and we proceed backward to age 30. The model features savings for retirement, precautionary savings against wage, health, and medical-expense risk, and accidental bequests, but no bequest motive. The solution consists of the consumption rules $c_t(X_t;\mathcal{P},\chi)$, the hours rules $n_t(X_t;\mathcal{P},\chi)$, and the claiming rules $b_t(X_t;\mathcal{P},\chi)$; participation P_t follows from the hours rule, and next-period assets a_{t+1} follow from the budget constraint (10).

Grids, quadrature, and interpolation. We discretize the state space with 30 asset points, 10 wage points, 10 AIME points, 2 claiming states ($b_{t-1} \in \{0,1\}$), 3 health states, and 2 states for previous participation ($P_{t-1} \in \{0,1\}$). We compute the value function directly at these grid points and integrate over the wage innovation ε_{t+1} with five-node Gauss–Hermite quadrature (Judd 1998). To evaluate the value function at points off the grid, we use linear interpolation within the grid and linear extrapolation outside it.

Simulation. We simulate life-cycle histories forward with the decision rules. Initial states at age 30 are drawn from the empirical joint distribution of wages, health, assets, and labor force participation of individuals aged 28 to 32 in the target cohort in the PSID, with the mean of log wages adjusted to match the estimated wage profile within each health group (Appendix B and Table B.6). The initial AIME is imputed as described in Appendix A.1, and no individual has claimed at age 30. Given the decisions at age 30, the state at age 31 is generated from the health transition probabilities, the survival probabilities, and draws of the wage innovation, and the process is repeated through age T, which yields complete simulated histories. The simulated histories deliver the life-cycle profiles used in estimation (Appendix C), the value at age 30 used to measure welfare (Section 4.2), and the per capita flows that enter the government accounts of Section 2.11. For each reform, we solve the model again under the reform's rules and simulate the cohort from the same initial distribution (Appendix D).

B Data and First-Step Estimation

This appendix describes the two samples, the measurement of health and disability, and the first-step estimation of the health, survival, medical-expense, and wage processes and of the initial conditions summarized in Section 3.

B.1 Samples

Panel Study of Income Dynamics. The PSID is a longitudinal survey of a representative sample of the US population that collects detailed information on labor market outcomes,

income, health, and wealth. It began in 1968 with interviews of about 4,800 families and has followed the original respondents and their descendants since, with interviews conducted annually until 1997 and biennially thereafter. The original sample has two components: 1,872 low-income families drawn from the Survey of Economic Opportunity (SEO) and 2,930 families selected by the Survey Research Center (SRC) at the University of Michigan to represent the US population.

We keep individuals from the SRC component interviewed between 1968 and 2015. We exclude the SEO component, most of whose families were dropped from the survey after 1997, which makes the sample more representative of the US population. We keep household heads and spouses, aged 20 to 90, with nonmissing labor and health information, and restrict the sample to individuals born between 1920 and 1940, a group of cohorts that faced similar Social Security rules and that the PSID observes over most of the life cycle. Table B.1 reports each selection step. The final sample has 2,248 individuals and 44,404 person–year observations. We use it to estimate health transitions, wage profiles and dynamics, survival probabilities, the initial distribution of the state variables, and the target profiles of the second step.

Medical Expenditure Panel Survey. The MEPS is a nationally representative survey of families, individuals, their medical providers, and employers across the United States, with detailed information on medical expenditures, sources of payment, insurance coverage, health status, and demographics. It covers individuals of all ages, with age top-coded at 85. Medical spending is reported at the individual level and cross-validated with information from providers and insurers. We use the 1999–2012 waves, restrict the sample to household heads aged 20 and older, and exclude observations with missing age, medical spending, insurance coverage during working years, or health status. Table B.2 reports each step; the final sample has 120,731 individuals and 211,709 person–year observations. We use the MEPS only to estimate out-of-pocket medical expenses.

B.2 Health and Disability Measures

PSID. Health and disability status in the PSID are constructed from three self-reported work-limitation questions asked in each wave:

(1) *Do you have any physical or nervous condition that limits the type of work or the amount of work that you can do?*

(2) *Does this condition keep you from doing some types of work?*

(3) *For work you can do, how much does it limit the amount of work you can do—a lot, somewhat, or just a little?*

Table B.1. PSID Sample Selection

Selection step	Individuals	Observations
Initial sample	76,880	3,075,200
Non-SEO sample	36,430	1,457,200
Household heads and spouse if present	21,225	355,314
Age 20 to 90	21,031	350,617
No missing labor or health information	19,433	222,319
Born between 1920 and 1940	2,248	44,404

Notes: Number of individuals and person–year observations remaining after each selection step, applied sequentially from top to bottom. Non-SEO sample: excludes the Survey of Economic Opportunity low-income subsample and keeps the nationally representative Survey Research Center component. *Data source:* Panel Study of Income Dynamics, 1968–2015 waves.

Table B.2. MEPS Sample Selection

Selection step	Individuals	Observations
Initial sample	259,263	491,795
Household heads	130,978	231,866
Age above 20	125,587	222,495
With insurance information at 25–65	121,385	213,005
Nonmissing health status	120,731	211,709

Notes: Number of individuals and person–year observations remaining after each selection step, applied sequentially from top to bottom. Insurance information is required at ages 25–65. *Data source:* Medical Expenditure Panel Survey, 1999–2012 waves.

The first question admits *Yes* or *No*. The second question, available from 1986, admits *Yes*, *No*, or *Can do nothing*. The response options to the third question changed over time: since 1976 they have been *Not at all*, *Somewhat*, *Just a little*, or *A lot*; before 1976 they were *We can't work*, *It limits me a lot*, *Some, not much*, or *Limitation but not on work*. For observations before 1986, health status is based on the first and third questions only.

Following Low and Pistaferri (2015), we classify individuals into three health states. An individual is in good health ($h_t = 0$) if she answers *No* to the first question or *Not at all* to the third; in poor health ($h_t = 1$) if she answers *Yes* to the first question and *Somewhat* or *Just a little* to the third; and has a disability ($h_t = 2$) if she answers *Yes* to the first question, *Can do nothing* to the second, and *A lot* to the third, in accordance with the Social Security Administration criterion for disability insurance qualification. Two alternative measures are not suitable for our purposes. The frailty index of Hosseini et al. (2022) relies on PSID questions available only since 2003, and self-rated health has been collected only since 1984. Neither provides consistent information on health over the entire life cycle of the 1920–1940 cohorts.

MEPS. In the MEPS, health status is based on self-rated health, following Pashchenko and Porapakkarm (2019). Each year includes three interview rounds, and in each round respondents rate their health as (1) excellent, (2) very good, (3) good, (4) fair, or (5) poor. An individual is in good health ($h_t = 0$) if her rating falls in categories 1 to 3 in all three rounds, has a disability ($h_t = 2$) if she reports poor health in any round, and is in poor health ($h_t = 1$) otherwise.

Comparison. Table B.3 compares the distribution of health by age group in the two surveys. The shares of people with a disability are similar at ages 50 to 70 and higher in the PSID after 70, and both surveys show the share in good health declining with age. The MEPS classifies a larger share of individuals in poor health at every age, so out-of-pocket medical expenses by health state are estimated from a population whose poor-health group is broader than in the PSID.

Table B.3. Health Status by Age in the MEPS and the PSID

	Ages 50–60		Ages 60–70		Ages 70 and older	
Health status (%)	MEPS	PSID	MEPS	PSID	MEPS	PSID
With a disability	10	9.3	11.7	12.9	15	19.7
Poor health	21	10.6	23.6	17.0	28.7	19.3
Good health	69	80.1	64.7	70.1	56.3	60.0

Notes: Percentage of individuals in each health state within each age group and survey. Health states are defined from self-rated health in the Medical Expenditure Panel Survey (MEPS) and from work-limitation questions in the Panel Study of Income Dynamics (PSID), as described in the text. *Data sources:* MEPS and PSID.

B.3 First-Step Estimation

Health transitions. We estimate the transition probabilities $\Pr(h_t = j \mid h_{t-1} = i)$, $i, j \in \{0, 1, 2\}$, by probit regressions of the indicator $\mathbf{1}\{h_t = j\}$ on a full set of age dummies, using the observations with $h_{t-1} = i$. The predicted values are the estimated transition probabilities, which we interpolate linearly across ages so that the transition matrices change smoothly over the life cycle.

Survival. We estimate age- and health-dependent survival probabilities in the PSID by a logistic regression of an indicator for survival on a polynomial in age, lagged health status, and their interactions, keeping the terms that are statistically significant. Table B.4 reports the estimated coefficients, and panel B of Figure 1 shows the implied mortality rates by age and health state.

Table B.4. Estimated Coefficients of the Survival Model

	Coefficient	Std. error
Age^2	-0.0006^{***}	(0.0000)
$\text{Health}_{t-1} = 1$	-1.3198^{***}	(0.2288)
$\text{Health}_{t-1} = 2$	-3.1419^{***}	(0.4189)
$\text{Health}_{t-1} \times \text{Age}^2$	0.0109^{***}	(0.0032)
Constant	7.8801^{***}	(0.1373)
Observations	132,070	

Notes: Logistic regression of an indicator of survival on the square of age, indicators of lagged health status (1: poor health; 2: disability), and the interaction of lagged health status with the square of age. Robust standard errors, clustered at the individual level, in parentheses; (0.0000) denotes a standard error that rounds to zero at four decimal places. Entries are reported at the precision of the original estimation output. *** $p < 0.01$. *Data source:* Panel Study of Income Dynamics.

Out-of-pocket medical expenses. Out-of-pocket medical expenses are total medical expenditures net of the amount covered by health insurance, measured in the MEPS. Following the procedure of Pashchenko and Porapakkarm (2019), we estimate the profiles of total medical expenditures by weighted regressions on age and year dummies, separately for each health state, using both the cross-sectional and the longitudinal weights. We estimate the profiles of insurance coverage in the same way. For working-age individuals, coverage is the amount paid by insurance; for individuals aged 65 and older, it is the amount paid by Medicare. Out-of-pocket expenditures are the product of the estimated coinsurance rate and total medical expenses at each age and health state. Following De Nardi et al. (2024), we multiply the estimated expenditures by 1.60 below age 65 and by 1.90 from age 65 so that medical spending is consistent with the aggregates of the National Health Expenditure Accounts. Finally, we smooth the profiles by regressing them on a quadratic function of age.

Wages. Hourly wages are annual earnings divided by annual hours worked, both reported in the PSID for the previous calendar year. We exclude observations with hourly wages below \$6.50 or above \$250 in 2016 dollars. Because wages are observed only for participants, we correct for selection with the two-stage estimator of Heckman (1976), as adapted by Guner et al. (2012). In the first stage, a probit regression of labor force participation on all PSID observations generates an inverse Mills ratio. In the second stage, we regress log hourly wages on an age polynomial, the interactions of health with the age polynomial, and the inverse Mills ratio. Family size and year of birth enter the selection equation but not the wage equation. Table B.5 reports the coefficients of both stages. Earlier studies instead estimate the wage equation inside the structural model by matching the

wage profiles of participants (French 2005; French and Jones 2011); the Heckman correction fits the targeted moments well and keeps the wage profiles outside the second-step search, which saves computation.

Table B.5. Estimated Coefficients of the Wage Equation with Selection

	Coefficient	Std. error
Wage equation (dependent variable: log hourly wage)		
Age	0.0582***	(0.00500)
Age^2	−0.00067***	(0.00006)
Health×Age	−0.0113***	(0.00127)
Health×Age^2	0.00013***	(0.00002)
Constant	1.447***	(0.104)
Selection equation (dependent variable: participation)		
Age	0.0977***	(0.00453)
Age^2	−0.00162***	(0.00004)
Health	−1.371***	(0.0682)
Health×Age	0.0122***	(0.00119)
Family size	0.0225***	(0.00369)
Birth year	−0.0123***	(0.00080)
Constant	23.79***	(1.581)
Inverse Mills ratio	0.182***	(0.0482)
Observations	64,941	

Notes: Two-stage Heckman selection model. Wage equation: dependent variable is the logarithm of the hourly wage. Selection equation: dependent variable is an indicator of labor force participation. Robust standard errors, clustered at the individual level, in parentheses. Entries are reported at the precision of the original estimation output. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. *Data source:* Panel Study of Income Dynamics.

Ability process. We estimate the autoregressive coefficient and innovation variance (ρ, σ^2) of the ability process (4) from the wage residuals of the second stage. We restrict the residuals to ages 25 to 65 and drop the top 0.5 percent to limit the influence of outliers on the variance. Estimation is by maximum likelihood with the standard Kalman filter recursions. The estimates are $(\rho, \sigma^2) = (0.981, 0.016)$.

Initial conditions. Simulated individuals enter the model at age 30. We draw their initial states from the empirical joint distribution of wages, health, assets, and labor force participation of individuals aged 28 to 32 in the PSID sample, adjusting the mean of log wages to match the estimated wage profile within each health state. To impute initial average indexed monthly earnings, we assume, following French (2005), that every individual has worked for five years, 2,000 hours per year, at her hourly wage at age 30. Table B.6 summarizes the initial distribution. Individuals in good health have higher

wages, assets, and participation, on average, than individuals in poor health or with a disability.

Table B.6. Initial Conditions

		By health status		
	Overall	Good health	Poor health	With a disability
Assets (2016 dollars)				
Mean	60,241	61,388	42,022	30,289
Standard deviation	146,975	149,670	91,894	19,310
Hourly wages (2016 dollars)				
Mean	18.95	19.19	15.13	12.50
Standard deviation	9.32	9.32	8.29	6.98
Participation				
Mean	0.96	0.97	0.89	0.45
Standard deviation	0.20	0.18	0.31	0.50
Share of sample (%)		94.74	4.17	1.09
Observations	4,849	4,594	202	53

Notes: Summary statistics of the empirical joint distribution from which initial states at age 30 are drawn: individuals aged 28 to 32 in the PSID sample. Participation is an indicator for working more than 800 hours in the year. Share of sample: percentage of observations in each health state. Monetary values in 2016 dollars. *Data source:* Panel Study of Income Dynamics.

C Second-Step Estimation and Alternative Preferences

This appendix describes the construction of the target profiles, the moment conditions and the estimation algorithm of the second step, and the standard errors, and it reports the estimates of an alternative specification with separable logarithmic preferences.

C.1 Target Profiles

The target profiles are labor force participation by health status, hours worked conditional on participation by health status, mean and median assets, and mean labor market reentry rates over the life cycle, all constructed from the PSID sample of Appendix B.

Variables. Labor force participation is an indicator equal to one if an individual's annual hours worked exceed 800, which captures both full-time and part-time employment. Hours worked are self-reported; respondents in survey year t report their hours worked in year $t-1$. Assets include real estate, farm or business equity, vehicles, stocks, mutual funds, individual retirement accounts, Keogh plans, liquid assets, bonds, and investment trusts, net of mortgages and other debts, plus home equity net of outstanding debts; they exclude

pension and Social Security wealth. Following Borella et al. (2018), we divide the assets of a couple equally between the two partners. Asset data are available only in selected survey years starting in 1984. For the other years, we impute assets from a fixed-effects regression of assets on an age polynomial, log wages, family size, education, the unemployment rate, health status indicators, the interactions of log wages and of health status with the age polynomial, and the interactions of health status with education, family size, the unemployment rate, and wages. The asset profiles use both observed and imputed values.

Fixed-effects profiles. Following French (2005), we estimate the labor-supply profiles from the fixed-effects regression

$$Z_{it} = f_i + \sum_k B_{gk}\mathbf{1}\{\text{age}_{it} = k\}\mathbf{1}\{h_{it} = 0\} + \sum_k B_{bk}\mathbf{1}\{\text{age}_{it} = k\}\mathbf{1}\{h_{it} \neq 0\} + \sum_f B_f\mathbf{1}\{\text{fs}_{it} = f\} + B_u U_t + u_{it},$$

where Z_{it} is participation or hours of individual i at age t, f_i is an individual effect, fs_{it} is family size, U_t is the unemployment rate, and the coefficients B_{gk} and B_{bk} trace the age profiles of healthy and unhealthy individuals. To remove family-size and business-cycle effects, we evaluate the profiles at a family size of 3 and an unemployment rate of 6.5 percent. To control for cohort effects and for the correlation between the individual effect and health, we use the mean individual effect of individuals born in 1935 who are in average health at age 50. The asset and reentry profiles are not conditioned on health. Healthy individuals work considerably more than unhealthy individuals over the life cycle, particularly after age 50, and reentry rates remain below 5 percent at every age (Figure 2 and Figure 3).

C.2 Moment Conditions and Criterion

Let $t \in \{30, \ldots, 69\}$ index the $J = 40$ target ages and $g \in \{\text{healthy}, \text{unhealthy}\}$ the two health groups, with unhealthy meaning $h \in \{1, 2\}$. Let $p_t(X; \mathcal{P}, \chi)$, $n_t(X; \mathcal{P}, \chi)$, $a_t(X; \mathcal{P}, \chi)$, and $RE_t(X; \mathcal{P}, \chi)$ denote the participation, hours, asset, and reentry outcomes implied by the model's decision rules in state X, and let $F_{g,t}(\cdot; \mathcal{P}, \chi)$ and $F_t(\cdot; \mathcal{P}, \chi)$ denote the simulated distributions of the state at age t within health group g and in the whole

population. The vector of moment conditions stacks, for every t and g,

$$\hat{\varphi}(\mathcal{P};\chi) = \begin{bmatrix} \hat{p}_{g,t} - \int p_t(X;\mathcal{P},\chi)\, dF_{g,t}(X;\mathcal{P},\chi) \\ \hat{n}_{g,t} - \int n_t(X;\mathcal{P},\chi)\, dF_{g,t}(X \mid n_t > 0;\mathcal{P},\chi) \\ \hat{a}_t - \int a_t(X;\mathcal{P},\chi)\, dF_t(X;\mathcal{P},\chi) \\ \hat{a}_t^{\text{med}} - \text{med}_{F_t}\left[a_t(X;\mathcal{P},\chi)\right] \\ \widehat{re}_t - \int RE_t(X;\mathcal{P},\chi)\, dF_t(X;\mathcal{P},\chi) \end{bmatrix},$$

where hats denote the data profiles of Section 3: $\hat{p}_{g,t}$ and $\hat{n}_{g,t}$ are participation and hours of participants in group g at age t, $\hat{a}_t$ and $\hat{a}_t^{\text{med}}$ are mean and median assets, and $\widehat{re}_t$ is the mean reentry rate. The vector has

(i) $2J$ elements for participation by health group,

(ii) $2J$ elements for hours conditional on participation by health group,

(iii) $2J$ elements for mean and median assets, and

(iv) J elements for the reentry rate,

for a total of $7J = 280$ moment conditions. The MSM estimator minimizes

$$Q(\mathcal{P}) = \frac{N}{1 + N/N_s}\, \hat{\varphi}(\mathcal{P};\chi)'\, \hat{\mathbf{W}}\, \hat{\varphi}(\mathcal{P};\chi),$$

where N is the number of observations in the data and N_s the number of simulated observations; the ratio N/N_s accounts for the additional variance introduced by simulation. The weighting matrix $\hat{\mathbf{W}}$ is the inverse of the $7J \times 7J$ diagonal matrix whose diagonal elements are the sampling variances of the data moments.

C.3 Estimation Algorithm

The estimation proceeds in five steps.

1. Estimate the empirical life-cycle profiles of participation and hours by health group, mean and median assets, and reentry, and their sampling variances.
2. Given the initial distribution of the state variables (Table B.6) and the first-step parameters χ that govern the data-generating process for the states, draw the health, survival, and wage shocks for the simulated individuals.
3. Given a value of $\mathcal{P}$, solve the model by backward induction (Appendix A.5) and use the decision rules to simulate the life cycles of participation, hours, assets, and reentry.

4. Construct the moment conditions $\hat{\varphi}(\mathcal{P};\chi)$ and evaluate the criterion $Q(\mathcal{P})$.

5. Choose a new value of $\mathcal{P}$ and repeat steps 3 and 4 until the criterion is minimized.

C.4 Identification and Standard Errors

Identification follows French (2005) and rests on two assumptions: individuals' work decisions do not affect their health or wages, and age changes the incentives to work and save but not preferences. Section 3 discusses which features of the data identify each parameter.

We compute standard errors from the asymptotic distribution of the MSM estimator. Let $\mathbf{D} = \partial\hat{\varphi}(\mathcal{P};\chi)/\partial\mathcal{P}'$ denote the 280×7 Jacobian of the moment conditions evaluated at $\hat{\mathcal{P}}$, and let $\mathbf{\Omega}$ denote the variance–covariance matrix of the data moments, scaled by N. Then

$$\sqrt{N}(\hat{\mathcal{P}} - \mathcal{P}_0) \xrightarrow{d} \mathcal{N}\left(0,\ \left(1 + \frac{N}{N_s}\right)(\mathbf{D}'\mathbf{W}\mathbf{D})^{-1}\mathbf{D}'\mathbf{W}\mathbf{\Omega}\mathbf{W}\mathbf{D}(\mathbf{D}'\mathbf{W}\mathbf{D})^{-1}\right),$$

where $\mathbf{W}$ is the probability limit of $\hat{\mathbf{W}}$, and the factor $1 + N/N_s$ inflates the variance for simulation error. Because $\mathbf{W}$ is diagonal rather than the inverse of $\mathbf{\Omega}$, the sandwich form does not simplify. The overidentification statistic reported in Table 2 is evaluated against a χ^2 distribution with $280 - 7 = 273$ degrees of freedom.

C.5 Alternative Preferences: Logarithmic Utility

The baseline specification has nonseparable preferences over consumption and leisure, with participation and reentry costs measured in hours. As a check on this functional form, we reestimate the model with preferences that are separable in consumption and hours, logarithmic in consumption, and subject to fixed utility costs of participation and reentry:

$$u(c,n) = \ln c - \kappa\,\frac{n^{1+1/\eta}}{1+1/\eta} - \tilde{\phi}(h)\,\mathbf{1}\{P=1\} - \tilde{\phi}_{RE}\,\mathbf{1}\{RE=1\},$$

where n is annual hours, κ scales the disutility of hours, η is the Frisch elasticity of hours, $\tilde{\phi}(h)$ is the utility cost of participation in health state h, with $\tilde{\phi}(1) = \tilde{\phi}(2)$ as in the baseline, and $\tilde{\phi}_{RE}$ is the utility cost of reentering the labor market. We fix $\eta = 0.5$, a common value for the intensive-margin Frisch elasticity of hours, and estimate $(\beta, \kappa, \tilde{\phi}(0), \tilde{\phi}(1), \tilde{\phi}_{RE})$ by the same MSM procedure. All other components of the model—the health and wage processes, medical expenses, policy rules, and estimation targets—are unchanged.

Table C.1 reports the estimates, and Figure C.1 compares the simulated and empirical profiles. The estimates are broadly consistent with the baseline. The utility cost of working

is higher for unhealthy individuals than for healthy ones, 0.59 against 0.53, as the time cost is higher in the baseline; and the reentry cost is large relative to the per-period cost of participation. The two specifications fit different parts of the data. The logarithmic specification fits mean assets at older ages better than the baseline, but it matches participation less accurately: simulated participation remains above the data until about age 60 for the healthy and until 64 for the unhealthy, so the model misses the earlier decline of unhealthy individuals. Simulated median assets exceed the data at nearly all ages, and simulated mean assets do so from age 39 to 59.

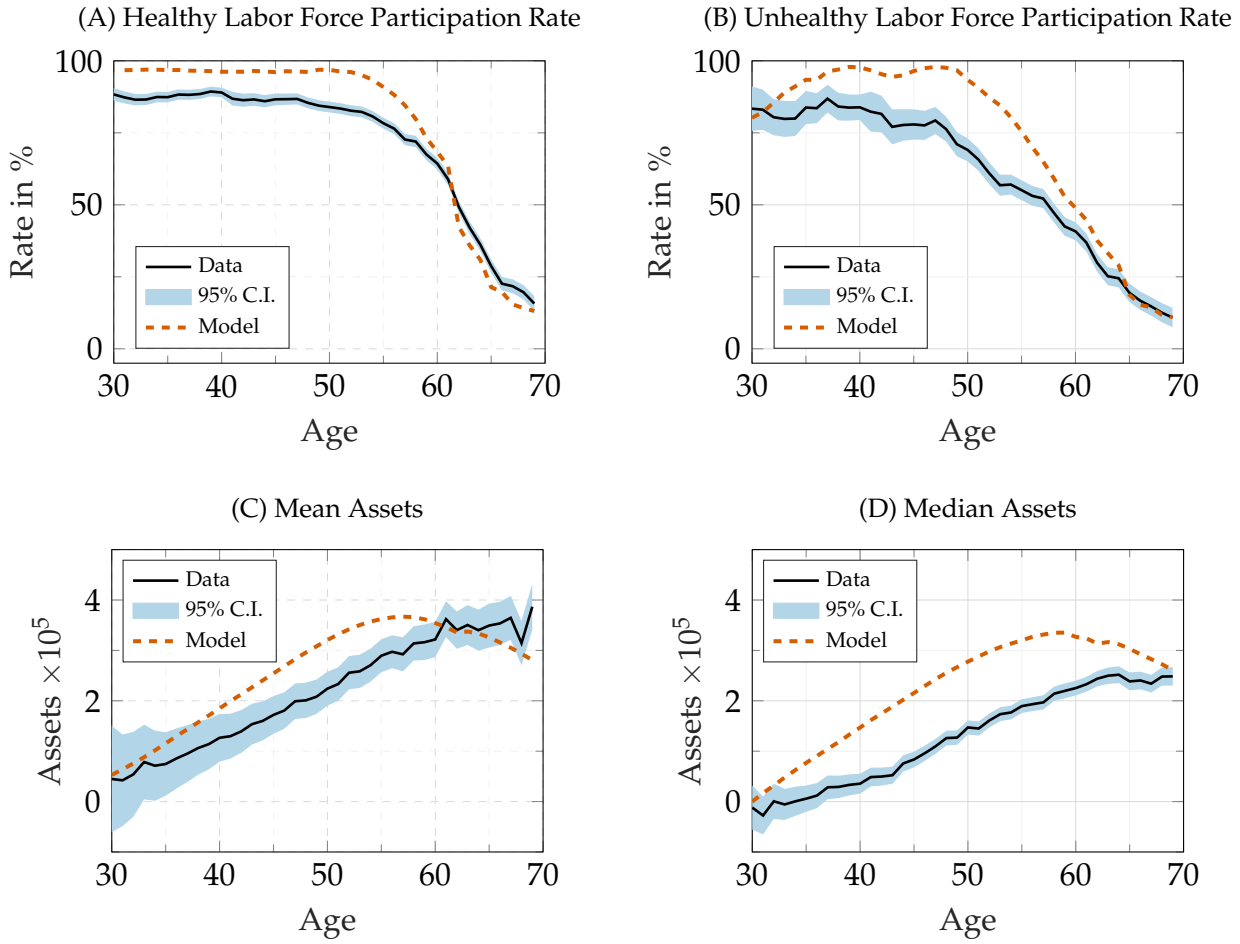
The reform analysis in Sections 4 to 6 uses the baseline nonseparable specification of Table 2. Participation and its decline by health status govern the behavioral responses and fiscal externalities of retirement reforms, and the baseline matches these profiles more closely.

Table C.1. Estimated Parameters: Logarithmic Utility

Parameter	Description	Estimate	Std. error
Estimated			
β	Time discount factor	0.97	0.0015
κ	Scale of the disutility of hours	1.37×10^{-10}	0.14×10^{-10}
$\tilde{\phi}(0)$	Utility cost of work, good health	0.53	0.0082
$\tilde{\phi}(1) = \tilde{\phi}(2)$	Utility cost of work, poor health or disability	0.59	0.0152
$\tilde{\phi}_{RE}$	Utility cost of labor market reentry	36.15	1.2686
Fixed			
η	Frisch elasticity of hours	0.5	

Notes: Method-of-simulated-moments estimates and standard errors for the specification $u(c, n) = \ln c - \kappa n^{1+1/\eta}/(1 + 1/\eta) - \tilde{\phi}(h)\mathbf{1}\{P = 1\} - \tilde{\phi}_{RE}\mathbf{1}\{RE = 1\}$. The costs $\tilde{\phi}(h)$ and $\tilde{\phi}_{RE}$ are in utils; poor health ($h = 1$) and disability ($h = 2$) share one cost of work. The Frisch elasticity η is fixed at 0.5, not estimated. Estimates and standard errors are reported at the precision of the original estimation output. Targets, first-step parameters, and policy rules are those of the baseline estimation. *Data source:* Panel Study of Income Dynamics.

Figure C.1. Data and Simulated Profiles: Logarithmic Utility



Notes: Panels A and B show labor force participation rates (percent) of healthy individuals ($h = 0$) and unhealthy individuals (poor health or disability, $h \in \{1, 2\}$); panels C and D show mean and median nonpension assets (hundreds of thousands of 2016 dollars). Solid lines are data profiles estimated from the Panel Study of Income Dynamics, shaded areas are 95% confidence intervals, and dashed lines are profiles simulated from the model with logarithmic utility at the estimates of Table C.1. Compared with the baseline (Figure 2 and Figure 3), the logarithmic specification fits mean assets at older ages better and participation worse.

D Construction of Reforms and Outcomes

This appendix describes how each instrument-elimination experiment and each reform is implemented and how its welfare, fiscal, MVPF, and participation outcomes are computed. Every reform is a permanent change in the rules of Section 2 that the cohort knows from age 30. For each reform, we solve the model again by backward induction under the new rules (Appendix A.5) and simulate the cohort forward from the same initial distribution as in the status quo. All other rules and parameters keep their status-quo values, and the wage profiles and the pretax return r_f are held fixed. Outcomes are computed from

the simulated histories.

D.1 Instrument-Elimination Experiments

The five experiments of Table 3 each remove one instrument from the estimated economy.

- *D1, no Social Security benefits:* Social Security benefits are set to zero. In a variant reported in Section 5.1, only retirement benefits are set to zero and disability benefits are kept.
- *D2, loss from incomplete coverage in Medicare:* out-of-pocket medical expenses after age 65 are set to zero for participants and nonparticipants, so that all medical expenditures after 65 are covered. The table reports the welfare change of the status quo relative to this economy, which, by the antisymmetry of the CE, is minus the welfare gain from full coverage.
- *D3, full savings subsidy:* the net return on assets r is set equal to the pretax return r_f. This experiment coincides with reform K3.
- *D4, no income tax:* the income tax $T_t(\cdot)$ of (9) is set to zero at every age.
- *D5, no disability benefits:* DI benefits db_t of (8) are set to zero.

These experiments change the government budget without financing the change. We report their CE and LE welfare changes only.

D.2 Claiming-Age Dependence

Let $\alpha(a) = b(a)/PIA$ be the benefit, as a fraction of the PIA, of an individual who claims at age $a \in \{62, \ldots, 70\}$. Under the rules of the status quo and of the AN and AD reforms,

$$\alpha(a) = \begin{cases} 1 - \frac{1}{15}\min\{NRA - a,\, 3\} - \frac{1}{20}\max\{NRA - a - 3,\, 0\} & \text{if } a < NRA, \\ 1 + \delta\,(a - NRA) & \text{if } a \geq NRA: \end{cases}$$

benefits fall by $6\frac{2}{3}$ percent of the PIA for each of the first three years by which the claim precedes the NRA and by 5 percent for each additional year, and they rise by the DRC δ for each year of delay past the NRA. The status quo (AN0) has $NRA = 65$ and $\delta = 4.5$ percent, approximately the average credit of the 1920–1940 cohorts, for whom the DRC rose with the year of birth from 3 to 7 percent.

- *AN1–AN3: NRA* $= 66$, 67, and 68, with $\delta = 4.5$ percent. The schedule shifts to the right by one, two, and three years, so that the benefit at 62 falls from 80 to 75, 70, and 65 percent of the PIA. Because DI benefits are paid until the NRA (8), they are paid until the new NRA.
- *AD1–AD5: NRA* $= 65$ and $\delta = 8$, 8.5, 9, 9.5, and 10 percent. Benefits for claims before the NRA are unchanged.
- *AL1–AL9:* $\alpha(a) = 1 + x\,(a - 65)$ at every claiming age, with $x = 7, 8, \ldots, 15$ percent. The reduction for early claiming and the credit for delay are equal in absolute value.

Table D.1 reports $\alpha(a)$, in percent, for every claiming age and every reform; Figure 4 plots selected schedules. The reductions for early claiming under the AN reforms follow the rules that the Social Security Administration applies when the NRA exceeds 65, extended to an NRA of 68.

D.3 Benefit Progressivity and Level

Write the PIA formula (6) with replacement rates $\omega = (\omega_1, \omega_2, \omega_3)$:

$$PIA_t = \omega_1 \min\{aime_t, aime_0\} + \omega_2 \min\left\{ \max\{aime_t - aime_0, 0\},\ aime_1 - aime_0 \right\} + \omega_3 \max\{aime_t - aime_1, 0\},$$

with $\omega = (0.90,\ 0.32,\ 0.15)$ in the status quo. The bend points $aime_0$ and $aime_1$ and the claiming-age adjustments are unchanged. The reforms set:

- *P1.1–P1.9:* $\omega = (0.90 + 0.01k,\ 0.32,\ 0.15)$ for $k = 1, \ldots, 9$, which raises the first rate by one percentage point per step, from 0.91 to 0.99.
- *P2.0–P2.9:* $\omega = \big(0.99,\ 0.32\,(1.01 + 0.01k),\ 0.15\big)$ for $k = 0, \ldots, 9$, which raises the second rate by 0.32 percentage points (1 percent of its initial value) per step, from 0.3232 to 0.352, on top of the first rate of P1.9.
- *P3.0–P3.9:* $\omega = \big(0.99,\ 0.352,\ 0.15\,(1.01 + 0.01k)\big)$ for $k = 0, \ldots, 9$, which raises the third rate by 0.15 percentage points (1 percent of its initial value) per step, from 0.1515 to 0.165, on top of the first two rates of P2.9.
- *L0–L9:* $\omega = (1.01 + 0.01k) \times (0.90,\ 0.32,\ 0.15)$ for $k = 0, \ldots, 9$, which raises all three rates proportionally by 1 to 10 percent; L0 sets $\omega = (0.909,\ 0.3232,\ 0.1515)$ and L9 sets $\omega = (0.99,\ 0.352,\ 0.165)$.

The P reforms trace a path from the status quo to a 10 percent increase in all three rates that raises the brackets one at a time, from the bottom up. The path ends at the formula of

Table D.1. Benefits by Claiming Age under the Claiming-Age Reforms

		Claiming age								
	Schedule	62	63	64	65	66	67	68	69	70
(AN0)	Status quo (NRA 65)	80	86.67	93.33	**100**	104.5	109	113.5	118	122.5
A. Raising the normal retirement age (NRA)										
(AN1)	NRA at 66	75	80	86.67	93.33	**100**	104.5	109	113.5	118
(AN2)	NRA at 67	70	75	80	86.67	93.33	**100**	104.5	109	113.5
(AN3)	NRA at 68	65	70	75	80	86.67	93.33	**100**	104.5	109
B. Raising the delayed retirement credit (DRC)										
(AD1)	DRC at 8%	80	86.67	93.33	**100**	108	116	124	132	140
(AD2)	DRC at 8.5%	80	86.67	93.33	**100**	108.5	117	125.5	134	142.5
(AD3)	DRC at 9%	80	86.67	93.33	**100**	109	118	127	136	145
(AD4)	DRC at 9.5%	80	86.67	93.33	**100**	109.5	119	128.5	138	147.5
(AD5)	DRC at 10%	80	86.67	93.33	**100**	110	120	130	140	150
C. Symmetric adjustment around the NRA of 65										
(AL1)	Linear $\pm$7%	79	86	93	**100**	107	114	121	128	135
(AL2)	Linear $\pm$8%	76	84	92	**100**	108	116	124	132	140
(AL3)	Linear $\pm$9%	73	82	91	**100**	109	118	127	136	145
(AL4)	Linear $\pm$10%	70	80	90	**100**	110	120	130	140	150
(AL5)	Linear $\pm$11%	67	78	89	**100**	111	122	133	144	155
(AL6)	Linear $\pm$12%	64	76	88	**100**	112	124	136	148	160
(AL7)	Linear $\pm$13%	61	74	87	**100**	113	126	139	152	165
(AL8)	Linear $\pm$14%	58	72	86	**100**	114	128	142	156	170
(AL9)	Linear $\pm$15%	55	70	85	**100**	115	130	145	160	175

Notes: Retirement benefit received by an individual who claims at each age from 62 to 70, in percent of the primary insurance amount (PIA). Bold entries mark the normal retirement age (NRA), at which the benefit equals the PIA. AN0: status quo of the 1920–1940 birth cohorts, with an NRA of 65, a reduction of $6\frac{2}{3}$ percent of the PIA per year of claiming before the NRA, and a delayed retirement credit (DRC) of 4.5 percent of the PIA per year of claiming after the NRA, approximately the average for these cohorts. AN1–AN3 raise the NRA to 66, 67, and 68; the reduction is $6\frac{2}{3}$ percent per year for the three years before the NRA and 5 percent per year before that. AD1–AD5 raise the DRC to 8, 8.5, 9, 9.5, and 10 percent. AL1–AL9 set the reduction for early claiming and the DRC to a common rate of 7 to 15 percent per year. *Source:* rules of the Social Security Administration for the status quo (with the cohort-average DRC) and the AN reforms (extended to an NRA of 68); authors' construction for the AD and AL reforms.

L9: P3.9 and L9 set the same rates, and their simulated outcomes in Table 5 and Table F.1 coincide.

D.4 Medicare

To implement the Medicare reforms, we first measure from the MEPS the average share of medical expenditures that Medicare covers for individuals aged 65 and older, by health status and working status. Denote this coverage rate by $q(h, P)$. At ages $t \geq 65$, out-of-pocket medical expenses are $m_t = \big(1 - q(h_t, P_t)\big)\, m_t^{tot}$, where m_t^{tot} is total medical expenditure, and Medicare pays $m_t^G = q(h_t, P_t)\, m_t^{tot}$ (Section 2.5). We simulate the estimated model with these coverage rates and use the simulated economy as the baseline for the two Medicare reforms.

- *Part A:* the coverage rate of every individual aged 65 and older rises by 20 percentage points, $q(h, P) \to q(h, P) + 0.20$, which represents expanded coverage of inpatient services. The targeted beneficiaries are all individuals aged 65 and older.
- *Part B:* the coverage rate of nonparticipants aged 65 and older rises by 20 percentage points, $q(h, 0) \to q(h, 0) + 0.20$, which represents expanded coverage of physician and outpatient services. Because most employer-provided health insurance plans already cover these services, the reform targets nonparticipants. The targeted beneficiaries are the nonparticipants aged 65 and older.

At unchanged behavior, each reform shifts 20 percent of the total medical expenditure of the targeted beneficiaries from out-of-pocket expenses to Medicare. Under Part B, the shift is conditional on not working, so the reform also changes the return to work after 65.

D.5 Old-Age Income Taxes

Reforms I1–I5 lower income taxes from age 62 by raising the level parameter of the after-tax income schedule $y - T_t(y) = \lambda_t\, y^{1-\tau_t}$:

$$\tilde{\lambda}_t = \begin{cases} \lambda_t & \text{if } t \leq 61, \\ \lambda_t + x\,(t - 61) & \text{if } t \geq 62, \end{cases} \qquad x = 0.01,\ 0.02,\ 0.03,\ 0.04,\ 0.05,$$

with the progressivity parameter τ_t unchanged. At age $t \geq 62$, after-tax income rises at every income level by the proportion $x(t - 61)/\lambda_t$, and the tax cut $x(t - 61)\, y^{1-\tau_t}$ rises with income and with age. Individuals anticipate the reform from age 30.

D.6 Savings Subsidies

Reforms K1–K3 raise the net return on assets at every age from r to $\tilde{r} = r + \Delta_K$, with $\Delta_K = 0.2$, 0.4, and 0.8 percentage points. The government pays the additional return $\Delta_K a_t$ on assets held at every age, and the present value of the subsidy is

$$K = \sum_{t=30}^{T} (1+r)^{-(t-62)} \, \overline{\Delta_K a_t},$$

where the bar denotes the per capita value in the simulated cohort under the reform. Since $r = (1-\tau_k)\, r_f$ with $r_f = 4$ percent and $\tau_k = 20$ percent, the three subsidies are equivalent to lowering the capital-tax wedge τ_k to 15, 10, and 0 percent (exact arithmetic: $(1-\tilde{\tau}_k) \times 4 = 3.4$, 3.6, and 4.0 percent). K3 sets the net return equal to the pretax return and coincides with experiment D3.

D.7 Construction of Outcomes

Welfare. Let $\bar{V}^j = \frac{1}{N}\sum_{i=1}^{N} V_{30}(X_{30}^i; j)$ be the average value at age 30 over the N simulated individuals under the rules j, where $j = 0$ denotes the baseline and $j = 1$ the reform; the initial states X_{30}^i are the same under both. Because flow utility satisfies $u(e^x c, l) = e^{\gamma(1-\nu)x} u(c,l)$, scaling consumption in the baseline allocation by e^x at every age and in every state multiplies $\bar{V}^0$ by $e^{\gamma(1-\nu)x}$. The consumption-equivalent welfare change is the x that solves $\bar{V}^1 = e^{\gamma(1-\nu)x}\bar{V}^0$, and the leisure-equivalent change the y that solves $\bar{V}^1 = e^{(1-\gamma)(1-\nu)y}\bar{V}^0$:

$$CE = \frac{\ln \bar{V}^1 - \ln \bar{V}^0}{\gamma(1-\nu)}, \qquad LE = \frac{\ln \bar{V}^1 - \ln \bar{V}^0}{(1-\gamma)(1-\nu)},$$

where, since $\bar{V}^0$ and $\bar{V}^1$ are negative when $\nu > 1$, the logarithms apply to their absolute values; the difference is $\ln(\bar{V}^1/\bar{V}^0)$, as in (13). The tables report $100 \times CE$ and $100 \times LE$, in percent. The two measures satisfy $LE = CE \times \gamma/(1-\gamma)$; at the estimates, $\gamma/(1-\gamma) = 0.5354/0.4646 \approx 1.15$, which reproduces the ratio of the LE and CE columns of Table 3 up to rounding. Both measures are antisymmetric: the welfare change of moving from rules j to rules k is minus that of moving from k to j. The willingness to pay for a reform is $WTP = 100 \times CE$.

Fiscal effects. The fiscal aggregates SS, DI, TR, IT, PT, and $G = SS + DI + TR - IT - PT$ are the per capita present values at age 62 of Section 2.11, discounted at the net return r. Medicare payments are not part of G; they enter only the outcomes of the two Medicare reforms. A superscript 0 denotes the baseline, and Δ the difference between the reform

and the baseline. In dollars, the direct cost DC, the indirect cost IC, and the net cost $NC = DC + IC$ of each family are those of Section 4.3. The results tables express each of them in percent, with positive entries for costs to the government and negative entries for savings. Each definition below is followed by the dollar change that the column measures.

- *Social Security reforms* (claiming age, progressivity, and level). Direct is the percentage change in the present value of total retirement benefits paid by the program, ΔSS. Indirect is the percentage change in the present value of all other taxes and transfers, DI benefits and consumption-floor transfers net of income- and payroll-tax revenue, $\Delta(DI + TR - IT - PT)$. Net combines the two and is the percentage change in the present value of net government spending, ΔG.

- *Medicare reforms.* Direct is the change in the present value of Medicare's payments to the targeted beneficiaries after age 65 (all individuals for Part A, nonparticipants for Part B), ΔMC^{tg}, as a share of baseline payroll-tax revenue, where $MC^{tg} = \sum_{t=65}^{T}(1 + r)^{-(t-62)}\,\overline{m_t^G\,\mathbf{1}\{\text{targeted at } t\}}$. Indirect is the percentage change in the present value of all taxes and transfers, including Social Security benefits, net of income- and payroll-tax revenue, ΔG. Net is the change in the present value of government spending plus the change in Medicare's payments to the targeted beneficiaries, $\Delta G + \Delta MC^{tg}$, as a share of baseline government spending. All baseline values refer to the Medicare baseline.

- *Income-tax reforms.* Direct is the percentage change in the present value of negative total income-tax revenue, that is, of the revenue forgone, $-\Delta IT$. Indirect is the percentage change in the present value of all other taxes and transfers, including Social Security benefits, net of payroll-tax revenue, $\Delta(SS + DI + TR - PT)$. Net combines the two and is the percentage change in the present value of net government spending, ΔG.

- *Savings subsidies.* Direct is the present value of the subsidy, K, as a share of baseline government spending. Indirect is the percentage change in the present value of government spending excluding the subsidy, ΔG. Net is the percentage change in the present value of government spending including the subsidy, $\Delta G + K$.

The three columns have different bases, so Net is in general not the sum of Direct and Indirect, although the dollar amounts add up in every family. The percentages add up only when the columns share a base, as for the savings subsidies, whose three columns are all expressed relative to baseline government spending.

Marginal value of public funds. For the reforms of Social Security, Medicare, and income taxes, the MVPF is the dollar value of the CE welfare gain divided by the dollar change in the present value of per capita government spending, direct and indirect effects included. The dollar value of the welfare gain is the CE times the present value of per capita baseline consumption from age 62, discounted to age 62:

$$MVPF = \frac{CE \times \bar{C}^0_{62}}{NC}, \qquad \bar{C}^0_{62} = \sum_{t=62}^{T} (1+r)^{-(t-62)}\, \bar{c}^0_t.$$

For the savings subsidies, which change returns from age 30, the dollar value of the welfare gain uses per capita baseline consumption from age 30, discounted to age 62, $\bar{C}^0_{30} = \sum_{t=30}^{T}(1+r)^{-(t-62)}\bar{c}^0_t$, and the denominator is the dollar change in government spending including the subsidy, $NC = \Delta G + K$. When the WTP is positive and the net cost zero or negative, the MVPF is infinite. For such reforms, we report as a lower bound the dollar value of the welfare gain divided by the dollar direct cost, $CE \times \bar{C}^0_{62}/DC$, when the direct cost is positive; when it is negative, the lower bound is infinite as well. For the savings subsidies, we also report the MVPF with the subsidy as the only cost, $CE \times \bar{C}^0_{30}/K$. The MVPF computed with the net cost is an upper bound, because the model taxes interest income as it accrues; the MVPF computed with the subsidy alone is a lower bound only if the fiscal externalities other than the taxation of interest are savings (Section 4.4). When the WTP and the net cost are both negative, the MVPF is positive and measures the welfare cost per dollar saved. Because the dollar welfare gain and the dollar net cost are the WTP and Net entries times fixed baseline aggregates, the ratio $MVPF \times Net/WTP$ depends on the baseline but not on the reform (Section 4.4).

Participation. The participation columns report the change in the labor force participation rate of the simulated cohort, in percentage points, between the reform and the baseline, for ages 30–49, 62–64, and 65–69.

E Derivations

This appendix proves Lemma 1, gives a sufficient condition under which the fair credit is higher in worse health, decomposes the change in retirement outlays caused by a change in the claiming-age schedule, and derives the relations between the net cost, the willingness to pay, and the fiscal multiplier used in Section 4.5.

E.1 Proof of Lemma 1

Let $b(\cdot)$ be any schedule of benefits by claiming age and $\Delta b(a) \equiv b(a+1) - b(a)$ its increment at age a. The definition of $A(\cdot)$ implies $A(a) = (1+r)^{-(a-62)}\mathcal{S}(a) + A(a+1)$. Hence

$$\begin{aligned} B(a+1) - B(a) &= b(a+1)\,A(a+1) - b(a)\,A(a) \\ &= \Delta b(a)\,A(a+1) - b(a)\,(1+r)^{-(a-62)}\,\mathcal{S}(a). \end{aligned}$$

For $a \geq NRA$, $\Delta b(a) = \delta\, PIA$, which gives (17). Since $a+1 \leq 70 \leq T$ and $\mathcal{S}(a+1) > 0$, $PIA\,A(a+1) > 0$, and $B(a+1) < B(a)$ if and only if $\delta < \delta^*(a)$ as defined in (18). Dividing the numerator and the denominator of $\delta^*(a)$ by $(1+r)^{-(a-62)}\mathcal{S}(a) > 0$ gives the second expression in (18), $\delta^*(a) = [b(a)/PIA]/\ddot{a}(a)$, where

$$\ddot{a}(a) \equiv \sum_{t=a+1}^{T} (1+r)^{-(t-a)}\,\frac{\mathcal{S}(t)}{\mathcal{S}(a)}$$

is the expected present value at age a, for an individual alive at a, of one dollar a year from age $a+1$ until death. The factor $b(a)/PIA = 1 + \delta(a - NRA)$ does not depend on survival, and $\ddot{a}(a)$ is strictly increasing in each $\mathcal{S}(t)/\mathcal{S}(a)$, $t > a$, and nondecreasing in T, since a later terminal age adds nonnegative terms. Therefore $\delta^*(a)$ is strictly decreasing in conditional survival at every age $t > a$ and nonincreasing in T. □

Three remarks qualify the lemma. First, the identity holds before the NRA with the corresponding increment: a one-year delay at any age a lowers the present value of benefits if and only if $\Delta b(a)/PIA < [b(a)/PIA]/\ddot{a}(a)$, the benefit forgone divided by the annuity factor. Under the schedules of Table D.1, $\Delta b(a)/PIA$ is $1/15$ in the three years before the NRA, $1/20$ in earlier years, δ after the NRA, and x at every age under the symmetric reforms. Second, only conditional survival $\mathcal{S}(t)/\mathcal{S}(a)$ enters $\delta^*(a)$, so the lemma holds with survival probabilities conditional on any information available at age a, in particular on health at a. Third, the lemma abstracts from the earnings test and from changes in the PIA. Additional earnings after age 60 raise the AIME only if they exceed its current value (5), and benefits withheld by the earnings test are credited as if claiming had been delayed (Section 2.6); both add terms to the fiscal value of a delay by a worker who keeps working.

E.2 Health and the Fair Credit

Order health states from best to worst, $h \in \{0,1,2\}$, and write conditional survival from age a to age $t > a$ as $\mathcal{S}(t)/\mathcal{S}(a) = \mathbb{E}\left[\prod_{k=a+1}^{t} S(h_{k-1},k) \,\middle|\, h_a\right]$, where the expectation is over the health paths generated by the transition probabilities $\pi_{j,i,k}$. Suppose that (i)

one-period survival $S(h,k)$ is nonincreasing in h at every age, and (ii) health transitions are stochastically monotone: for every age k and state j, $\Pr(h_k \geq j \mid h_{k-1} = i)$ is nondecreasing in i. Then an individual in health h_a has weakly higher conditional survival at every horizon than an individual in health $h'_a > h_a$, and hence, by Lemma 1, a weakly lower fair credit $\delta^*(a)$.

To see this, construct both health paths from common uniform draws U_k, setting h_k equal to the smallest j such that $\Pr(h_k \leq j \mid h_{k-1}) \geq U_k$. By (ii), the conditional distribution functions are ordered, so $h_{k-1} \leq h'_{k-1}$ implies $h_k \leq h'_k$; by induction, the path that starts in better health is never in worse health. By (i), each factor $S(h_{k-1}, k)$ is then weakly larger along that path, and so is the product. Taking expectations gives the result. In the estimated model, one-period survival is lower in poor health and with a disability at every age (panel B of Figure 1), which is condition (i).

E.3 Retirement Outlays under a Change in the Claiming-Age Schedule

Consider a population of N individuals indexed by i, with primary insurance amounts PIA_i, survival probabilities $\mathcal{S}_i(\cdot)$, and annuity values $A_i(\cdot)$ as in Lemma 1. Under the status-quo schedule $b_i^0(\cdot)$, individual i claims at age a_i^0; under the reformed schedule $b_i^1(\cdot)$, at age a_i^1. Write $B_i^j(a) = b_i^j(a) A_i(a)$. Holding each PIA fixed and abstracting from the earnings test, the change in the per capita present value of retirement outlays is, in expectation over survival,

$$\Delta SS = \underbrace{\frac{1}{N}\sum_i \left[b_i^1(a_i^0) - b_i^0(a_i^0)\right] A_i(a_i^0)}_{\text{mechanical}} + \underbrace{\frac{1}{N}\sum_i \left[B_i^1(a_i^1) - B_i^1(a_i^0)\right]}_{\text{behavioral}}.$$

For an individual who delays, $a_i^1 > a_i^0$, the behavioral term telescopes into one-year steps, $B_i^1(a_i^1) - B_i^1(a_i^0) = \sum_{a=a_i^0}^{a_i^1-1} \left[B_i^1(a+1) - B_i^1(a)\right]$, each valued by the identity of the proof of Lemma 1 under the new schedule; for an individual who claims earlier, it is minus the corresponding sum from a_i^1 to $a_i^0 - 1$.

For an increase in the DRC from δ to $\delta + d\delta$, $b_i^1(a) - b_i^0(a) = (a - NRA)\, PIA_i\, d\delta$ for $a > NRA$ and zero otherwise, so the mechanical term is

$$\frac{1}{N}\sum_i \mathbf{1}\{a_i^0 > NRA\}\,(a_i^0 - NRA)\, PIA_i\, A_i(a_i^0)\, d\delta \;\geq\; 0,$$

the additional benefits of the inframarginal late claimers. Each one-year delay after the NRA contributes a negative amount to the behavioral term if and only if $\delta + d\delta < \delta_i^*(a)$, evaluated under the new schedule, and each delay before the NRA if and only if $\Delta b_i^1(a) <$

$b_i^1(a)/\ddot{a}_i(a)$, where $\Delta b_i^1(a)$ is the increment of the new schedule at age a. The direct cost of the reform is negative if and only if the behavioral term is negative and larger in absolute value than the mechanical term. For an increase in the NRA, $b_i^1(a) < b_i^0(a)$ at every claiming age from 62 to 70 (Table D.1), so the mechanical term is negative whatever the claiming ages, and behavioral responses add to or subtract from this saving. In both cases, the earnings test and the effect of additional earnings on the PIA add terms to ΔSS, and the effects on taxes, DI, and transfers enter the indirect cost.

E.4 Mechanical Effects, Fiscal Externalities, and Program Accounts

Let ζ^0 and ζ^1 denote the rules of the status quo and of the reform, and $\mathcal{B}^0$ and $\mathcal{B}^1$ the simulated histories of the cohort under them: hours, claiming dates, and asset holdings at every age, together with the health, survival, and wage draws, which are the same under both. Let $\Gamma(\zeta, \mathcal{B})$ be the per capita present value at age 62 of the cost aggregate of the reform family—G, $G + MC^{tg}$, or $G + K$—computed by applying the rules ζ to the histories $\mathcal{B}$. Then

$$NC = \Gamma(\zeta^1, \mathcal{B}^1) - \Gamma(\zeta^0, \mathcal{B}^0) = \underbrace{\Gamma(\zeta^1, \mathcal{B}^0) - \Gamma(\zeta^0, \mathcal{B}^0)}_{ME} + \underbrace{\Gamma(\zeta^1, \mathcal{B}^1) - \Gamma(\zeta^1, \mathcal{B}^0)}_{FE},$$

which is the first equality in (16): the mechanical effect applies the new rules to unchanged behavior, and the fiscal externality is the change in cost caused by the change in behavior under the new rules.

The cost aggregate is a sum of signed program accounts, $\Gamma = \sum_p \Gamma_p$, with outlays entering positively and revenues negatively. Let o denote the program the reform changes. The direct cost is $DC = \Delta\Gamma_o = ME_o + FE_o$ and the indirect cost $IC = \sum_{p \neq o} \Delta\Gamma_p = ME_{-o} + FE_{-o}$, with each term defined as above for the corresponding accounts. Summing gives the second equality in (16). The mechanical effect on other programs, ME_{-o}, is nonzero when programs interact at given behavior. For example, the consumption floor (11) falls one for one with benefits for an individual whose transfer is positive both before and after the reform, so a benefit increase lowers transfers mechanically.

For a small reform, the fiscal externality can be attributed to margins to first order: it is the sum, over the choices that the reform changes, of the induced change in each choice times its fiscal value, the change in the cost aggregate Γ that the choice causes when the individual's other choices are held fixed. The fiscal value of a one-year delay of the claim is the change in outlays of Lemma 1, together with the changes in transfers and in benefits withheld by the earnings test. The fiscal value of participation at age t is minus

the participation tax,

$$\tau_t^P = \left[T_t(y_t^1) - T_t(y_t^0)\right] + T_t^p(w_t n_t) + \left(db_t^0 - db_t^1\right) + \left(tr_t^0 - tr_t^1\right) + \left(ss_t^0 - ss_t^1\right) - \Delta B_t^{aime},$$

where superscripts 1 and 0 denote values when the individual works n_t hours and when she does not work, the term in ss_t is the withholding by the earnings test of a beneficiary, and ΔB_t^{aime} is the present value of the increase in future retirement and disability benefits caused by the year's earnings, through the AIME and through the crediting of benefits withheld by the earnings test (see Kleven and Kreiner 2006, for participation tax rates). The fiscal value of an additional dollar of assets is minus the present value of the income taxes on the additional interest income and of the reduction in means-tested transfers. For nonmarginal reforms, such an attribution depends on the order in which margins are changed; the program accounts do not.

E.5 The Willingness to Pay for a Small Reform

Index the rules by a scalar ζ that affects only the resources available at each age,

$$R_t \equiv a_t + Y_t + b_t\, ss_t + db_t + tr_t - m_t,$$

so that $c_t = R_t - a_{t+1}$ by (10); every reform of Section 4 has this form, since benefits, taxes, out-of-pocket medical expenses, and the return on assets enter only R_t. Treat hours n_t, claiming b_t, and next-period assets a_{t+1} as the choices. Holding them fixed, a change in ζ changes consumption one for one with resources, $dc_t = (\partial R_t/\partial\zeta)\, d\zeta$, where $\partial R_t/\partial\zeta$ is the mechanical change in resources at age t, including the offsetting change in consumption-floor transfers. Because individuals choose optimally, the envelope theorem gives

$$\frac{dV_{30}(X_{30})}{d\zeta} = \mathbb{E}_{30} \sum_{t=30}^{T} \beta^{t-30} \left(\prod_{k=31}^{t} s_k \right) u_c(c_t, l_t)\, \frac{\partial R_t}{\partial \zeta},$$

evaluated at the optimal choices. The theorem applies with discrete participation and claiming choices because V_{30} is the upper envelope of the values of the discrete alternatives and is differentiable at almost every ζ (Milgrom and Segal 2002); individuals who switch alternatives in response to the reform are indifferent between them at the margin, so their switch has no first-order effect on welfare.

Homogeneity of (1) implies $u_c(c,l)\, c = \gamma(1-\nu)\, u(c,l)$, so that $\gamma(1-\nu)\, \bar{V}^0$ equals the average over individuals of $\mathbb{E}_{30} \sum_t \beta^{t-30} (\prod_k s_k)\, u_c(c_t, l_t)\, c_t$. Differentiating (13) at the status

quo, the CE of a small reform $d\zeta$ is

$$dCE = \frac{d\bar{V}/d\zeta}{\gamma(1-\nu)\,\bar{V}^0}\,d\zeta = \frac{\overline{\mathbb{E}_{30}\sum_t \beta^{t-30}\big(\prod_k s_k\big)\,u_c(c_t,l_t)\,\partial R_t/\partial\zeta}}{\overline{\mathbb{E}_{30}\sum_t \beta^{t-30}\big(\prod_k s_k\big)\,u_c(c_t,l_t)\,c_t}}\,d\zeta,$$

where the bars denote averages over individuals. The numerator is the marginal-utility-weighted mechanical transfer, the willingness to pay of Hendren (2016) expressed in utility units; the denominator is the marginal-utility-weighted consumption stream, which converts it into a uniform proportional change in consumption. Behavioral responses do not enter. The dollar willingness to pay of (14), $WTP^{\$} = \bar{C}_s^0\,dCE$, and the value of consumption smoothing per mechanical dollar, $WTP^{\$}/ME$, depend both on how the transfer is distributed across states and ages relative to consumption and on the base $\bar{C}_s^0$.

E.6 The Fiscal Multiplier and the Desirability of a Reform

Combine a reform with net cost NC with a change in another policy that balances the budget: if $NC > 0$, the other policy is cut so as to raise NC; if $NC < 0$, it is expanded so as to spend $-NC$. Let $MVPF_f \geq 0$ be the MVPF of that policy at the margin, so that each dollar raised by cutting it costs its beneficiaries $MVPF_f$ dollars of willingness to pay and each dollar spent on it delivers $MVPF_f$ dollars. In both cases, the combined change raises welfare, in dollars, by

$$WTP^{\$} - MVPF_f\,NC.$$

For $NC > 0$, this is positive if and only if $MVPF > MVPF_f$. With $FM = NC/ME$ and $ME > 0$, $MVPF = (WTP^{\$}/ME)/FM$, so the condition is

$$\frac{WTP^{\$}}{ME} > MVPF_f \times FM.$$

When the financing policy costs its payers one dollar of willingness to pay per dollar raised, $MVPF_f = 1$, and the condition compares the value of consumption smoothing with the fiscal multiplier, the two terms of the Baily–Chetty form used by Haller (2022). For $WTP^{\$} > 0$ and $NC \leq 0$, the combined change raises welfare for every $MVPF_f \geq 0$; for $ME > 0$, this is the case $FM \leq 0$, in which the behavioral responses finance the entire mechanical cost.

F Additional Results

This appendix reports the reforms of the upper replacement rates of the PIA formula and the participation profiles under the instrument eliminations of Section 5.1.

Upper replacement rates. Table F.1 reports reforms that raise the second replacement rate of the PIA formula from 0.32 to 0.352 (P2.0–P2.9) and then the third from 0.15 to 0.165 (P3.0–P3.9), in steps of 1 percent of the initial rate. All of them keep the first rate at 0.99, the value under P1.9, so each row, MVPF included, reports the total effect relative to the status quo, and the effect of the upper rates is the difference between a row and P1.9 in Table 5. Across P2.0–P2.9, the cumulative MVPF falls from 0.56 to 0.29: WTP rises from 0.83 to 0.95 percent, while the net cost more than doubles, from 0.57 to 1.27 percent. Relative to P1.9, P2.9 adds 0.15 percent to WTP and 0.83 percent to the net cost. Raising the third rate changes neither WTP nor the net cost by more than a few hundredths of a percentage point, within the precision of the simulations, and the MVPF stays at 0.29–0.30. Participation at 30–49 rises by 0.24 to 0.28 points, and participation at 62–64 and 65–69 falls by 0.83 to 2.41 points, more than under any bottom-bracket reform. P3.9 and L9 set the same three rates, (0.99, 0.352, 0.165), and their rows coincide.

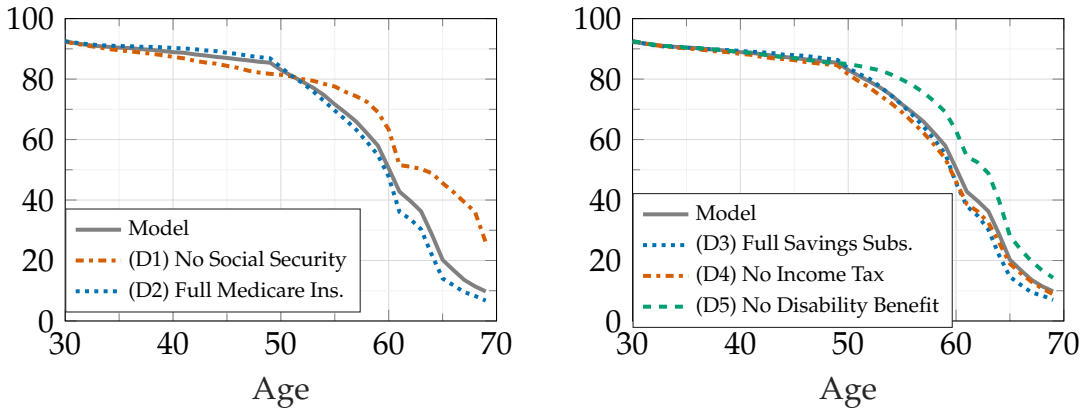
Participation under the instrument eliminations. Figure F.1 shows the participation profiles of the whole cohort under the five eliminations of Table 3, and Figure F.2 shows them by health status. Eliminating Social Security lowers participation before age 50, by up to about 4 percentage points in the aggregate and by up to about 6 and 9 points in poor health and with a disability, and raises it afterward in every health state: from age 50 for individuals with a disability, from the early 50s for those in poor health, and from the mid-50s for those in good health. Eliminating DI leaves participation unchanged before age 50 and raises it afterward in every health state, most for individuals with a disability, for whom DI is the main pathway out of the labor force. Full Medicare coverage, the full savings subsidy, and the abolition of the income tax lower aggregate participation at older ages.

Table F.1. Reforms of the Upper Replacement Rates

	PIA rates	Welfare		Fiscal cost (% change)			Δ Participation (pp)		
		MVPF	WTP (%)	Net	Direct	Indirect	30–49	62–64	65–69
A. Raising the second PIA rate (first rate at 0.99)									
(P2.0)	(0.99, 0.3232, 0.15)	0.56	0.83	0.57	4.75	−0.07	0.25	−1.10	−0.83
(P2.1)	(0.99, 0.3264, 0.15)	0.50	0.86	0.67	5.28	−0.05	0.24	−1.29	−1.03
(P2.2)	(0.99, 0.3296, 0.15)	0.49	0.88	0.70	5.99	−0.11	0.25	−1.05	−1.01
(P2.3)	(0.99, 0.3328, 0.15)	0.44	0.90	0.78	6.56	−0.10	0.24	−1.37	−1.15
(P2.4)	(0.99, 0.336, 0.15)	0.38	0.92	0.93	7.21	−0.05	0.24	−1.60	−1.63
(P2.5)	(0.99, 0.3392, 0.15)	0.37	0.93	0.98	7.95	−0.10	0.27	−1.63	−1.58
(P2.6)	(0.99, 0.3424, 0.15)	0.36	0.95	1.03	8.46	−0.11	0.28	−1.47	−1.68
(P2.7)	(0.99, 0.3456, 0.15)	0.33	0.94	1.11	8.99	−0.10	0.27	−1.74	−1.99
(P2.8)	(0.99, 0.3488, 0.15)	0.31	0.95	1.19	9.49	−0.10	0.28	−2.29	−2.10
(P2.9)	(0.99, 0.352, 0.15)	0.29	0.95	1.27	9.97	−0.09	0.28	−2.26	−2.05
B. Raising the third PIA rate (first two rates at 0.99 and 0.352)									
(P3.0)	(0.99, 0.352, 0.1515)	0.29	0.95	1.26	10.04	−0.10	0.28	−2.22	−2.05
(P3.1)	(0.99, 0.352, 0.153)	0.29	0.96	1.27	10.13	−0.10	0.28	−2.30	−2.07
(P3.2)	(0.99, 0.352, 0.1545)	0.29	0.96	1.28	10.19	−0.10	0.28	−2.30	−2.07
(P3.3)	(0.99, 0.352, 0.156)	0.29	0.96	1.26	10.26	−0.13	0.28	−2.25	−2.03
(P3.4)	(0.99, 0.352, 0.1575)	0.29	0.96	1.27	10.34	−0.13	0.28	−2.23	−2.03
(P3.5)	(0.99, 0.352, 0.159)	0.29	0.97	1.29	10.41	−0.12	0.28	−2.41	−2.02
(P3.6)	(0.99, 0.352, 0.1605)	0.29	0.97	1.27	10.48	−0.14	0.28	−2.39	−2.02
(P3.7)	(0.99, 0.352, 0.162)	0.30	0.97	1.27	10.55	−0.15	0.28	−2.39	−2.02
(P3.8)	(0.99, 0.352, 0.1635)	0.30	0.97	1.25	10.63	−0.18	0.28	−2.37	−2.01
(P3.9)	(0.99, 0.352, 0.165)	0.30	0.98	1.24	10.70	−0.20	0.28	−2.37	−1.99

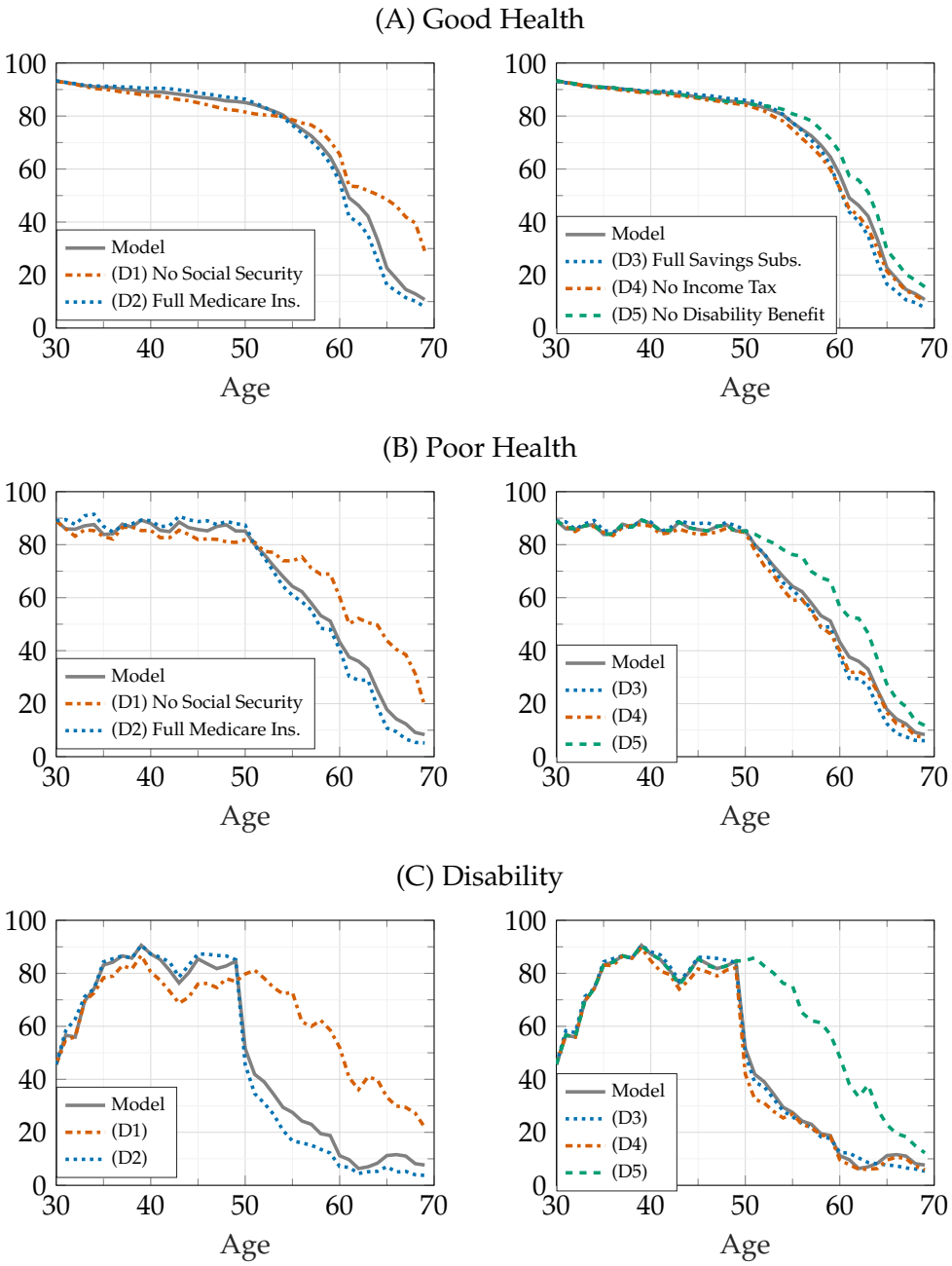
Notes: Each row reports the outcomes of simulating the estimated model for the cohort under the reform, announced before age 30 and in force over the life cycle, with prices held fixed (partial equilibrium). The second column gives the three replacement rates of the PIA formula; the status quo is (0.90, 0.32, 0.15). P2.0–P2.9 raise the second rate in steps of 0.32 percentage points with the first rate at 0.99; P3.0–P3.9 raise the third rate in steps of 0.15 percentage points with the first two rates at 0.99 and 0.352. Every row includes the increase of the first rate to 0.99 (P1.9 in Table 5) and reports the change from the status quo. WTP: ex ante consumption-equivalent welfare change at age 30, in percent. Direct: percentage change in the present value of Social Security retirement benefits. Indirect: percentage change in the present value of all other taxes and transfers (DI benefits and consumption-floor transfers, net of income- and payroll-tax revenue). Net: percentage change in the present value of government spending. Present values are per capita and discounted to age 62 (Section 4.3); because the three fiscal columns have different bases, Net is not the sum of Direct and Indirect. MVPF: WTP valued with the present value of per capita baseline consumption from age 62, divided by the dollar change in government spending (equation (14)). Δ Participation: change in the labor force participation rate at the indicated ages, in percentage points. Abbreviations: AIME, average indexed monthly earnings; DI, disability insurance; PIA, primary insurance amount.

Figure F.1. Participation under the Elimination of Policy Instruments



Notes: Labor force participation rate (percent) by age, simulated from the estimated model (Model) and from counterfactual economies that eliminate one policy instrument. Left panel: the *No Social Security* experiment (D1) sets Social Security benefits to zero; the *Full Medicare Insurance* experiment (D2) sets out-of-pocket medical expenses after age 65 to zero for participants and nonparticipants. Right panel: the *Full Savings Subsidy* experiment (D3) sets the net return on assets equal to the pretax return; the *No Income Tax* experiment (D4) sets the income tax to zero; the *No Disability Benefit* experiment (D5) sets disability benefits to zero. The *No Social Security* experiment produces the largest increase in participation, followed by the *No Disability Benefit* experiment. Welfare effects are in Table 3.

Figure F.2. Participation under the Elimination of Policy Instruments, by Health Status



Notes: Labor force participation rate (percent) by age and health status, simulated from the estimated model (Model) and from counterfactual economies that eliminate one policy instrument. Rows (A), (B), and (C) show individuals in good health, in poor health, and with a disability. Left panels: D1 sets Social Security benefits to zero; D2 sets out-of-pocket medical expenses after age 65 to zero for participants and nonparticipants. Right panels: D3 sets the net return on assets equal to the pretax return; D4 sets the income tax to zero; D5 sets disability benefits to zero.